When Are Robust RAG Results Comparable? A Survey with an Empirically Tested Taxonomy

Anonymous ACL submission

Abstract

Robust retrieval-augmented generation (RAG) has produced a literature in which papers share labels—noise robustness, faithfulness, security, calibration—while answering different evaluation questions. Existing surveys organise this literature conceptually, but none tests whether its own categories can be recovered from published prose. We survey the field across four failure surfaces (evidence, model, adversary, system control) and reduce them to five *reporting coordinates* that decide when two numbers are comparable: evidence state, origin and capability, intervention, outcome and unit, and policy. On constructed passages that state these coordinates, independent judges recover them with 0.95 accuracy (κ 0.74). Applying the same instrument at scale, a deterministic screen over 259 full texts yields 1,703 candidate cross-paper comparisons (704 eligible, 610 classified by provenance), of which 533 re-implement baselines inside one harness rather than transporting published numbers—a structural property automated judges recover reliably (0.91 [0.86, 0.96] accuracy). Deciding whether a quoted gap survives comparison is far harder: model panels reach only κ 0.06 against a ceiling of 0.46, and the gap does not close with full paper text, an abstention option, 9 open-weight models, or non-generative classifiers. On 80 published claims annotated over 210 person-hours, human coders reach κ 0.35 [0.16, 0.49] (ceiling 0.49), no better than automated panels (+0.15 [-0.00, 0.32]). The bottleneck is underspecified reporting, not evaluator quality. We release the screening frame, evidence map, human annotations, model codings, and verification tools.

1 Introduction

Retrieval-augmented generation conditions a generator on evidence fetched at inference time (Lewis et al., 2020; Karpukhin et al., 2020; Guu et al., 2020). Because that evidence is chosen by a retriever rather than by the model, a large literature now studies what happens when it is irrelevant, insufficient, conflicting, stale, weakly sourced, or deliberately malicious, and several recent surveys organise this work by mechanism (Gao et al., 2024; Yu et al., 2024; Zhou et al., 2024; Xu et al., 2026, 2024a; Wen et al., 2025a). Those taxonomies are useful for finding related work, but each is offered as a conceptual scheme: none reports whether its categories can in fact be recovered from the papers it classifies.

Whether they can matters, because readers of this literature do not only browse it—they compare its numbers. Consider one juxtaposition. PoisonedRAG reports a "97% ASR by injecting 5 malicious texts for each target question" on Natural Questions and, in its Table 4, sets that beside four baselines, the weakest of which—the corpus poisoning attack of Zhong et al. (2023)—appears at 0.01 (Zou et al., 2025). Read as a ranking, the message is that corpus poisoning barely works. Yet the corpus poisoning paper reports getting an adversarial passage into the retrieved set for 84.2% of Natural Questions queries using a single injected passage.

Both numbers are correct, and neither paper misreports anything. They share a corpus and a Contriever retriever, but they differ in the size of the retrieved set (top-20 versus five texts) and, decisively, in the event being counted. Corpus poisoning scores retrieval success—"the percentage of queries for which at least one adversarial passage is retrieved in the top-k results" (Zhong et al., 2023, l. 333–337)—whereas PoisonedRAG scores end-to-end generation success, that is, whether the model emits the attacker's target string. PoisonedRAG says as much, noting that the weakest baselines "are ineffective because they cannot achieve generation, generation, and retrieval condition, respectively" [sic] (Zou et al., 2025, l. 1338–1341); we reproduce the sentence as printed and, together with six further checks behind Figure 1, verify it

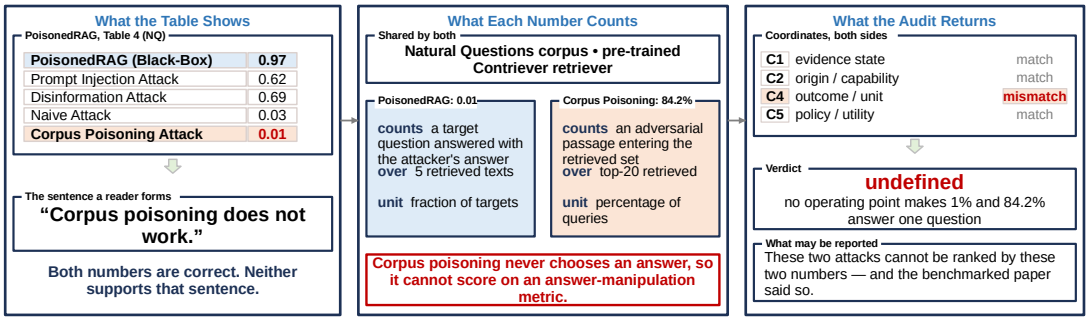


Figure 1: One case, in full. Two numbers that share a corpus and a retriever and still cannot be ranked, because they count different events in different units over different retrieved sets. Every value and every quoted condition is re-opened at its cited line range by a released verifier (7/7 checks pass).

against the source with a released tool (7/7). Inside PoisonedRAG's own harness the 0.01 is exactly right; carried across the harness boundary it is a comparison between two different events. Cases like this are why we think a survey of this field should say when two published numbers answer the same question, instead of ranking them.

Contributions. **(i)** We survey robust RAG along four failure surfaces—evidence, model, adversary and system control—organised around the contrasts between papers rather than around a list of methods (§4–§7). **(ii)** We reduce those surfaces to five *reporting coordinates* and an explicit three-way decision procedure that says when a quoted gap survives, narrows, or is undefined (§8). **(iii)** We then *test the taxonomy as an instrument*: coordinates are recoverable from prose that states them (0.95 accuracy, κ 0.74), and we characterise where recovery breaks down, including an arm on which a constant baseline beats every model judge (§9). **(iv)** We apply it at scale to 1,703 candidate cross-paper comparisons screened from 259 full texts, finding that 533 of 610 classified pairs re-implement their baselines locally rather than transporting numbers, and that adjudicating whether a gap survives stays underdetermined for automated panels and human experts alike (§10). **(v)** We derive reporting guidelines from those measurements rather than from intuition (§11).

Relation to prior audits. Asking whether published numbers are comparable has precedent outside RAG. Marie et al. (2021) hand-coded 769 machine translation papers and found comparisons drawn across incomparable configurations to be routine, and Bowman and Dahl (2021), Gehrmann et al. (2023) and Howcroft et al. (2020); van der Lee et al. (2021); Clark et al. (2021) document analogous problems in benchmarking and human evaluation. What is new here is the combination: no RAG survey has tested whether its own categories are extractable from paper text, and few audits report the point at which adjudication becomes underdetermined rather than merely difficult. Since our pipeline leans on LLM judges, whose reliability is itself contested (Zheng et al., 2023; Panickssery et al., 2024; Wang et al., 2024a), we treat evaluator reliability as a quantity to be measured—with human expert panels and developer-independence checks—rather than as an assumption.

2 Scope, Protocol and Evidence Base

Scope. We survey work in which a retrieval-augmented generator has to behave sensibly when the evidence it receives is imperfect. The relevant conditions are irrelevant retrieval, insufficient context, conflicts between sources or between a source and the model's own parameters, stale or weakly attributed material, and deliberately corrupted passages. Two neighbouring literatures fall outside this scope: retrieval studies that never generate an answer, and generation studies that never retrieve. Neither exhibits the interaction we care about, namely a generator committing to an answer on the basis of evidence it did not choose and cannot fully trust.

Search and screening. We ran 55 arXiv queries together with five venue and repository sources at snapshot 2026-06-03, returning 1,904 records. Of these, 1,445 had open PDFs, 1,399 survived deduplication, 536 passed relevance screening, and 259 full texts were extracted and indexed into overlapping windows. The screening rules and the scripts that apply them are released, and the PRISMA flow is reproduced as Appendix Figure 7. One stage is

documented but not re-executable: per-query hit counts were not logged at the snapshot date and live indices drift, so everything from the 259 extracted texts downward can be re-run, while the population those texts were drawn from cannot be reconstructed exactly. We therefore keep two activities apart throughout. The narrative synthesis of §4–§7 draws on the broader literature and cites it directly; every number we report is stated for the screening frame, never for the field.

Frame coverage and audit. Because a frame that missed most of the field would make even frame-local numbers hard to read, we audited it against outside evidence. We took the bibliographies of 6 prior surveys (§8), title-screened them for robust-RAG scope, and matched 95 unique references against our index. Recall is moderate rather than high: 47% [0.38, 0.57] of those references reach our 259 full texts. The misses follow a systematic pattern—they name a specific system architecture in the title without using any scope term—which is why Sections 4–7 cite the wider literature rather than the frame alone (Appendix Table 26). The narrative rests on a released 342-row evidence map recording, for each paper, the failure surface it addresses, the intervention it makes, the benchmarks it uses, and the metrics it reports. Because that map is preprint-heavy, we re-check every empirical claim on the peer-reviewed stratum alone (137 rows; Appendix Table 13).

3 The Four Failure Surfaces

A RAG system is best read as an evidence-conditioned decision process: a query arrives, a retriever assembles a context, and a generator commits to an action on the basis of that context. Failures occur when the assembled evidence is incomplete, irrelevant, misleading, mutually contradictory, corrupted, or simply insufficient to support any answer (Appendix I). Sorting the literature by *where* in this process a paper intervenes yields four failure surfaces, which we use as the top-level organisation of the survey and summarise in Figure 2.

Evidence-centric work (§4) treats the composition of the context as the object of study, asking which passages reach the generator and in what order. **Model-centric** work (§5) holds the context fixed and asks how the generator uses it, which is where faithfulness, attribution and knowledge conflict live. **Adversarial** work (§6) replaces the assumption of benign noise with an adaptive attacker who chooses the evidence. **System-centric** work (§7) steps outside the single forward pass and asks what the system should *do*—retrieve again, verify, abstain, or answer—under uncertainty and cost.

The four surfaces partition the literature, not the phenomena, and a single event routinely touches several at once. A poisoned passage is an evidence state, an adversary capability, and—if the system declines to answer because of it—a policy decision. That overlap is exactly why placing two papers on the same surface does not make their numbers comparable, and it motivates the reduction to five reporting coordinates in §8. Evaluation practice cuts across all four surfaces and is treated separately in Appendix G; the rules used to assign a paper to a surface are in Appendix L.

4 Evidence-Centric Robustness

Evidence-centric work takes the context window itself as the object of study: what the retriever puts in front of the generator, in what order, and at what budget (Appendix C). The stakes are higher here than in classical retrieval, because the retrieved set is not a ranked list a user inspects but a prompt a model reads, and a single misleading passage can move the answer, the stated confidence, and the citations all at once.

Noise is not one condition. The most consequential distinction in this literature is between failing to fetch the right evidence and failing to ignore the wrong evidence. Yoran et al. (2024) separate the two explicitly, and Fang et al. (2024) show that the second is far harder than it looks: clearly off-topic passages are discarded easily, whereas semantically close counterfactuals and hard negatives are not. RGB makes the same point by construction, scoring noise robustness, negative rejection and counterfactual handling as three separate abilities rather than one (Chen et al., 2024a). The practical consequence is that a single robustness-under-noise number is uninterpretable on its own, because two papers reporting it may be stressing different abilities.

Context construction has negative marginal utility. Adding passages is not monotone. Cuconasu et al. (2024) find that topically related but wrong passages hurt generation while purely random text can occasionally stabilise it, and Liu et al. (2024) show that the same passage helps or hurts depend-

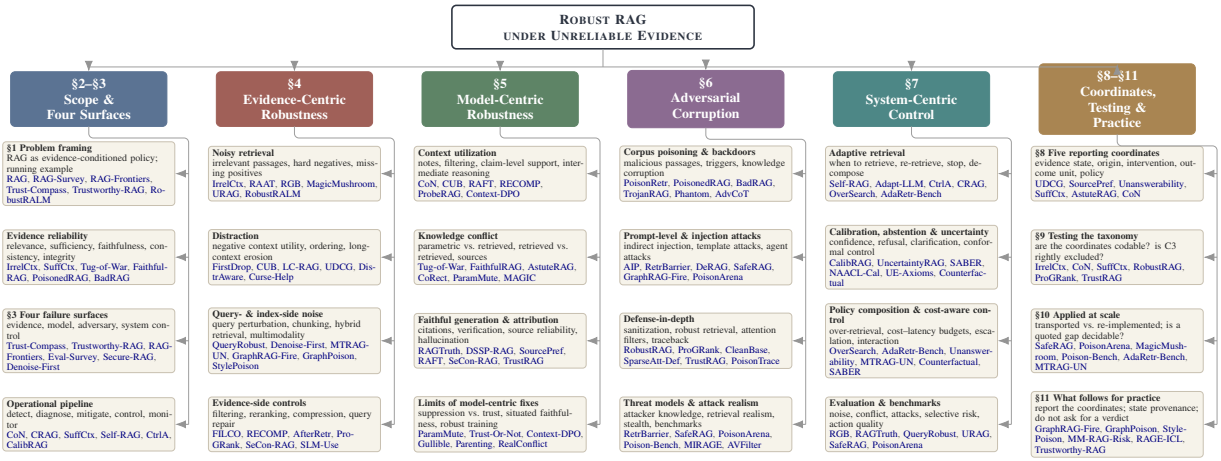


Figure 2: **Outline of this survey.** A top-down tree of the survey structure. The root is robust RAG under unreliable evidence; the trunks are the four failure surfaces of §3 together with the scope (§2) that fixes them and the reduction (§8–§11) they support. Each leaf lists representative cited work using compact aliases; the full-length treatment of every surface is in Appendix Part I. Colours denote themes: ■ scope, ■ evidence, ■ model, ■ adversary, ■ system control, ■ reduction.

ing on where it sits in the prompt. Longer contexts do not resolve this: hard negatives and misleading content continue to degrade reasoning as the window grows (Jin et al., 2025; Gao et al., 2026). Benchmarks such as CUB (Hagström et al., 2026) and UDCG (Trappolini et al., 2026) follow this observation to its conclusion and score retrieval by downstream utility rather than by relevance, treating the retriever as an active constructor of the model's input rather than as a passive supplier of documents.

Mitigations trade one failure for another. Evidence-side defences—filtering (Wang et al., 2023), compression and reordering (Jin et al., 2025), utility-aware reranking (Trappolini et al., 2026), and noise-aware training (Fang et al., 2024)—all buy precision with recall, so a defence that improves accuracy under distraction may lose answers that a noisier pipeline would have found. Interventions on the query and index side, such as rephrasing (Percin et al., 2025) or chunking and graph construction (Rosenthal et al., 2026; Liang et al., 2026), add failure surfaces of their own. None of these results transfers across papers unless the corpus snapshot, retriever, and top-k are stated, which is the content of coordinate C1 in §8.

5 Model-Centric Robustness

Model-centric work holds the retrieved set fixed and asks what the generator does with it (Appendix D). Given identical context, models differ in how closely they follow instructions, how strongly they defer to their parametric priors, how much of the prompt they actually consult, and how well their stated confidence tracks correctness.

Availability is not utilisation. A passage being present in the prompt does not mean it was used, and an unsupported passage may drive the answer more than the gold one. Chain-of-Note makes this explicit by summarising each passage, judging its relevance, and abstaining when the set is insufficient (Yu et al., 2023), while benchmarks such as CUB score whether a model leans on gold evidence while ignoring plausible distractors (Hagström et al., 2026). Reporting only end-task accuracy collapses the two, so a system that answers correctly for the wrong reason is scored as robust.

Knowledge conflict is structural, not incidental. Retrieval augmentation puts a model's memory, its retrieved context, and multiple retrieved sources in the same prompt, and these can disagree. Longpre et al. (2021) formalised the first case through entity substitution, and Xie et al. (2024a) showed that models comply with coherent counter-evidence but resist incoherent claims, favouring what they already believe. Two opposite failures follow—over-compliance with false retrieved content and over-reliance on stale memory—and a study that fixes one can worsen the other (Jin et al., 2024). Mitigations differ in where they act: localising the disagreement (Zhang et al., 2025d), editing activations (Ma et al., 2026), or aggregating evidence weighted by source reliability (Wang et al., 2025b). Because *resolving conflict* can mean answering correctly, following the context, or reconciling sources,

papers that share the label often optimise different objectives.

Citing is not grounding. Source-preference frameworks ask which evidence wins under conflict (Schuster et al., 2026; Pham et al., 2024), and a recurring finding is that a produced citation is weak proof of support: models cite passages that are topically aligned but do not entail the claim (Niu et al., 2024). Reference-free evaluation suites accordingly separate faithfulness, relevance, and precision instead of merging them (Es et al., 2024; Saad-Falcon et al., 2024). Since all of these quantities are defined relative to the evidence the model saw, comparing two generators is only meaningful when the evidence state is held constant on both sides.

6 Adversarial Corruption and Security

Adversarial work removes the assumption that unreliable evidence arrives by accident and replaces it with an attacker who chooses the evidence (Appendix E). Because RAG externalises knowledge into corpora, indices, prompt templates and tools, every one of these becomes an attack surface that a purely parametric model does not have.

Poisoning turns small corpus edits into large output changes. Zhong et al. (2023) showed that a handful of crafted passages is enough to distort a dense retriever's neighbourhood structure, and PoisonedRAG carried the same idea through to the generated answer (Zou et al., 2025). Later work varies the trigger and the persistence model: BadRAG studies triggered retrieval (Xue et al., 2024), while TrojanRAG and Phantom induce attacker-chosen behaviour through retrieved context alone, without touching model weights (Cheng et al., 2024; Chaudhari et al., 2024). What these share is an amplification effect: retrieval converts a marginal perturbation of the corpus into a first-class input to the model.

Injection and poisoning are different threats. Indirect prompt injection plants *executable instructions* rather than false evidence, and the payload must first clear a retrieval barrier before it can act (Chang et al., 2026); AIP instead exploits prompts shared across users (Chaturvedi et al., 2025). The distinction matters for defence design, because a filter tuned to detect implausible claims will not detect a plausible document that ends with an instruction. Treating templates, ingestion pipelines and retrieved text as untrusted boundaries is therefore a stronger design stance than treating the corpus as the only thing to sanitise.

Defences are not comparable across threat models. Proposed defences span certified aggregation (Xiang et al., 2024), adversarial reranking (Yin et al., 2026), attention-level filtering (Choudhary et al., 2026), and post-hoc traceback of the responsible passage (Zhang et al., 2025a), and each is evaluated against the threat its authors assumed. Two further sources of divergence are less visible. Papers differ in what the attacker is allowed to know and do, and they differ in the unit that the attack budget is counted in—documents injected into the corpus versus documents actually retrieved into the context—so nominally identical budgets can differ by an order of magnitude. Reporting adversary access and budget scope, coordinate C2 in §8, is what makes two attack success rates commensurable. Recent evaluation suites move this way, fixing corpus, retriever and injection budget before pitting attacks against one another (Liang et al., 2025; Chen et al., 2026; Zhang et al., 2025c).

7 System-Centric Control

System-centric work treats RAG as a policy rather than a fixed pipeline (Appendix F). The system decides whether to retrieve at all, how much context to admit, whether to verify, when to fall back on parametric memory, and—most consequentially—whether to answer or decline.

Sufficiency and confidence answer different questions. The negative-rejection condition in RGB shows that systems routinely answer when the evidence does not license an answer (Chen et al., 2024a), and Sufficient Context reframes evaluation around whether the retrieved set can support an answer at all (Joren et al., 2025), a framing that unanswerability benchmarks extend to multi-turn and open-domain settings (Peng et al., 2025; Rosenthal et al., 2026). Keeping the two apart is essential: *evidence uncertainty* asks whether the context supports the query, *model uncertainty* asks whether the generator can use it correctly. High confidence under low sufficiency is hallucination on top of false context, whereas low confidence under adequate evidence calls for verification, not refusal.

The action space is wider than answer-or-abstain. Self-RAG, adaptive-retrieval classifiers,

CtrlA and Corrective RAG all gate retrieval, using reflection tokens, learned predictors, internal confidence signals, or evidence-quality thresholds respectively (Asai et al., 2024; Labruna et al., 2024; Liu et al., 2025b; Yan et al., 2024); because each buys accuracy with calls, they are only comparable on an accuracy–cost plane rather than at a single point (Xie et al., 2026). Beyond gating, a deployed system can ask a clarifying question, re-retrieve, or surface the contradiction instead of resolving it (Rosenthal et al., 2026). This is why a reported "abstention rate" is not yet a result: without the action set and the loss attached to each action—coordinate C5 in §8—the same number can describe a cautious system and a useless one. The difficulty is empirical rather than hypothetical: C5 is not extractable at all from 53% of the text windows in our census.

Beyond one modality and one turn. These questions return in harder form once retrieval spans modalities, tools, or several interacting agents, where one unreliable passage can be reused across turns and propagated between sub-agents (Yu et al., 2025; Park et al., 2024; Liang et al., 2026). That literature is still thin, and it inherits every reporting problem above: the action set grows, error costs become path-dependent, and the unit of measurement stops being a single question.

8 From Four Surfaces to Five Reporting Coordinates

The four surfaces tell a reader where a paper sits, but they do not tell a reader whether two papers' numbers may be placed side by side. For that we need a description of each *side* of a comparison rather than of each paper. We use five reporting coordinates, each one a compression of a question the surfaces above already raise.

C1 *Evidence state* (§4): what the context looks like at the decision point—supported, distracting, insufficient, conflicting, stale, weakly sourced, or manipulated—together with the corpus snapshot, retriever, top-k, and whether a gold passage is present at all.

C2 *Origin and capability* (§6): how that state came about, whether through benign retrieval error, a synthetic stressor, distribution drift, or an adaptive adversary; for attacks, the adversary's access, persistence, budget unit (documents injected versus documents retrieved), and whether the score is over all queries or only triggered ones.

C3 *Intervention* (§5): where the change is made, from the corpus and retriever through the generator to a verifier module or the evaluation artifact.

C4 *Outcome and unit*: what is scored—answer correctness, retrieval quality, claim support, attack success, leakage, calibration error, or cost—and per what unit: a question, a response, or a span.

C5 *Policy and utility* (§7): the recourse available under uncertainty—answer, verify, re-retrieve, expose the conflict, quarantine, or abstain—and the costs attached to being wrong, refusing, and following up.

A decision procedure for a quoted gap. Once both sides are recorded, a gap between them is adjudicated in three steps. **(1)** If C1, C2, C4 and C5 match, the gap *survives* and may be reported as stated. **(2)** If they do not match but one paper reports an operating point matching the other's configuration, the comparison *narrows* to that point and must be restated there. **(3)** Otherwise the gap is *undefined*: the claim must be confined to the common subset or withdrawn, not computed across the mismatch. The third outcome is the one current practice lacks a vocabulary for, and §10 shows how often it is the correct one.

Why C3 is recorded but not gated. The intervention coordinate is deliberately excluded from the rule above, and the reason is definitional rather than empirical: two systems being compared differ in their intervention by construction, so requiring C3 to match would declare every cross-system comparison invalid. We nonetheless record C3 and report in §9 exactly what this choice costs.

Relation to existing survey taxonomies. Prior surveys organise robust RAG by mechanism—noise types (Gao et al., 2024), trust dimensions (Zhou et al., 2024), attack vectors (Xu et al., 2026), conflict types (Xu et al., 2024a), or abstention strategies (Wen et al., 2025a)—and those structures are effective for finding related work. None of them structures the act of comparison, and none tests whether paper text carries enough detail for its own categories to be recovered. Comparability audits of this kind exist outside RAG, in translation, benchmarking and human evaluation (Marie et al., 2021; Howcroft et al., 2020; Gehrmann et al., 2023), and our coordinates adapt long-standing ideas from evidence synthesis, selective prediction and security evaluation (Richardson et al., 1995;

Chow, 1970; Papernot et al., 2018) to this setting (Appendix Table 52).

9 Testing the Taxonomy

A taxonomy that cannot be applied by anyone other than its authors is a description of the authors' intuitions. We therefore treat the coordinates of §8 as an instrument and measure it: can independent coders recover the coordinates from prose, what does excluding C3 actually cost, and are five coordinates doing work that two would not? The coders in this section are LLM judges from several providers; §10 adds a human expert panel.[1]

Coordinates are recoverable when they are stated. We wrote 432 short passages in the style of an experimental-setup paragraph, each generated from a ground-truth specification of whether two settings agree on C1, C2, C4 and C5 and of which of them the text actually discloses; the correct verdict then follows deterministically from the rule in §8. Disclosure varies across three regimes—everything stated; something withheld but the verdict still fixed by what remains (baseline 0.34); and something withheld so that the correct answer is *underspecified* (a degenerate arm, baseline 1.00)—crossed with a checklist-versus-prose presentation factor (Appendix O). On the 180 fully disclosing passages, 6 judges recover coordinate labels with 0.95 accuracy and agree on the verdict at κ 0.74. Over all 428 items the composed rule reaches 0.861 against a constant baseline of 0.336, and writing the same content as natural prose rather than as a checklist costs almost nothing (0.93 to 0.89). The instrument is therefore sound; what follows is about the text it is applied to.

What excluding C3 costs. Since dropping C3 is a definitional choice, we report its consequences rather than defend it. Across 704 eligible windows, majority vote marks C3 a mismatch on 692 instances (99.6% of majority-classified items). Gating on C3 would flip 285 of 704 composite verdicts and cut surviving comparisons from 285 to 0, which is to say it would abolish cross-paper comparison rather than discipline it. One statistic needs care here: inter-judge κ on C3 is near zero, but 87.1% of its labels fall in a single class, and under that skew κ is uninformative. Raw agreement is 0.75, Gwet's AC_1 is 0.72, and $\kappa_{\max}$ is only 0.42: this is the prevalence paradox (Sim and Wright, 2005), not an inability to read intervention descriptions.

Five coordinates versus two. On non-degenerate items the five-coordinate rule reaches 0.861 against 0.832 for the best two-coordinate estimator (paired difference +0.029, exact McNemar $p = 0.00146$). On the degenerate arm ($N = 71$) five coordinates give 0.33 and two give 0.11, yet both lose to a constant predictor that always answers "underspecified" (1.00 by construction)—the clearest single sign that the bottleneck is the source text rather than the schema. On real corpus data a reduced C1/C4 codebook performs comparably (+0.03 difference, [-0.05, +0.12]). We read this as C1 and C4 being the minimal pair a comparison cannot do without, while C2 and C5 earn their place by naming the ambiguity that remains.

10 The Taxonomy Applied at Scale

We now run the same instrument over published text. A deterministic screen across 259 full texts isolates 1,703 passages in which one paper's number is placed beside another's, drawn from 171 papers (Appendix N); majority vote over three model judges keeps 704 of them as genuine cross-paper comparisons (κ 0.51). A second pass asks a structural question about each survivor—were the compared baselines *transported* from their original papers, or *re-implemented* locally?—and a third asks the substantive one: does the quoted gap survive?

Most comparisons are made inside one harness. Of the 610 eligible candidates whose provenance could be classified, 533 re-implement their baselines inside a single corpus, retriever, generator and metric harness, and only 77 carry numbers across a harness boundary (87.4% re-implemented; 86.1% under unanimous gating, 85.8–87.2% across leave-one-out sensitivity). This is a reassuring finding about the field, and it is one automation can be trusted with: judges agree at κ 0.55 and reach 0.91 [0.86, 0.96] accuracy against reference adjudicators who read the full papers (170 windows, baseline 0.79; Appendix Table 27).

Whether a gap survives is not decidable from the text. The substantive question behaves entirely

[1] Throughout, κ is Fleiss' κ over the panel, $\kappa_{\max}$ the largest value attainable given the raters' marginals (Umesh et al., 1989; Sim and Wright, 2005), AC_1 is Gwet's prevalence-robust statistic, and "observed" is raw pairwise agreement. Under skewed labels κ is deflated by construction, so we report all four. Intervals are percentile bootstrap over the stated cluster unit.

differently. Over the same 704 windows, verdict agreement is κ 0.06—13% of the attainable ceiling 0.46, at observed agreement 0.45—and accuracy against reference adjudicators is 0.82 [0.74, 0.89] over 114 quote-verified windows (baseline 0.77). Bounding the 51 windows with no majority gives worst and best cases of 0.58 and 0.87. The failures are not subtle: the weakest judge contradicts its own coordinate assignments on 162 of 704 items (rule fidelity 0.77 versus 1.00 for the strongest), and 21 of 1,879 quoted pieces of evidence (1.1%) cannot be found in the cited source.

The bottleneck is the prose, not the evaluator. To locate the cause we varied one thing at a time with the codebook and items fixed (Figure 11). Supplying the whole paper instead of a window raises agreement by +0.20 [0.09, 0.36], confirming that windowing truncates relevant context but not closing the gap. The remaining interventions barely move it: offering an explicit *underspecified* option shifts agreement by +0.01 [-0.07, 0.11], composing the rule deterministically by +0.02 [0.00, 0.05], 9 open-weight models reach κ 0.09 ($\kappa_{\max} = 0.46$), and a non-generative structural classifier trained on reference verdicts reaches 0.83 against a 0.77 baseline (+0.06 [-0.05, 0.17]).

Finally, we asked whether trained people do better. Three NLP and RAG researchers annotated 80 published claims over 210 person-hours with senior adjudication, reaching κ 0.35 [0.16, 0.49] against a ceiling of 0.49; the model panel scored κ 0.21 on the same items, a difference of +0.15 [-0.00, 0.32]. A power analysis over 9 clusters bounds any human advantage below 0.22, and the experts spread as widely as the models in how many claims they were willing to call valid (36%–78% versus 0%–90% across 9 models). Underdetermination therefore lives in what published papers say, not in who is reading them.

11 What Follows for Practice

Three recommendations follow directly from the measurements above; extended versions are in Appendices J and K, and a one-page form is the checklist in Appendix T.

State all five coordinates, and state C1 and C4 first. Evidence state and outcome unit are the pair that no comparison survives without: when they are missing, adjudicating a cross-paper gap is underdetermined for every reader we tested. That the single most effective intervention was simply giving judges the full paper (+0.20 [0.09, 0.36]) says the information is often present but scattered outside the results section, where a table caption should have carried it. C2 and C5 matter for a different reason: they do not decide comparability so much as name the ambiguity that would otherwise be silently resolved, and C5 is currently unrecoverable in 53% of published cases.

Declare where each baseline number came from. Provenance is cheap to disclose and reliably recovered once disclosed (0.91 [0.86, 0.96], stable at 85.8–87.2 across eligibility gates). It is worth disclosing because transported and locally re-implemented baselines are not interchangeable evidence, and silently mixing them biases a comparison in a direction the reader cannot estimate. One sentence in the table caption is enough.

Use automation to extract coordinates, not to issue verdicts. Model panels reach only 0.82 verdict accuracy against reference adjudicators, barely above the 0.77 constant baseline, so a pipeline that emits pass/fail judgements is reporting its own noise. The same panels are accurate at retrieving comparisons and filling in coordinate metadata, which is the role we recommend: surface the candidates and their coordinates, and leave the verdict to a human. Any study that then aggregates such judgements must treat raters as random assignment units rather than fixed effects, since rater thresholds vary substantially ($\hat{\sigma}_r = 0.50$ [0.33, 0.62]; Appendix Q). All codebooks, prompts, logs and verification tools are released (Appendix U).

12 Conclusion

We surveyed robust RAG along four failure surfaces and reduced them to five reporting coordinates describing each side of a quantitative comparison. The reduction is validated, not asserted: on prose that states its coordinates, coders recover them at 0.95 accuracy (κ 0.74), with C1 and C4 minimal throughout. Applied to 1,703 candidate comparisons from 259 full texts, it shows 533 of 610 classified pairs re-implementing baselines locally rather than transporting published numbers—a fact automation recovers well. Whether a quoted gap survives is not recoverable: model panels and human experts (κ 0.35) fail alike, because papers do not say enough. Frameworks should ask authors for coordinates and provenance, not for verdicts.

Limitations

Corpus coverage and search bounds. Our empirical evaluations operate over a defined screening frame rather than the entire literature; reference-based recall auditing (§2) establishes that 47% of in-scope references from six prior surveys reach our indexed full texts. The corpus emphasizes open-access English publications; papers presenting extended arguments outside standardized results sections may be under-represented in windowed screening (§10). All composition statistics are reported strictly for the screening frame or the re-weighted universe of 6,710 windows.

Evaluation models and human annotation panel. Corpus-scale annotations in §9 and §10 utilize LLM panels evaluated against an evidence-privileged reference (adjudicators examining full paper texts with verbatim quote verification). To anchor automated models, 80 published comparative claims were annotated by three independent NLP/RAG researchers over 210 person-hours (Appendix R). Human agreement bounds confirm that verdict underdetermination reflects source prose ambiguity rather than evaluator artifact.

Synthetic benchmark and variance estimation. Synthetic benchmark items (§9) measure coordinate extractability under explicit prose specifications rather than arbitrary unstructured text. Diagnostic interventions (§10) confirm that providing full paper text increases agreement without achieving complete inter-rater reproducibility. The estimated rater variance parameter ($\hat{\sigma}_r = 0.50$ [0.33, 0.62]) indicates that empirical evaluations must treat raters as random assignment units to maintain valid Type-I error control (Appendix Q).

Ethics Statement

Human subject annotation protocol. Human evaluations (§10) involved three expert annotators labelling 80 published comparative claims, adjudicated by a senior researcher across 210 person-hours. Annotators were recruited via professional academic networks, possessed formal research backgrounds in NLP, RAG, or experimental methodology, and were compensated at standard professional rates. All participants provided explicit consent for de-identified annotation release. Institutional review determined formal IRB oversight was exempt or non-human-subjects research. Released datasets comprise labels, source text locations, rationales, and adjudicated consensus; participant identities and consent forms are withheld for privacy. Annotators operated independently without access to model outputs prior to label finalization.

Reproducibility and automated execution. All corpus-scale annotations (§9 and §10) were produced via deterministic algorithms or LLM judges under fully released prompts and raw log files, enabling full re-execution. Excerpted paper quotes are checked verbatim against source line numbers by a released verifier.

Dual use considerations and licensing. Our literature audit relies exclusively on public academic publications and introduces no personal contact. Regarding dual-use security considerations: our survey discusses published threat models and defenses strictly at a conceptual level to assess experimental assumptions, releasing no attack payloads, poisoned corpora, or exploit code. All open-weight model runs utilize deterministic decoding parameters with full raw output releases.

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What to Read in the Appendix

The appendix provides detailed empirical artifacts backing every numerical claim in the main text. To assist navigation, the core tables and figures addressing key experimental questions are highlighted below:

- **Screening frame coverage:** Table 26 (recall against six prior survey bibliographies) and Figure 7 (PRISMA flow diagram).
- **Coordinate extractability:** Tables 41 and 42 (synthetic benchmark evaluations, including baseline comparison arms).
- **Provenance classification robustness:** Table 27 (accuracy against reference adjudicators), Table 36 (eligibility gate sensitivity), and Table 40 (frame re-weighting analysis).
- **Verdict underdetermination diagnostics:** Table 48 (human expert panel evaluation), Table 37 (non-generative structural classifier), Table 38 and Table 39 (developer-independence and self-preference audits), Table 28 (unresolved window bounds), Table 30 ($\kappa_{\max}$ upper bounds), and Figure 11 (diagnostic intervention ladder).
- **Evaluator diagnostic forensics:** Table 29 (judge-level rule fidelity and quote verification analysis, including 21 unverified quotes).
- **Author reporting guidelines:** Table 56 (reporting checklist).

Part I: Extended Survey

A Substantive Comparability in Robust RAG

We use the literature map as a source of boundary cases, not as prevalence evidence. Five *operating questions* organise discovery: how evidence is constructed, how a model uses it, how deliberate corruption changes the game, how signals become actions, and how those actions are evaluated. They are a different cut of the same literature from the four failure surfaces of §3, and both are different again from the five *reporting coordinates* of §8, which describe not a literature but a single proposed contrast. The boundary cases below are what made the coordinates necessary.

Evidence availability is not evidence use. Irrelevant-context studies test whether retrieved distractors redirect a generator (Yoran et al., 2024); noise-aware training changes the generator's training distribution (Fang et al., 2024); RGB supplies diagnostic stress conditions (Chen et al., 2024a); and Chain-of-Note changes how retrieved documents are read (Yu et al., 2023). A filter, a trained reader, and a benchmark may share a harness, but their native scores do not rank interchangeable interventions. Insufficient evidence creates another boundary: when no passage can support an answer, re-retrieval or abstention can dominate better distractor resistance (Joren et al., 2025).

Support, authority, conflict, and attack are distinct. RAGTruth annotates unsupported spans, while evaluators estimate claim or response support (Niu et al., 2024; Es et al., 2024; Saad-Falcon et al., 2024); knowledge-conflict methods instead study competition between parametric memory and retrieved evidence (Jin et al., 2024; Zhang et al., 2025d; Wang et al., 2025b). A response can be correct but unsupported, or faithful to a weak source. Provenance and timestamp determine which source should govern (Petroni et al., 2021; Vu et al., 2024; Kasai et al., 2023). Deliberate corruption changes the game again: corpus poisoning, indirect prompt injection, and backdoors differ in corpus write access, query knowledge, retriever visibility, insertion budget, persistence, and objective (Zou et al., 2025; Chaturvedi et al., 2025; Chaudhari et al., 2024), and their defenses prevent, detect, certify, trace, quarantine, or recover under different assumptions (Xiang et al., 2024; Si et al., 2025; Zhang et al., 2025a). An attack-success comparison is transportable only within a matched capability envelope.

A signal is not a policy. Calibration predicts a property of a score (Guo et al., 2017); selective prediction trades coverage against risk (Geifman and El-Yaniv, 2017); sufficient-context classification asks whether available passages can support an answer; adaptive retrieval chooses whether to search, verify, answer, or abstain (Jang et al., 2025; Joren et al., 2025; Labruna et al., 2024). Self-RAG, CRAG, and CtrlA expose this policy layer explicitly (Asai et al., 2024; Yan et al., 2024; Liu et al., 2025b). Comparing them requires a common action set, budget, and loss: accuracy alone cannot price extra retrieval, coverage, or refusal.

Comparison ladder. The strongest contrast reruns alternatives with the same corpus snapshot, retriever, generator, evidence state, capability envelope, action set, and outcome. When that is impossible, synthesis should retain only the common scope supported by both studies. If a decision-critical field is absent, the comparison is underspecified rather than evidence that either method is better. Appendix B gives paper-level contrasts and the documentary-map boundary; the next section turns these recurring boundaries into a claim-level audit.

B Extended Synthesis by Operating Question

Sections 4–7 organise robust RAG by failure surface, which is where a paper sits. This appendix cuts the same literature the other way, by the *operating question* a study answers, and develops the paper-level contrasts the main text compresses. It is a synthesis, not an exhaustive bibliography: the released 342-row evidence map holds the long-tail records, source strata and intervention labels.

Two limits carry over from §2 and govern how the contrasts below may be read. The screening flow's first stage is documented but not re-executable, so this synthesis identifies recurring scientific distinctions rather than estimating prevalence or ranking evidence families; and the map carries no study-level risk-of-bias weighting. A review-directed recall check also surfaced older, pre-freeze publications absent from the frozen map, which we cite as qualitative context and log as post-freeze discoveries without entering them into any denominator. Where this survey states a number

rather than a distinction, it does so only for the frame of §10.

B.1 When Does Retrieved Context Help?

Irrelevant-context and noise-aware-training studies intervene at different points. Yoran et al. (2024) shows that retrieved distractors can actively redirect generation and tests filtering and targeted fine-tuning. Fang et al. (2024) instead trains a generator across random, relevant, and counterfactual retrieval conditions. RGB supplies stress conditions and diagnostic abilities rather than a competing mitigation (Chen et al., 2024a). These papers can share an end-to-end harness, but their native results do not rank a filter, a trained reader, and a benchmark as interchangeable objects.

Context construction creates a second boundary. Chain-of-Note changes how the generator reads retrieved documents, whereas compression and reranking alter what reaches it (Yu et al., 2023; Wang et al., 2023; Trappolini et al., 2026). Long-context studies show that retaining more passages can improve evidence availability while worsening position sensitivity or distraction (Jin et al., 2025; Gabín et al., 2026). A fair comparison therefore fixes the candidate evidence, generator, decoding, and budget when the intended claim is about the intervention rather than the entire system.

Insufficient evidence is not merely severe distraction. Sufficient Context labels whether the available passages can support an answer and evaluates selective behavior (Joren et al., 2025). A reader that resists a hard negative and a controller that abstains when support is absent solve different problems. Reports should state both answer-bearing support availability and the scored action.

B.2 Which Knowledge Should Govern an Answer?

Knowledge-conflict work separates context–memory disagreement from conflict among retrieved sources. Tug-of-War measures how models trade parametric and contextual knowledge; FaithfulRAG models fact-level discrepancies; Astute RAG reconciles imperfect retrieval with internal knowledge (Jin et al., 2024; Zhang et al., 2025d; Wang et al., 2025b). Although all three concern "conflict," their experimental units, source assumptions, and desired responses differ. The appropriate outcome may be answer correctness, contextual faithfulness, calibrated disclosure, or source reconciliation. Earlier entity-substitution experiments and PopQA isolate when parametric memory resists or benefits from retrieved evidence (Longpre et al., 2021; Mallen et al., 2023). ConflictBank broadens the benchmark unit to millions of claim–evidence pairs across misinformation, temporal conflict, and semantic ambiguity (Su et al., 2024); its scale does not make those conflict types interchangeable.

Grounding resources address another layer. RAGTruth provides span-level hallucination labels, while RAGAS and ARES estimate response or context support (Niu et al., 2024; Es et al., 2024; Saad-Falcon et al., 2024). These evaluators can identify an unsupported answer, but support alone does not establish that the cited source is current or authoritative. A system can be faithful to weak evidence or correct for a reason absent from its citations.

Source reliability and time therefore belong in the evidence record. Recent work models source reliability directly (Hwang et al., 2025), and other studies examine source preference and contextual authority (Schuster et al., 2026; Li et al., 2026). The comparison-relevant question is not only whether the model used retrieved text, but whether the benchmark specified which source should govern, at what timestamp, and under what corroboration rule. KILT standardizes provenance against a fixed knowledge source, while FreshQA and RealTimeQA test knowledge whose answer changes over time (Petroni et al., 2021; Vu et al., 2024; Kasai et al., 2023). ALCE and generative-search verifiability work separately score citation attribution and whether cited sources actually support generated statements (Gao et al., 2023; Liu et al., 2023).

Multi-agent methods add yet another intervention boundary. MAIN-RAG uses agent voting to filter retrieved documents, MADAM-RAG assigns agents to conflicting documents before aggregation, and DRAG introduces debate during retrieval and generation (Chang et al., 2025; Wang et al., 2025c; Hu et al., 2025). General multi-agent debate supplies methodological background (Du et al., 2024), but these RAG systems should be compared only when the evidence allocation, agent budget, aggregation rule, and target outcome match.

B.3 What Changes Under an Attacker?

Corpus poisoning makes evidence unreliability intentional. PoisonedRAG inserts passages to steer target answers; earlier corpus-poisoning work manipulates retrieval through adversarial passages

(Zou et al., 2025; Zhong et al., 2023). Attack success is conditional on corpus write access, insertion budget, query knowledge, retriever visibility, and target scope. Omitting these fields can make two attack rates numerically similar but scientifically incomparable.

Indirect prompt injection and backdoors target a different interface. Retrieved text may contain instructions that override the user task, or a trigger may activate behavior learned during training (Chaturvedi et al., 2025; Chaudhari et al., 2024; Cheng et al., 2024). These attacks should not inherit claims from factual poisoning merely because both place malicious text in the context. Retrieval realism, instruction-following behavior, and trigger access define distinct games.

Defenses likewise have different roles. Certified RAG gives guarantees under a bounded corruption model; filtering and semantic isolation demote suspicious documents; traceback identifies likely poisoned sources (Xiang et al., 2024; Si et al., 2025; Zhang et al., 2025a). Prevention, certified accuracy, detection, and recovery require different outcomes. Any cross-defense comparison should pair the same capability envelope with clean utility and state whether the defense prevents, detects, traces, quarantines, or recovers.

B.4 How Does a Signal Become an Action?

Calibration, answerability, and retrieval usefulness are related but not identical signals. Calibration-oriented RAG predicts reliability, whereas Sufficient Context asks whether the retrieved evidence can support an answer (Jang et al., 2025; Joren et al., 2025). A well-calibrated score can still feed a poor threshold or action policy, and an insufficient-context detector does not determine whether retrieval, clarification, or abstention has the highest utility.

Adaptive systems make the policy explicit. Self-RAG learns when to retrieve and critique; CRAG adds corrective retrieval; CtrlA separates uncertainty sources to select control actions (Asai et al., 2024; Yan et al., 2024; Liu et al., 2025b). Their reported gains depend on the available action set and resource budget. Answer-only accuracy cannot capture the benefit or cost of extra retrieval, verification, or refusal.

Selective evaluation should therefore retain risk, coverage, action, and cost rather than collapse them into one score. Trust-or-abstain methods, conformal controllers, and uncertainty benchmarks instantiate different operating points (Zhu et al., 2026; Chen et al., 2024b; Nguyen et al., 2026). Comparisons are strongest when systems share evidence and actions and are evaluated across a risk–coverage or utility curve.

B.5 What Can the Evidence Map Establish?

The map documents that robust-RAG research spans evidence construction, evidence use, adversarial integrity, and action control. Existing surveys provide architectural, evaluation, trustworthiness, security, conflict, and abstention views (Gao et al., 2024; Yu et al., 2024; Zhou et al., 2024; Xu et al., 2026, 2024a; Wen et al., 2025a). Controlled searches for RAG best practices provide an empirical component comparison under shared tasks (Wang et al., 2024b), complementary to survey taxonomies and to the claim-level audit studied here. Table 52 states the narrower claim-level increment tested here.

The map cannot establish that these families occur at representative rates, that its taxonomy is complete, or that one family has stronger evidence. Accepted-paper additions strengthen several historical lines, while 205 of 342 mapped rows remain preprints against 137 peer-reviewed, and coding depth varies across legacy strata. Map counts therefore provide documentary context only. Every measured claim in this survey rests elsewhere: the taxonomy test on the constructed benchmark of §9, the composition figures on the screened frame of §10, and the design claim on the assignment-unit simulation.

C Evidence-Centric Robustness in Full

Evidence-centric robustness concerns the quality, integrity, and sufficiency of the evidence that enters the context window. In classical information retrieval, irrelevant results are often treated as a ranking problem. In RAG, however, the retrieved set becomes part of the model input; a single misleading passage can change the generated answer, confidence, and citations. Evidence-centric work therefore asks three questions: what kinds of bad evidence appear, how do they affect generation, and which pre-generation controls can reduce their harm?

Retrieval Noise: Beyond Relevant vs. Irrelevant
The earliest robust-RAG studies show that retrieval errors do not merely reduce recall; they alter the generator's behavior. Yoran et al. (2024) demon-

Theme	Representative work	Core idea	What it reveals for robust RAG
Irrelevant context	IrrelCtx	Analyze and mitigate irrelevant retrieved context with filtering and robust training	Even answerable questions degrade when irrelevant passages enter the context; robustness is not equivalent to retrieval recall.
Retrieval-noise training	RAAT	Categorize retrieval noise and train with adaptive adversarial objectives	Noise types should be separated: irrelevant, relevant-but-insufficient, and counterfactual evidence have different effects.
Generator note-taking	CoN	Generate per-passage notes before answering	Robust generation benefits from intermediate relevance judgments rather than direct answer generation.
Fine-grained noise benchmark	MagicMushroom	Configurable benchmark for distracting, low-quality, inconsequential, and irrelevant noise	"Noise" is not monolithic; evaluation should vary type, amount, and placement of bad evidence.
Nonlinear distraction	FirstDrop	Study disproportionate effect of small amounts of misleading information	A few strong distractors can dominate long-context reasoning, so average noise rate is insufficient.
Long-context RAG	LC-RAG	Analyze long-input RAG and hard negatives	More retrieved context can hurt unless ordering, filtering, or training adapts to long-context failure modes.
Sentence-level filtering	FILCO	Filter retrieved context before generation	Filtering can be more fine-grained than document reranking and can remove local distractors inside otherwise useful passages.
Retrieval evaluation	UDCG	Utility-aware retrieval metric for LLMs	RAG retrieval should be evaluated by downstream utility, including negative utility from distracting passages.

Table 1: Representative evidence-centric work on noisy retrieval and distraction.

strate that retrieval-augmented language models can be brittle to irrelevant context and propose filtering and robust training as mitigation. This work is important because it separates two effects that are often conflated: a retriever may fail to retrieve enough relevant evidence, and a generator may fail to ignore irrelevant evidence once it is retrieved. Robustness therefore depends on both retriever quality and generator selectivity.

Subsequent work expands the typology of noise. Fang et al. (2024) distinguishes multiple retrieval-noise types and trains models with adaptive adversarial objectives and explicit noise awareness. The distinction matters: an irrelevant passage that is obviously off-topic may be easy to ignore, while a counterfactual or hard-negative passage may be semantically close to the query and therefore much more dangerous. RGB similarly evaluates noise robustness, negative rejection, information integration, and counterfactual robustness as separate abilities (Chen et al., 2024a). Taken together, these works suggest that "RAG robustness" cannot be measured by a single noisy-context accuracy number. A strong evaluation should vary the type, quantity, ranking position, and semantic hardness of negative evidence.

Recent benchmarks make this point more explicit. Magic Mushroom introduces fine-grained retrieval noise categories, including distracting and low-quality evidence (Zhang et al., 2025e). URAG formulates uncertainty quantification in RAG and studies the effect of irrelevant and noisy retrieval on conformal prediction (Nguyen et al., 2026). PRGB and other placeholder or perturbation-based benchmarks test whether models remain robust when retrieval context is systematically manipulated (Tan et al., 2025). The trajectory of this literature is clear: robust RAG evaluation is moving from randomly injected distractors to controlled stress tests that expose distinct failure mechanisms.

Distraction and Context Utility Distraction is stronger than irrelevance. A passage is distracting when it has negative marginal utility: adding it changes the answer, confidence, attribution, or reasoning path in a harmful direction. This explains why more retrieval is not always better. In long-context settings, retrieving more passages can increase the chance that a hard negative appears before or near useful evidence. Jin et al. (2025) study long-context RAG and show why ordering and hard negatives become central once the generator can consume many passages. Gao et al. (2026), whose title names the *first drop of ink*, identify a

nonlinear effect in which a small amount of misleading information disproportionately degrades long-context reasoning. CUB evaluates context-utilisation techniques across gold, conflicting, and irrelevant contexts (Hagström et al., 2026), while UDCG redefines retrieval evaluation around downstream utility rather than traditional relevance alone (Trappolini et al., 2026).

This line of work changes the role of the retriever. The retriever is not only a recall engine; it is a context constructor. A retriever optimised for lexical or embedding similarity may return passages that are highly related to the query but harmful to the generator. A robust retriever should consider downstream answer utility, redundancy, conflict, and the interaction between passages. Similarly, a robust RAG system should not treat the top-k list as a fixed input. It should be able to filter, reorder, compress, diversify, or abstain when the context set is unreliable.

Evidence-Side Mitigations Evidence-side defenses operate before or during context construction. They include retriever retraining, reranking, filtering, passage compression, and context ordering. Filtering can be document-level or sentence-level. FILCO-style sentence filtering removes irrelevant local content inside otherwise useful passages (Wang et al., 2023); this is important because a retrieved document can contain both answer evidence and distracting details. Utility-aware retrieval metrics and rerankers attempt to optimise what helps the generator, not what merely matches the query (Trappolini et al., 2026). Context ordering and long-context tuning address cases where the same evidence set has different effects depending on position (Jin et al., 2025).

Training-based approaches expose the generator to noise so that it learns to separate relevant from irrelevant context (Fang et al., 2024; Ding et al., 2025). This can improve robustness but also raises a scalability question: robust training may be less effective or less necessary as base LLMs become stronger, and adversarially generated noise can overfit to a particular retriever or generator. Training-free approaches, such as filtering and note-taking, are easier to add to existing systems but depend on reliable relevance or sufficiency signals. A recurring pattern is that no single pre-generation control is sufficient: retrieval filtering reduces easy noise, reranking handles ordering, compression improves context budget, and downstream calibration remains necessary for residual uncertainty.

Query-Side and Index-Side Sources of Noise Evidence noise does not begin at retrieval scoring. Query formulation is itself a brittle interface. Query-level robustness work shows that minor changes in wording can substantially affect retrieval and downstream answers (Percin et al., 2025). This suggests that robust RAG should include query normalisation, query expansion, and query uncertainty estimation. In multi-turn settings, the query may be non-standalone or underspecified, making retrieval quality depend on dialogue state tracking (Rosenthal et al., 2026). The retriever may also inherit biases from chunking and indexing: relevant facts can be split across chunks, qualifiers can be separated from claims, and embeddings can overemphasise topical similarity over answer sufficiency.

Index-side design also changes robustness. Smaller chunks improve retrieval granularity but can remove context needed for factual interpretation. Larger chunks preserve context but increase the chance of including distracting material. Hybrid sparse–dense retrieval can reduce some embedding failures but may increase redundancy. Graph-based, multimodal, and agentic/process-level RAG add structure but introduce new poisoning surfaces, as shown by GraphRAG poisoning, graph poisoning, and reasoning-style poisoning work (Liang et al., 2026; Wen et al., 2025b; Zhou and Wang, 2025). Thus, evidence-centric robustness should be studied as a corpus construction problem, not only as a retriever ranking problem.

Design Tradeoffs Evidence-side controls have three recurring tradeoffs. First, filtering improves precision but can remove weakly relevant evidence needed for multi-hop reasoning. Second, reranking can demote distractors but may also amplify model-specific biases if the reranker is trained against a narrow generator. Third, context compression saves tokens but can delete qualifiers, source information, or uncertainty markers. Robust RAG systems should therefore expose confidence about filtering decisions and allow later stages to recover from false negatives. A practical design is multi-stage: retrieve broadly, rerank cautiously, filter locally, preserve provenance, and let the generator or verifier request more evidence when sufficiency is low.

D Model-Centric Robustness in Full

Model-centric robustness concerns how the generator interprets, weighs, and integrates evidence once it appears in the prompt. Even with the same retrieved context, different generators can answer differently because they vary in instruction following, parametric prior strength, context utilisation, and calibration. Model-centric work therefore studies the internal and behavioral mechanisms that determine whether retrieved evidence is used faithfully or misleadingly.

Context Utilisation and Intermediate Reasoning One model-centric strategy is to force the generator to process evidence more explicitly before answering. Chain-of-Note asks the model to write notes for each retrieved passage, assess relevance, and abstain when evidence is insufficient (Yu et al., 2023). This moves the generator from direct answer synthesis to a read–judge–answer pipeline. The value of this design is not simply better prompting; it externalises intermediate evidence judgments that can be inspected, scored, or used to trigger abstention.

Other approaches manipulate the context itself. Context compression and filtering reduce the amount of irrelevant information passed to the model, while context-utilisation benchmarks test whether models actually rely on gold evidence when it is present and ignore conflicting or irrelevant evidence when it is not (Hagström et al., 2026). The key distinction is between *context availability* and *context use*. A passage can be present in the prompt but ignored; a misleading passage can be unsupported but still dominate the answer. Thus, robust RAG must evaluate how models allocate attention, probability mass, and rationales across sources.

Parametric–Retrieved Knowledge Conflict RAG introduces a conflict structure that does not exist in standard closed-book generation. The model has parametric memory, the retriever supplies external evidence, and multiple retrieved passages may disagree. When these sources conflict, models can exhibit two opposite failures: over-compliance with false retrieved evidence or stubborn reliance on stale parametric knowledge. Tug-of-War constructs and studies internal–external conflict in retrieval-augmented language models (Jin et al., 2024). Adaptive Chameleon and related work show that LLMs can be highly receptive to coherent counter-evidence but resistant under other conflict conditions (Xie et al., 2024b). Real-document conflict studies show that behavior under conflict is not only a synthetic benchmark artifact (Kortukov et al., 2024).

Conflict-aware methods differ in granularity. FaithfulRAG extracts parametric self-facts, aligns them with retrieved facts, localises discrepancies, and uses a self-thinking step before generation (Zhang et al., 2025d). CoRect applies hidden-state and logit rectification to resolve conflicts without full model retraining (Ma et al., 2026). Astute RAG elicits internal knowledge and consolidates it with retrieved passages through source-aware reliability comparison (Wang et al., 2025b). These methods share a principle: conflict should be represented explicitly before the final answer is produced.

From Faithfulness to Source Reliability Many faithful RAG methods implicitly assume that retrieved context should be trusted. Robust RAG cannot make this assumption. Retrieved context can be stale, poisoned, incomplete, or irrelevant; parametric knowledge can be outdated or hallucinated. The correct policy depends on source reliability. Source preference studies ask which sources win under conflict (Schuster et al., 2026); the *Who's Who* evaluation of Pham et al. (2024) studies behaviour under practical entity-level conflicts. These works suggest that future systems should represent not only claims but also provenance, timestamp, authority, and consistency. In a high-stakes setting, the most robust answer may not be a single synthesised fact; it may be a statement that exposes disagreement and explains why one source is more credible.

Citation, Attribution, and Verification RAG systems are often expected to cite sources, but citations do not guarantee faithfulness. A model can cite a passage that is topically related but does not support the generated claim, or cite the correct document while generating a claim from parametric memory. RAGTruth provides fine-grained hallucination annotations for RAG outputs (Niu et al., 2024). Related grounding methods train models to quote source spans or decompose external and parametric semantics, but these are not substitutes for claim-level citation verification (Zhang et al., 2024; Sui et al., 2025). The key challenge is granularity. Document-level citation is too coarse when a passage contains multiple claims, qualifiers, or contradictions. Claim-level or sentence-level attribution is more informative but requires reliable

Problem	Representative work	Mechanism	Limitation or open issue
Parametric–retrieved conflict	Tug-of-War	Construct and analyze conflicts between internal and external knowledge	Conflict behavior depends on model, prompt, evidence coherence, and task framing.
Fact-level conflict modeling	FaithfulRAG	Extract parametric self-facts, align them with retrieved facts, and reason before answering	Requires reliable fact extraction and alignment; complex claims remain difficult.
Source-aware consolidation	AstuteRAG	Elicit internal knowledge and consolidate it with retrieved sources using reliability comparison	Training-free but still depends on LLM self-assessment and source-quality cues.
Hidden-state rectification	CoRect	Locate layers where parametric suppression occurs and contrast contextual/non-contextual logits	Promising for model-centric control, but model internals and transferability remain open.
Context utilisation benchmarks	CUB	Test gold, conflicting, and irrelevant contexts across models	Benchmarks reveal whether a model uses context, but not always why a model chooses one source.
Real-document conflict	RealConflict	Study behavior under context-memory conflicts with real documents	Realistic evidence can be messier than synthetic conflicts and harder to label.
Source preference	SourcePref	Analyze which sources win under conflict	Source preference may encode bias, recency effects, or authority priors.

Table 2: Representative model-centric work on knowledge conflict and contextual faithfulness.

entailment, alignment, and segmentation.

Verification can be pre-generation, post-generation, or iterative. Pre-generation verification asks whether the context is sufficient and trustworthy before answering. Post-generation verification checks whether each generated claim is supported by the evidence. Iterative verification alternates between generation and retrieval, using detected gaps or conflicts to form new queries. The limitation is cost: verification often requires additional model calls, retrieval calls, or entailment models. Robust RAG therefore needs cheap first-pass signals for routine cases and stronger verification for high-risk cases.

Limits of Model-Centric Fixes Model-centric methods can improve evidence use, but they cannot fully solve evidence unreliability. A generator cannot reliably infer provenance if the corpus does not preserve it. It cannot resolve a conflict if all sources are equally stale or poisoned. It cannot calibrate answerability if retrieval never exposes missing evidence. Conversely, evidence-side controls cannot guarantee faithful generation. This complementarity motivates the layered framework: evidence-centric controls shape the input distribution, model-centric controls shape context use, and system-centric controls decide when the remaining uncertainty is too high to answer.

E Adversarial Corruption and Security in Full

Adversarial corruption is evidence unreliability under an adaptive attacker. RAG systems are vulnerable because they externalise memory into corpora, indices, prompts, and tools. An attacker can target any of these surfaces: the corpus before indexing, the retriever during matching, the prompt template, the retrieved context, or the downstream agent that executes actions.

Corpus and Retriever Poisoning Corpus poisoning attacks insert malicious passages into the retrieval database so that they are returned for target queries. Early work on poisoning retrieval corpora demonstrates that small numbers of adversarial passages can manipulate dense retrieval (Zhong et al., 2023). PoisonedRAG extends this threat to end-to-end generation: retrieved malicious passages can cause the LLM to output attacker-chosen answers (Zou et al., 2025). BadRAG, UniC-RAG, and related methods explore different trigger structures, optimisation objectives, and attack settings (Xue et al., 2024; Geng et al., 2025; Zhang et al., 2026). The important lesson is that retrieval augmentation can amplify small corpus changes into large

Attack/defense surface	Representative work	Threat or defense	Implication
Retriever poisoning	PoisonRetr	Inject adversarial passages to manipulate dense retrievers	Retrieval security can fail before generation begins.
RAG knowledge corruption	PoisonedRAG	Craft malicious passages that steer generated answers	End-to-end RAG needs corpus integrity and generation safeguards.
Backdoor attacks	TrojanRAG; Phantom	Triggered retrieval/generation behavior via poisoned contexts	RAG can act as a backdoor carrier without modifying model weights.
Prompt template attacks	AIP	Weaponize shared instructional prompts to steer retrieval and generation	Prompt templates are part of the attack surface, not just implementation details.
Indirect prompt injection	RetrBarrier	Optimize trigger fragments so malicious instructions are retrieved in the wild	Real attacks must overcome the retrieval barrier, not merely place malicious text in a corpus.
Certified defense	RobustRAG	Isolate-and-aggregate answers for robustness certificates	Certification gives stronger guarantees but typically assumes specific attack budgets.
Reranking defense	ProGRank	Use probe-gradient instability to demote poisoned passages	Defense can be inserted retriever-side without retraining the LLM.
Attention defense	AVFilter	Detect poisoning via normalised attention variance	Internal attention signals may expose poisoned evidence, but adaptivity remains a concern.

Table 3: Representative adversarial corruption attacks and defenses.

Threat class	Corpus	Query	Retr.	Gen.	Objective	Boundary
Dense-retriever poisoning (Zhong et al., 2023)	write	train set	white/gray	no	retrieve bad passages	Not always end-to-end.
RAG knowledge corruption (Zou et al., 2025; Xue et al., 2024)	write	target topic	black/gray	black	answer steering	Stealth/dynamics vary.
Backdoor-style RAG (Cheng et al., 2024; Chaudhari et al., 2024)	write	trigger	gray	black	latent behavior	Normal queries may pass.
Indirect prompt injection (Chaturvedi et al., 2025; Chang et al., 2026)	write/share	natural	black	black	instruction hijack	Must overcome retrieval.
Opinion/fairness manipulation (Chen et al., 2025c; Bagwe et al., 2025)	write	group/topic	black/gray	black	bias steering	QA metrics miss social harm.
Defense setting (Xiang et al., 2024; Yin et al., 2026; Choudhary et al., 2026; Zhang et al., 2025a)	bounded	varies	assumed	black/gray	detect/demote/trace	Depends on budget/visibility.

Table 4: Attacker-capability matrix for adversarial RAG. The same "attack success rate" can mean different things depending on corpus write access, query control, retriever visibility, generator access, and objective.

generation effects.

Backdoor-style attacks make the threat more persistent. TrojanRAG and Phantom show that triggers can activate malicious behavior through retrieved contexts without changing the generator weights (Cheng et al., 2024; Chaudhari et al., 2024). Other attacks optimise token-level substitutions, influential tokens, or attention-steering patterns (Wang et al., 2025d; Chen et al., 2025b; Wang et al., 2025a). These works expose a system risk: a RAG system may appear safe under normal queries but fail under targeted concepts, phrases, or document manipulations.

Indirect Prompt Injection and Prompt-Level Attacks Indirect prompt injection differs from factual poisoning. Instead of inserting false evidence, the attacker inserts instructions that are retrieved and interpreted by the LLM as commands. In practice, an indirect prompt injection must overcome the retrieval barrier: the malicious document must actually be retrieved under natural queries. Chang et al. (2026) explicitly study this barrier and propose trigger fragments that improve retrieval of attack fragments. AIP broadens the surface further by showing that shared instructional prompts can be weaponised to steer RAG retrieval and generation (Chaturvedi et al., 2025). These attacks imply that prompt templates, document ingestion rules, and retrieved text should be treated as untrusted inputs.

Defenses and Their Assumptions Defenses against adversarial corruption operate at multiple

Comparison	Why it looks comparable	Hidden mismatch	Metric pitfall	Normalised reading
PoisonedRAG vs. dense-retriever poisoning	Both insert malicious evidence into a retrieval corpus.	One evaluates end-to-end answer steering; the other may evaluate retriever ranking before generation.	Retrieval success and answer attack success are not the same target.	Compare only after recording whether poisoned evidence is retrieved and whether the generator changes the final answer.
Certified RAG defense vs. prompt-injection defense	Both are RAG defenses against corrupted retrieved content.	Certification assumes a bounded number of corrupted passages; prompt injection exploits instruction following rather than only false facts.	A certificate under bounded corruption may not imply instruction-hijack resistance.	Treat certification, instruction isolation, and document sanitisation as complementary defenses.
Traceback vs. prevention	Both reduce harm after poisoning.	Traceback identifies influential poisoned documents after failure; prevention attempts to block harmful output before response.	Lower post-hoc traceback error does not equal lower user-facing attack success.	Report prevention, detection, demotion, traceback, and recovery as separate outcome variables.
Backdoor trigger vs. natural query attack	Both can produce targeted malicious behavior.	Trigger attacks may rely on rare phrases; natural-query attacks must survive ordinary retrieval and ranking.	Attack success under attacker-crafted triggers overstates risk under natural user queries.	Require query-control and trigger-frequency fields before comparing attack strength.

Table 5: Concrete examples of threat-model normalisation. These cases show how reported attack or defense success can reverse when attacker access, retrieval realism, and evaluation target differ.

layers. Certified approaches such as RobustRAG isolate and aggregate outputs to provide guarantees under bounded retrieval corruption (Xiang et al., 2024). Reranking defenses such as ProGRank use instability or probe-gradient signals to demote suspicious passages (Yin et al., 2026). Attention-based filters detect abnormal attention patterns caused by poisoned evidence (Choudhary et al., 2026). Clustering and conflict-resolution defenses such as TrustRAG and SeCon-RAG filter semantically inconsistent or suspicious evidence before generation (Zhou et al., 2025; Si et al., 2025). Traceback methods attempt to identify poisoned texts after an attack has occurred (Zhang et al., 2025a).

The main limitation is threat-model fragmentation. A defense designed for corpus poisoning may not stop prompt-template attacks; a filter trained on obvious malicious instructions may not detect fact-level poisoning; a certified aggregation method may assume a corruption budget that is hard to estimate in open corpora. Robust RAG security therefore needs compositional evaluation: combine poisoned facts, retrieval triggers, prompt-bearing documents, and benign noise in the same pipeline.

Threat Model Dimensions Security papers differ in attacker knowledge, access, and objective. Some assume the attacker can insert a few documents into a public corpus but cannot see user queries. Others assume black-box query access, white-box retriever gradients, control over prompt templates, or access to agent memory. Objectives also differ: targeted answer steering, denial of service, refusal induction, opinion manipulation, privacy extraction, fairness manipulation, or guardrail bypass. Comparing attacks without specifying these dimensions is misleading. A poisoning method that works with five injected documents under black-box access has stronger practical relevance than a method requiring full index control; a defense evaluated only against keyword triggers may fail against semantic triggers.

Table 4 makes this comparison explicit. We separate corpus write access, query control, retriever visibility, generator access, and objective because they determine whether two attack results are comparable. This matrix also clarifies why benign retrieval noise and adversarial corruption should not be collapsed into a single robustness score: the former tests system tolerance to accidental evidence errors, while the latter tests adaptive pressure against the retrieval-generation interface.

Attack Success Is Not Enough Attack success rate is necessary but insufficient. A strong RAG attack must also be stealthy, durable, and retrieval-realistic. Stealth means poisoned passages should not be trivially detected by lexical filters or human inspection. Durability means attacks should persist under chunking, reranking, context filtering,

and prompt changes. Retrieval realism means the malicious content must be retrieved under natural user queries, not only under attacker-crafted queries. Work on indirect prompt injection in the wild explicitly highlights this retrieval barrier (Chang et al., 2026). Recent stealth and attention-aware work similarly argues that defenses should evaluate whether poisoned content can be distinguished from benign evidence (Choudhary et al., 2026).

Defense-in-Depth The surveyed defenses suggest a defense-in-depth architecture. At ingestion time, systems should validate sources, scan documents for prompt-like instructions, preserve provenance, and detect anomalous clusters. At retrieval time, systems can rerank with robustness signals, compare retrieved documents for contradictions, and demote passages with suspicious trigger behavior. At generation time, systems should isolate evidence from instructions, verify answer support, and refuse to execute instructions found in retrieved documents. During operation, systems need logging and traceback to identify which corpus items influenced harmful outputs. No layer is sufficient alone, but layered controls reduce the chance that one bypass compromises the entire pipeline.

F System-Centric Control in Full

System-centric control treats RAG as a policy rather than a fixed pipeline. A robust system must decide whether to retrieve, how much to retrieve, whether to re-retrieve, whether to verify, whether to trust internal knowledge, and whether to answer at all. This layer is where evidence reliability becomes an action.

Answerability and Sufficient Context The most basic control question is whether the available evidence is sufficient. RGB's negative rejection task and counterfactual robustness task show that RAG systems often answer when they should reject unsupported or false evidence (Chen et al., 2024a). Sufficient Context reframes the problem: rather than asking whether the model can answer, ask whether the retrieved context contains enough information to answer (Joren et al., 2025). Unanswerability benchmarks extend this idea to underspecified and unsupported queries (Peng et al., 2025; Rosenthal et al., 2026).

This line of work separates two forms of uncertainty. *Evidence uncertainty* asks whether the corpus and retrieved context support the answer. *Model uncertainty* asks whether the generator can integrate the evidence correctly. A system can have high model confidence but low evidence sufficiency, which is dangerous. Conversely, a system may have adequate evidence but low model confidence, suggesting verification or decomposition rather than refusal.

Adaptive Retrieval and Retrieval Budgets Always retrieving is inefficient and can be harmful; never retrieving leaves the model dependent on stale parametric knowledge. Adaptive retrieval methods learn or infer when external evidence is needed. Self-RAG trains models to retrieve, generate, and critique with reflection tokens (Asai et al., 2024). Adapt-LLM-style methods use explicit retrieval tokens to signal when retrieval is needed (Labruna et al., 2024). CtrlA uses representation-level honesty and confidence directions to guide retrieval timing (Liu et al., 2025b). Corrective RAG uses a retrieval evaluator to trigger additional retrieval or web search when the initial evidence is weak (Yan et al., 2024).

Recent work also shows that retrieval can be over-used. Over-searching studies introduce metrics such as tokens-per-correctness and benchmarks that measure when search-augmented systems retrieve unnecessarily (Xie et al., 2026). Adaptive retrieval without self-knowledge benchmarks many uncertainty-estimation methods and shows that retrieval policies must be evaluated for both accuracy and efficiency (Moskvoretskii et al., 2025). The emerging view is that retrieval is a budgeted decision: each retrieval step trades cost, latency, exposure to noise, and potential accuracy gain.

Calibration, Abstention, and Interaction Calibration asks whether confidence matches correctness. In RAG, confidence depends on both model uncertainty and evidence uncertainty. CalibRAG optimises for calibrated decision-making (Jang et al., 2025); UncertaintyRAG uses span-level uncertainty to improve retrieval training and robustness under distribution shift (Li et al., 2024); noise-aware verbal confidence calibration targets overconfidence under irrelevant or counterfactual retrieval (Liu et al., 2026). SABER-style trust-or-abstain approaches estimate whether parametric knowledge and contextual knowledge paths are reliable, then map them to trust and abstention decisions (Zhu et al., 2026).

Abstention should not be treated as a single

Control/evaluation question	Representative work	Signal or decision	Why it matters
When to retrieve	Self-RAG; Adapt-LLM; CtrlA	Reflection tokens, special retrieval tokens, or representation-based confidence/honesty directions	Retrieval is an action with costs and risks, not a fixed preprocessing step.
When retrieval is weak	CRAG; SuffCtx	Retrieval evaluator or sufficient-context signal	A system should repair or abstain rather than answer from insufficient evidence.
Calibration	CalibRAG; UncertaintyRAG; NAACL-Cal	Decision calibration, span uncertainty, verbal confidence calibration	Robustness requires confidence to reflect both model and evidence uncertainty.
Unanswerability	Unanswerability; MTRAG-UN	Unanswerable and underspecified RAG benchmarks	Real users often ask questions the corpus cannot answer; refusal and clarification need direct evaluation.
RAG capabilities	RGB	Noise robustness, negative rejection, integration, counterfactual robustness	Multi-axis evaluation exposes failures hidden by answer accuracy.
Hallucination detection	RAGTruth	Word-level hallucination annotations for RAG outputs	Fine-grained labels support detection and attribution beyond answer-level scoring.
Query robustness	QueryRobust	Perturb query wording and measure pipeline sensitivity	Robustness must include the query-to-retrieval interface.

Table 6: Representative work on system-centric control and evaluation.

Lineage	Classical question	RAG-specific version	Why it matters for robust RAG
Selective prediction	When should a model abstain to control risk?	Abstain when retrieved evidence is insufficient, conflicting, stale, or corrupted.	Selective risk must condition on evidence state, not only answer confidence.
Calibration	Does confidence match correctness?	Does confidence match answer correctness, evidence sufficiency, citation support, and retrieval usefulness?	A calibrated answer probability can still hide unreliable evidence.
Active clarification	When should the system ask a user for more information?	Ask clarification before retrieval when the query is ambiguous or underspecified.	Clarification can be the correct action even when retrieval could produce a plausible answer.
Cost-sensitive retrieval	When is additional computation worth it?	Decide whether to retrieve, re-retrieve, verify, rerank, or escalate under latency/cost budgets.	Robustness gains should be reported with action cost and coverage, not only accuracy.
Decision-theoretic control	Which action minimizes expected risk?	Choose among answer, abstain, expose conflict, cite uncertainty, retrieve again, or quarantine evidence.	Action quality is the bridge between evidence reliability estimates and user-facing behavior.

Table 7: Action-control lineage. The framework imports ideas from selective prediction, calibration, clarification, and cost-sensitive decision-making, but makes retrieved evidence state the object being controlled.

fallback string. A RAG system can refuse, ask a clarifying question, retrieve more, answer partially with caveats, or expose conflicting evidence. Multi-turn unanswerability benchmarks show that robust systems must distinguish between genuinely unanswerable questions, underspecified questions requiring clarification, and answerable questions requiring retrieval (Rosenthal et al., 2026). The next step is interactive calibration: a system should not only know when it is uncertain but also choose the user-facing action that best reduces risk.

Policy Composition Most control methods study one decision at a time: retrieve or not, abstain or answer, verify or skip. Deployed RAG systems need policy composition. For example, if a query is ambiguous, the system should ask clarification before retrieval; if retrieval is low-confidence, it should re-query or broaden the corpus; if evidence conflicts, it should verify sources and expose disagreement; if a document contains instructions, it should treat them as data rather than commands. These actions have dependencies. Abstention after one weak retrieval may be premature if re-retrieval is cheap; re-retrieval after detecting poisoning may be unsafe if the corpus itself is compromised. A full control policy should represent state: query ambiguity, retrieval confidence, evidence sufficiency, conflict level, adversarial risk, and user risk tolerance.

Cost, Latency, and Deployment Constraints Control policies are constrained by cost and latency. Multi-query retrieval, reranking, entailment verification, and self-consistency can improve reliability but increase token cost and response time.

Over-searching work makes this explicit by measuring the cost of unnecessary search (Xie et al., 2026). In organizational settings, retrieval may also be constrained by access control, privacy, and logging requirements. A useful research direction is risk-adaptive computation: cheap pipelines for low-risk queries, stronger verification for high-risk domains, and human review for cases with high uncertainty or high potential harm.

G Evaluation and Benchmarks

Evaluation must reflect the full hierarchy. Standard exact match, F1, and retrieval recall remain useful but insufficient. A RAG system can retrieve relevant evidence and still answer from a distractor; it can answer correctly but be overconfident; it can refuse correctly but fail to explain what evidence is missing; it can pass benign noise tests but fail under poisoned corpora. Robust evaluation must therefore measure both failure resistance and action quality.

We group robust RAG evaluation into six families:

1. **Noise robustness**: performance as irrelevant or misleading passages are injected, removed, reordered, or ranked higher.

2. **Distraction sensitivity**: whether answer or confidence changes when distractors are added while gold evidence remains available.

3. **Conflict sensitivity**: whether the system detects and resolves parametric–retrieved or retrieved–retrieved contradictions.

4. **Adversarial robustness**: attack success rate, poisoned retrieval rate, trigger robustness, defense bypass rate, and recovery after filtering.

5. **Uncertainty and selective prediction**: selective risk, abstention precision/recall, expected calibration error, and clarification quality.

6. **End-to-end action quality**: whether the system chooses the right policy among answer, retrieve, re-retrieve, filter, abstain, or ask.

Benchmarking work such as RGB, RAGTruth, robustness-at-query-level analyses, poisoning benchmarks, and uncertainty benchmarks provides pieces of this landscape (Chen et al., 2024a; Niu et al., 2024; Percin et al., 2025). RGB is useful because it separates RAG abilities rather than reporting one score (Chen et al., 2024a). RAGTruth contributes fine-grained hallucination annotations that support detection and attribution (Niu et al., 2024). Query-level robustness work shows that the query–retrieval interface is itself fragile (Percin et al., 2025). PoisonArena-style benchmarks and LIAR-style attack evaluations stress adversarial settings rather than benign retrieval noise (Chen et al., 2026; Tan et al., 2024). URAG and uncertainty benchmarks evaluate whether systems know when their answers are reliable (Nguyen et al., 2026; Qiu et al., 2026).

The main gap is joint evaluation. Most benchmarks isolate one failure: noise, conflict, poisoning, or unanswerability. In plausible high-risk document-grounded settings, these failures can interact: a query may be ambiguous, retrieval may surface partially relevant evidence, a document may be stale, and one source may be malicious or misleading. A future benchmark should therefore cross product failure types: benign hard negatives, conflicting evidence, poisoned passages, query perturbations, and insufficient context. It should also report action metrics: when the system retrieves, when it filters, when it verifies, when it abstains, and whether these actions improve risk–cost tradeoffs.

Table 9 turns this observation into a proposed benchmark schema. We treat it as a research agenda, not as a validated benchmark contribution: no leaderboard or empirical re-ranking is claimed. The key design choice is to annotate not only answers but also evidence states and action traces. Without evidence annotations, a model can receive credit for an answer supported by parametric memory rather than retrieved evidence. Without action traces, a system that repeatedly retrieves, verifies, and self-consistently samples can look more robust while hiding its cost. Without attacker capability fields, security benchmark results become incomparable.

To make the proposal concrete, we include a 12-item synthetic action-trace pilot with three toy policies (Table 11). The pilot is not a benchmark release, but it demonstrates the diagnostic difference: an answer-only policy can receive some answer credit while failing action quality, whereas action-aware scoring distinguishes answer correctness, control behavior, and their joint success.

Table 12 makes the comparison problem more explicit. Many reported gains depend on retriever choice, generator scale, context length, poisoning

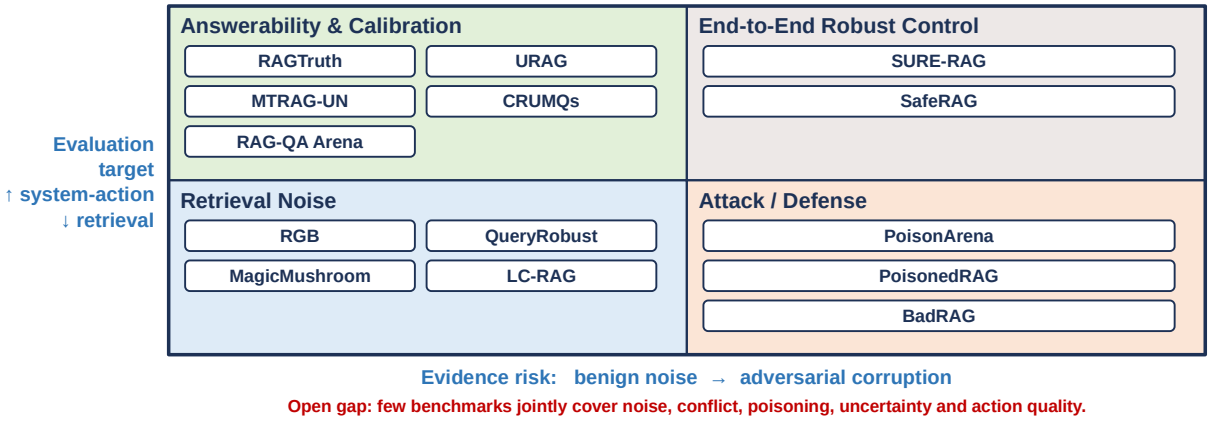


Figure 3: Conceptual benchmark landscape. Placement is illustrative rather than data-derived; the figure organises differences in evidence risk and evaluation level, not empirical distance or benchmark quality. The two axes are coordinates C1 and C4, and the open gap is the reason a single "robustness" number rarely transfers between quadrants.

budget, and metric target. A method can improve EM while worsening citation support; a defense can reduce attack success while reducing answer utility; a retrieval policy can increase accuracy while hiding cost. We therefore treat robust-RAG evaluation as a multi-axis reporting problem rather than a leaderboard scalar.

A concrete benchmark protocol. An executable robust-RAG benchmark should instantiate the following template. Each item contains a user query, a gold answer or abstention/clarification target, a corpus snapshot with document IDs, a retrieved-context candidate set, and labels for supporting, distracting, conflicting, stale, poisoned, and insufficient evidence. The benchmark should define four splits: benign retrieval noise, knowledge conflict, adversarial corruption, and answerability uncertainty. For each split, systems must output both an answer and an action trace: whether they retrieved, filtered, verified, re-retrieved, abstained, or asked for clarification. The metric suite should report answer utility, evidence support precision, attack success or defense bypass where applicable, calibration/coverage, and cost-adjusted action utility. This protocol makes it possible to compare systems that answer directly against systems that spend additional computation on verification or abstention.

Metric Design Robust RAG metrics should be diagnostic. For retrieval, recall and nDCG should be complemented by negative-utility metrics that penalise distracting evidence (Trappolini et al., 2026). For generation, exact match should be complemented by faithfulness, citation support, and claim-level entailment. For attacks, attack success rate should be paired with stealth, retrieval rate, injected-document budget, and transferability. For calibration, expected calibration error should be paired with selective risk and coverage. For control, the unit of evaluation should be an action trace: retrieve, filter, verify, answer, abstain, or ask clarification. This makes it possible to distinguish a lucky correct answer from a robust decision process.

Benchmark Construction Principles We identify four principles for future benchmark construction. First, benchmarks should include both benign and adversarial evidence unreliability. Second, they should provide source-level annotations: which passage supports which claim, which passage is distracting, which passage is poisoned, and which source conflicts with which. Third, they should include unanswerable and underspecified questions so that abstention and clarification can be evaluated. Fourth, they should report cost and latency, because robust RAG methods often trade reliability for additional computation. Benchmarks that omit these dimensions risk rewarding systems that answer more often rather than systems that answer more responsibly.

Benchmark	Risk focus	Task / data	Perturbation or threat	Metrics	Boundary
RGB (Chen et al., 2024a)	noise, rejection, integration, counterfactuality	English/Chinese QA	noisy passages, false evidence, negative evidence	accuracy, ability-specific scores	Broad diagnostic benchmark but not adaptive attack-oriented.
RAGTruth (Niu et al., 2024)	hallucination / faithfulness	QA, data-to-text, summarization	naturally generated unsupported spans	word-level hallucination labels	Strong annotation granularity; less focused on retrieval-side threat models.
Sufficient Context (Joren et al., 2025)	answerability	QA with context sufficiency labels	insufficient retrieved evidence	selective generation / abstention	Captures answerability but not adversarial corruption.
MagicMushroom (Zhang et al., 2025e)	retrieval noise	configurable RAG stress tests	distracting, low-quality, inconsequential, irrelevant noise	robustness under noise types	Fine-grained noise; limited conflict/security coverage.
URAG (Nguyen et al., 2026)	uncertainty	RAG as MCQA / conformal prediction	irrelevant/noisy retrieval and uncertainty sets	coverage, set size, calibration	Good for uncertainty; less direct for free-form generation.
QueryRobust (Percin et al., 2025)	query robustness	QA with query perturbations	wording variation at query–retrieval interface	retrieval and end-task degradation	Focuses on query perturbation, not corpus corruption.
PoisonArena (Chen et al., 2026)	adversarial poisoning	RAG attack competition setting	competing poisoning attacks	attack success, retrieval effects	Strong security focus; benign uncertainty is secondary.
SafeRAG (Liang et al., 2025)	RAG security	security benchmark for RAG	poisoning / malicious retrieval contexts	attack and defense metrics	Security-oriented; not a full evidence reliability suite.
MTRAG-UN (Rosenthal et al., 2026)	multi-turn unanswerability	multi-turn RAG conversations	unanswerable, underspecified, non-standalone questions	abstention / clarification quality	Covers interaction; less focused on adversarial evidence.
CRUMQs (Liu et al., 2025a)	unanswerable multi-hop RAG	realistic multi-hop queries	unanswerable and uncheatable questions	answerability and robustness	Useful for abstention; not a broad attack/noise benchmark.

Table 8: Benchmark-level comparison for robust RAG evaluation. The table makes explicit what each benchmark perturbs, what it measures, and which failure modes remain outside its scope.

Benchmark component	Required annotation	Metric suite	Why it is needed
Evidence reliability	relevance, sufficiency, freshness, provenance, support span	retrieval recall, sufficient-context accuracy, support precision	Distinguishes merely related passages from answer-supporting evidence.
Distraction and conflict	distractor type, conflicting claim pair, source agreement	answer flip rate, conflict sensitivity, context-utilisation score	Reveals whether models ignore harmful context or silently pick a source.
Adversarial corruption	attacker access, insertion budget, trigger type, poisoned document ID	attack success, poisoned retrieval rate, stealth, defense bypass	Makes security results comparable across threat models.
Unanswerability and ambiguity	answerability label, ambiguity type, clarification target	selective risk, abstention precision/recall, clarification accuracy	Evaluates robust action selection, not just answer generation.
Cost and system action	number of retrievals, verifier calls, latency, final action trace	cost-adjusted accuracy, action utility, coverage–risk curve	Prevents methods from hiding robustness gains behind unbounded computation.

Table 9: A proposed benchmark blueprint for robust RAG under unreliable evidence. This is a research agenda rather than a validated leaderboard; existing benchmarks cover parts of the schema.

Pilot item type	Evidence state label	Correct action trace	Answer metric alone would say	Action-aware diagnosis
Benign hard negative	Gold evidence present; hard negative ranked above it	retrieve → rerank/filter → answer with support	Correct/incorrect answer	Separates retriever ranking error from generator distraction.
Insufficient context	Retrieved context related but lacks answer-bearing span	retrieve → detect insufficiency → re-retrieve or abstain	Wrong answer or refusal	Rewards abstention/retrieval control rather than forced guessing.
Retrieved conflict	Two sources disagree with different dates/provenance	retrieve → compare sources → expose conflict or prefer fresher source	One answer may match gold	Checks whether the system hides unresolved conflict.
Poisoned evidence	Malicious document retrieved with plausible support	retrieve → detect/demote/quarantine → answer or abstain	Attack success/failure	Distinguishes poisoned retrieval, defense detection, and final answer steering.

Table 10: Illustrative action-trace pilot for the benchmark blueprint. This is a worked annotation example, not a released leaderboard or empirical system ranking.

Toy policy	Ans.	Act.	Joint
Answer-only	0.25	0.25	0.25
Abstain-on-missing	0.67	0.50	0.50
Evidence-action oracle	1.00	1.00	1.00

Table 11: Action-trace pilot over 12 synthetic diagnostic items. Answer accuracy, action accuracy, and joint answer-action accuracy diagnose different behavior.

Axis	Typical variation across papers	Why conclusions fail to transfer	Minimum reporting requirement
Retriever/generator setting	BM25 vs dense vs hybrid retrieval; small vs frontier generators; frozen vs fine-tuned models	A defense may work by compensating for one retriever or one model scale rather than solving the failure mode.	Retriever, index, top-k, reranker, generator, prompt, decoding, and context length.
Attack budget and access	Poisoned-document count, corpus write access, query control, retriever visibility, generator access	Attack success rates are incomparable when one attacker edits a corpus and another only controls the query.	Capability matrix, insertion budget, target set, stealth objective, and whether malicious evidence is retrieved.
Metric target	EM/F1, faithfulness, citation support, attack success, calibration error, selective risk, cost	Improving answer accuracy can worsen citation faithfulness; improving abstention can reduce coverage; reducing attacks can lower utility.	Report utility, evidence support, risk, calibration, cost, and security as a Pareto surface.
Statistical and reproducibility evidence	Single seed/model vs multiple seeds/models; code/data availability varies	Small prompt or model changes can reverse robustness rankings, especially for LLM-as-judge or attack benchmarks.	Seeds, confidence intervals or bootstrap tests where feasible, released prompts, data splits, and code or exact API settings.
Long-context baseline	RAG-only comparisons vs long-context LLMs with the same documents	A robustness gain may reflect retrieval compression rather than evidence reliability control.	Include a long-context or full-context baseline when context length permits, and report latency/cost.
Attribution vs answer accuracy	Some tasks score only final answer; others require evidence-supported citations	A model can answer correctly for the wrong reason, cite irrelevant evidence, or ignore retrieved support.	Claim-level support labels, citation precision/recall, and answer correctness reported separately.

Table 12: Evaluation-assumption matrix for robust RAG. The table summarises why benchmark results often cannot be compared without aligning system setup, threat model, metric target, reproducibility evidence, and long-context baselines.

H Cross-Cutting Analysis

Peer-Reviewed Backbone and Preprint Trends The main synthesis is anchored in the venue-linked backbone summarised in Table 13. Under that stricter evidence stratum, five conclusions remain stable: retrieved evidence can harm generation; conflict and poisoning require different assumptions; security results require attacker-capability normalisation; calibration and abstention are separate control decisions; and answer accuracy alone is insufficient for robust-RAG evaluation. The broader 268-paper adjudicated map is still useful, but mainly for trend discovery and auditability. Its preprint-heavy cross-tabs are moved to Appendix H.1; they diagnose where the map is dense or sparse, but they are not used as field-wide prevalence estimates.

Critical judgments from the re-analysis. The framework weakens three common readings of the literature. First, many "robust to retrieval noise" claims are over-broad: once distractor hardness, gold-evidence availability, and intervention point are aligned, several methods address complementary mechanisms rather than competing on one robustness axis. Second, many RAG security defenses are narrower than their abstracts suggest: a defense against bounded factual corruption is not evidence of robustness to prompt-bearing documents, trigger-based backdoors, or post-hoc traceback settings. Third, answer-only benchmark gains are often under-specified: without citation support, evidence sufficiency, and action-cost labels, a higher score can mean better answering, riskier coverage, or simply more computation. These are the places where the framework changes interpretation rather than merely renaming categories.

Where Robustness Interventions Act Across the literature, robustness interventions cluster by where they act in the RAG pipeline. **Corpus-level interventions** sanitise, filter, watermark, or validate documents before indexing. They are attractive because they prevent bad evidence from entering the system, but they require corpus governance and may not be possible for open-web retrieval. **Retriever-level interventions** improve ranking, detect poisoned passages, or diversify evidence. They are efficient and model-agnostic, but they only control what enters the context, not how the generator uses it. **Context-level interventions** compress, reorder, or filter the retrieved context. They are flexible, but can remove qualifiers or multi-hop evidence. **Generator-level interventions** use prompting, fine-tuning, note-taking, self-reflection, or hidden-state control. They directly address evidence use but may be model-specific. **System-level interventions** decide when to retrieve, verify, abstain, or ask. They are closest to deployment needs but require calibrated signals from all earlier stages.

This intervention map helps explain why papers sometimes appear to solve the same problem while actually operating at different layers. For example, irrelevant-context filtering (Yoran et al., 2024), Chain-of-Note (Yu et al., 2023), and Sufficient Context (Joren et al., 2025) all address unreliable evidence, but the first filters evidence, the second changes generator reasoning, and the third changes answer policy. Similarly, PoisonedRAG-style attacks (Zou et al., 2025) target the corpus-to-generation path, while AIP attacks prompt templates (Chaturvedi et al., 2025). A comprehensive survey must therefore compare methods by intervention point, not only by reported task accuracy.

Critical Case Studies of Taxonomy Use

Case 1: irrelevant evidence is not one problem. IrrelCtx, RAAT, RGB, Chain-of-Note, and Sufficient Context are sometimes all described as robustness-to-noise work, but they instantiate different experimental objects. IrrelCtx tests whether a generator can ignore irrelevant context; RAAT treats noise as a training distribution; RGB decomposes reliability into benchmark abilities; Chain-of-Note changes the generator's intermediate reading behavior; Sufficient Context changes the answer policy when evidence is inadequate. A direct leaderboard comparison would mix retrieval perturbation, generator selectivity, and answerability control. The taxonomy changes the question from "which method is best under noise?" to "which evidence state is present, which module intervenes, and what action should be scored?"

Case 2: poisoning papers differ by attacker capability. Poisoning Retrieval Corpora, PoisonedRAG, BadRAG, TrojanRAG, Phantom, AIP, and newer one-shot or traceback studies are all adversarial, but they assume different access and objectives. Some manipulate dense retrieval, some steer final answers, some rely on triggers, some attack shared prompt templates, and some evaluate whether poisoned evidence can be traced after the fact. Attack success rate is therefore not a portable metric un-

Backbone conclusion	Venue-linked evidence base	What remains stable	What remains preprint/trend evidence
Irrelevant or noisy evidence can change generation behavior.	Irrelevant-context robustness, noise-aware training, counterfactual-noise analyses, and robust fine-tuning papers in ICLR/ACL/NAACL/SIGIR-style venues.	Robustness is not just retriever recall; generator selectivity and context construction matter.	Fine-grained noise taxonomies, long-tail stress tests, and model-specific mitigation rankings.
Knowledge conflict is distinct from poisoning.	Conflict-resolution and imperfect-retrieval work in TACL/COLING/EMNLP-style records, plus venue-linked conflict benchmarks.	Natural contradiction requires source reconciliation and evidence sufficiency reasoning.	Many automatic conflict-resolution methods and large-scale conflict benchmark variants.
RAG security comparisons require capability normalisation.	USENIX/WWW/CCS/KDD-style poisoning, traceback, membership-inference, and backdoor records.	Attack success cannot be interpreted without corpus write access, query control, retriever visibility, insertion budget, and stealth.	The detailed ranking of attacks/defenses and many newly proposed bypasses.
Calibration and abstention are action-control decisions.	Venue-linked relevance-assessment, answerability, calibration, and RAG evaluation records.	Confidence, evidence sufficiency, retrieve/answer decisions, and abstention are separable evaluation targets.	Specific adaptive-control policies and frontier-model calibration behavior.
Benchmark design needs evidence and action labels.	RGB, RAGTruth, RAG evaluation libraries, long-form/context benchmarks, and challenge-style records.	Answer accuracy alone is insufficient for robust RAG evaluation.	The full action-trace blueprint remains a proposal until implemented and validated.

Table 13: Peer-reviewed / venue-linked backbone of the analysis. These are the conclusions the paper treats as stable when preprint-only evidence is downweighted.

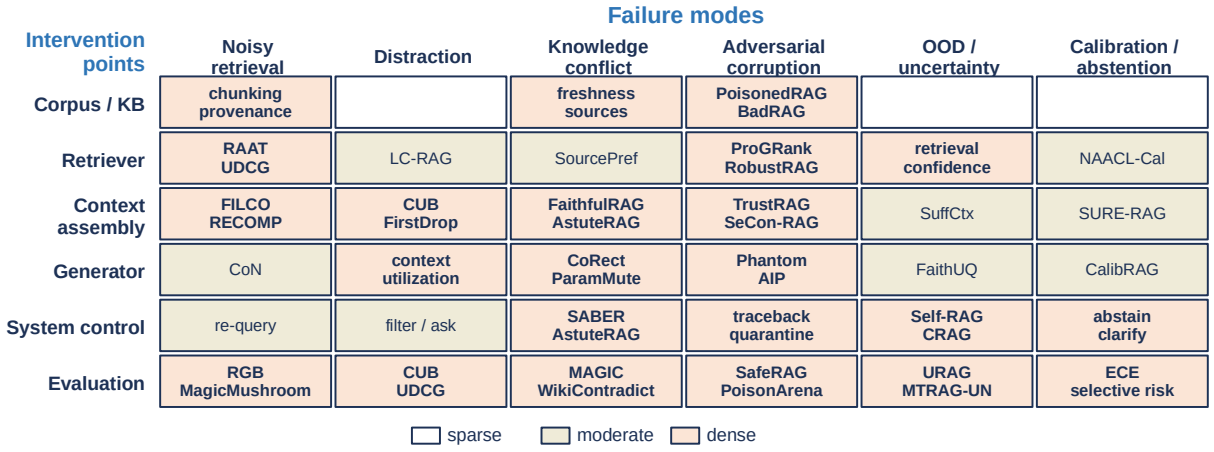


Figure 4: Schematic literature-family × intervention matrix. Shading and example labels are illustrative design aids, not counts from Map B; quantitative Map-B distributions appear in Table 21. Rows are coordinate C3 and columns are coordinate C1, which is why a cell, not a column, is the unit at which two systems become comparable.

less corpus write access, query control, retriever visibility, generator access, insertion budget, and stealth requirements are aligned. This is why Table 4 treats capability fields as part of the result, not just implementation detail.

Case 3: calibration is not the same as abstention. UncertaintyRAG, CalibRAG, URAG, Self-RAG, CtrlA, Sufficient Context, and SABER-style trust/abstain methods all relate to uncertainty, but they optimise different objects. Some estimate uncertainty in spans or answer sets; some calibrate decision confidence; some decide when to retrieve; some decide when to abstain; some predict whether the retrieved context is sufficient. Expected calibration error, selective risk, retrieval-trigger accuracy, and abstention precision cannot be collapsed into one number. The taxonomy forces a distinction between belief estimation and action selection, explaining why a method can improve calibration without repairing evidence or improve retrieval triggering without improving final answer faithfulness.

Case 4: poisoning defenses do not share one threat model. Certified defenses, isolate-and-aggregate defenses, poisoned-document detectors, activation-based traceback, and prompt-injection defenses respond to different adversarial assumptions. A certificate can be meaningful under a bounded corruption budget but irrelevant when the attacker controls prompt-like instructions; a traceback method may be useful after an incident but may not prevent answer steering; a prompt-injection defense may not address embedding-space poisoning. The framework therefore asks defense papers to report the same capability fields

Case	Naive reading	Normalisation coordinate that changes it	Revised conclusion
IrrelCtx (Yoran et al., 2024) vs. RAAT (Fang et al., 2024)	Both are "noise-robust RAG" methods.	IrrelCtx targets naturally retrieved irrelevant context and filtering/back-off; RAAT trains on constructed relevant/irrelevant/counterfactual mixtures.	They are complementary context-filtering vs. noise-training regimes, not one leaderboard.
IrrelCtx (Yoran et al., 2024) vs. Chain-of-Note (Yu et al., 2023)	Both reduce distractor harm.	Intervention point differs: context filtering/fine-tuning vs. generator-side document reading.	A valid comparison must hold context state fixed and report whether the action is filter, answer, or unknown.
Sufficient Context (Joren et al., 2025) vs. Self-RAG (Asai et al., 2024)	Both decide when retrieval is useful.	Evidence-sufficiency labeling vs. retrieve/generate/critique action policy.	Sufficiency prediction is an evidence-state signal; adaptive retrieval is a policy over actions and costs.
RAGTruth (Niu et al., 2024) vs. FaithfulRAG/AstuteRAG (Zhang et al., 2025d; Wang et al., 2025b)	All are faithfulness/conflict papers.	Unit of support differs: generated span support vs. fact/source conflict resolution.	Detector F1 and conflict-resolution accuracy should not be pooled as one faithfulness score.
Tug-of-War (Jin et al., 2024) vs. ParamMute (Huang et al., 2025)	Both study parametric-vs-contextual knowledge.	Source-preference analysis vs. parametric-knowledge suppression/intervention.	The conclusion shifts from "trust context or model" to reporting source reliability and update policy.
Poisoning corpora (Zhong et al., 2023) vs. PoisonedRAG (Zou et al., 2025)	Both are RAG poisoning attacks.	Corpus-write access, query control, retriever visibility, and target objective differ.	Attack success rates transfer only inside a shared capability envelope.
PoisonedRAG (Zou et al., 2025) vs. CertifiedRAG (Xiang et al., 2024)	Attack and defense appear directly comparable.	Defense guarantee assumes bounded retrieved corruption and an aggregation/abstention policy.	Certification is meaningful only under the stated budget; it is not a universal defense to open-web insertion.
AgentPoison (Chen et al., 2024d) vs. document poisoning (Zou et al., 2025)	Both corrupt retrieved evidence.	Persistent memory/tool state vs. static corpus passages.	Agentic poisoning needs memory-update and tool-call action labels, not only answer steering.
Spill-Beans/privacy (Qi et al., 2025) vs. steering attacks (Zou et al., 2025)	Both are security failures.	Objective differs: data extraction or privacy leakage vs. answer manipulation.	Privacy, extraction, steering, traceback, and recovery require separate outcome columns.
RAGAS/ARES (Es et al., 2024; Saad-Falcon et al., 2024) vs. RAGTruth (Niu et al., 2024)	All evaluate grounded RAG.	LLM-as-judge metric toolkit vs. human-annotated hallucination/support resource.	Automated evaluation and corpus labels are different evidence artifacts; agreement with one does not validate the other.
RGB (Chen et al., 2024a) vs. Sufficient Context (Joren et al., 2025)	Both benchmark unreliable evidence.	Diagnostic ability axes vs. answerability/sufficiency labels and selective risk.	A benchmark score is interpretable only after stating the evidence state and required action.
Lost-in-the-Middle (Liu et al., 2024) vs. long-context retrieval	More context seems to solve retrieval misses.	Evidence availability differs from evidence selection and citation.	Full-context access can still fail if the model cannot select or ground the supporting span.
ColPali/Table/Graph RAG (Faysse et al., 2025; Chen et al., 2024c; He et al., 2024) vs. passage RAG	Extension papers look like new application areas.	Evidence object changes: page image, table cell/schema, graph neighborhood, or tool memory.	They are retained as extension-layer stress tests of typed evidence states, not as broad co-equal survey scope.

Table 14: Concrete conclusions that change after five-coordinate normalisation. Each row is grounded in released case-study or Map-B records and shows why a common citation-list comparison becomes invalid, partial, or more specific once failure, assumption, intervention, metric, and action are aligned.

as attack papers and to distinguish prevention, detection, demotion, traceback, and recovery.

Strengths and Weaknesses by Method Family Filtering and reranking are attractive robustness methods because they can be added around existing LLMs. Their weakness is false negatives: an aggressive filter may discard the only passage that supports a rare or multi-hop answer. Robust training can teach models to ignore noise, but training data may not cover future attacks or domain-specific distractors. Conflict modeling provides a principled way to handle contradictions, but fact extraction and alignment remain brittle for long, compositional, or numerical claims. Certified defenses provide valuable guarantees, but their assumptions are often narrower than realistic attack surfaces. Adaptive retrieval and abstention improve risk control, but their performance depends on calibration signals that can themselves be miscalibrated under distribution shift.

The synthesis suggests that method families should be composed according to risk profile rather than stacked indiscriminately. A low-risk FAQ system may only justify lightweight filtering and citation checks, while a high-stakes document-grounded setting may require source validation, contradiction detection, selective abstention, and escalation. These are design implications, not claims that the evidence map validates any particular enterprise, medical, or legal deployment.

What Counts as Progress? The corpus currently reports progress through heterogeneous metrics: EM/F1, hallucination rate, retrieval recall, attack success rate, calibration error, abstention accuracy, and human preference. These metrics are not directly comparable. A defense that reduces attack

Work	Failure	Task / setup	Key assumption	Metric	Boundary
IrrelCtx	irrelevant context	Open-domain QA	Irrelevance can be filtered or learned against	Robust accuracy	No adversarial/conflict setting.
RAAT	retrieval noise	Noisy QA training	Noise-aware training improves generation	F1/EM	Needs deployment-like noise.
RGB	multi-ability RAG	Bilingual benchmark	RAG reliability decomposes into abilities	Ability accuracy	Not adversarially adaptive.
FaithfulRAG	conflict	Fact-level conflict	Conflicts can be localised and reasoned over	Faithfulness	Needs reliable fact extraction.
AstuteRAG	imperfect retrieval	Training-free consolidation	Internal/external knowledge can be reliability-compared	Robust QA	Depends on LLM self-assessment.
PoisonedRAG	poisoning	Injected passages	Few crafted passages can dominate retrieval	Attack success	Targeted corruption focus.
RobustRAG	defense	Bounded corruption	Isolate-and-aggregate can certify robustness	Certificate, accuracy	Assumes attack budget.
Self-RAG / CtrlA	adaptive retrieval	Retrieve-or-answer	Model state can trigger retrieval	Trigger quality	Calibration can shift.
CalibRAG / URAG	calibration	Decision/MCQA	Confidence should reflect evidence uncertainty	ECE, coverage	Does not repair evidence.
PoisonArena / SafeRAG	security eval.	Poisoning benchmark	Threat models must be explicit	ASR, bypass	Still isolates attacks.

Table 15: Analytical matrix of representative robust RAG work. Compared with a citation list, the matrix highlights each work's task setup, central assumption, metric focus, and boundary of applicability.

success may reduce answer utility; a calibration method may improve selective risk while lowering coverage; a retriever that improves recall may increase distraction. Robust RAG progress should therefore be reported as a Pareto frontier across utility, robustness, calibration, cost, and security. This mirrors the broader lesson from trustworthy ML: robustness is not a scalar property but a tradeoff surface.

What Is Still Missing from the Literature Several gaps recur across categories. First, benchmarks rarely combine failures. Noise, conflict, poisoning, and unanswerability are usually tested separately, even though realistic systems may encounter them jointly. Second, many papers evaluate with one retriever and one generator, making transfer difficult to assess. Third, few works evaluate long-term corpus dynamics: documents are updated, poisoned documents are removed, indices are rebuilt, and user feedback changes future retrieval. Fourth, most papers treat users as passive recipients rather than interactive participants who can clarify intent or verify sources. Fifth, multilingual and multimodal robust RAG remains underdeveloped relative to text-only English QA; because our corpus is also text-heavy, this is a boundary of the present survey rather than a resolved conclusion about those areas.

Work	System setup	Failure construction	Metric / cost	Main claim	Failure boundary
IrrelCtx	RALM / open-domain QA; retrieved context injected into generator	Irrelevant passages with varying plausibility	Robust accuracy / answer degradation	Models can be brittle to irrelevant context.	Does not cover adversarial corruption or answerability policy.
RAAT	Noise-aware training for retrieval-augmented LMs	Multiple retrieval-noise types during training/evaluation	EM/F1 under noisy retrieval	Adaptive adversarial training improves noise robustness.	Transfer depends on whether deployment noise matches training noise.
RGB	Benchmark over English/Chinese QA-style RAG tasks	Noise, negative rejection, integration, counterfactual evidence	Ability-specific accuracy	RAG reliability decomposes into distinct abilities.	Benchmark abilities are diagnostic but not a unified control policy.
Sufficient Context	QA contexts labeled by sufficiency	Insufficient vs sufficient retrieved evidence	Selective answering / abstention quality	Evidence sufficiency should govern answerability.	Does not repair retrieval and does not model adversarial evidence.
PoisonedRAG	Corpus-injection attack against RAG	Crafted poisoned passages inserted into retrieval corpus	Attack success, retrieval of poison	Few poisoned documents can steer answers.	Strength depends on corpus write access, topic targeting, and retrieval success.
Certified / isolate defenses	Bounded-corruption RAG defense setting	Assumed corruption budget over retrieved evidence	Certificate, robust accuracy	Aggregation can certify robustness under budget assumptions.	Guarantees narrow when attacker violates budget or attacks instructions/tools.
Self-RAG / CtrlA	Adaptive retrieve-or-answer policies	Uncertainty about whether retrieval is needed	Trigger quality, answer utility, cost	Retrieval should be a conditional action.	Calibration shifts under distribution change; action cost often underreported.
RAGTruth	Annotated hallucination/faithfulness resource	Naturally generated unsupported spans across tasks	Span-level hallucination labels	Fine-grained support labels improve evaluation.	Faithfulness labels do not fully specify retrieval failure or system action.

Table 16: Method-level comparison of representative robust-RAG work. The table makes explicit the setup, failure construction, metric target, main claim, and failure boundary that must be aligned before comparing methods.

Case	Released rows / papers	Audited comparison fields	Metric or failure conflict	Synthesis / artifact boundary
Noise / distraction robustness	5 rows: IrrelCtx, RAAT, Chain-of-Note, Sufficient Context, RGB	Assumption, QA dataset or benchmark slice, generator / reader setup, negative type, rank and gold-evidence availability.	EM/F1, rejection, selective risk, and ability-specific benchmark scores answer different questions.	Treat filtering, noise-aware training, generator reading, sufficiency prediction, and benchmark diagnosis as different mechanisms; artifact status is recorded per paper in the released worksheet.
Faithfulness, conflict, and attribution	5 rows: RAGTruth, FaithfulRAG, AstuteRAG, Tug-of-War, Chain-of-Note	Unit of support, source-conflict assumption, attribution granularity, model-side reasoning setup, artifact availability.	Span-level support, fact-level conflict resolution, source preference, and final answer accuracy can disagree.	Citation faithfulness should not be collapsed into conflict robustness; the worksheet records where artifact evidence is dataset-level, code-level, or paper-only.
Adversarial corruption and RAG security	6 rows: PoisonedRAG, corpus poisoning, BadRAG, certified defense, TrojanRAG, traceback	Corpus write access, insertion budget, retriever visibility, query control, generator access, stealth, retrieved malicious evidence, defense timing.	Attack success, certified robust accuracy, detection, traceback, and clean utility are not one defense score.	Compare only inside a capability envelope; prevention, demotion, detection, traceback, recovery, and certification remain separate outcomes.
Calibration, abstention, and adaptive control	5 rows: Sufficient Context, Self-RAG, CRAG, CalibRAG, uncertainty RAG	Confidence target, evidence sufficiency labels, retrieval trigger, action set, cost budget, model/policy setup.	ECE, selective risk, coverage, retrieval-trigger quality, and answer utility can move in different directions.	Separate belief estimation from action selection; report coverage, cost, and final action alongside correctness.
Evaluation and benchmark comparability	5 rows: RGB, RAGTruth, Sufficient Context, query robustness, RAG security benchmark	Evidence-state labels, perturbation source, model outputs scored, threat model, benchmark artifact availability.	Harder scores may reflect retrieval instability, unsupported spans, unanswerability, or adversarial steering.	Evaluation resources are diagnostic slices, not a universal robustness scale; benchmark comparisons require aligned evidence state and action policy.

Table 17: Deeper case-study synthesis. The released 26-row worksheet compares assumptions, datasets or benchmark settings, model/evidence setup, metrics, failure modes, artifact availability, and comparison implications for the same five C1–C5 cases.

Apparent conclusion	What the framework changes	Non-trivial finding	Implication
"Method A is more robust to noise than method B."	Aligns noise type, rank position, gold-evidence availability, and intervention point.	Some methods solve generator distraction while others solve retriever defects; their scores are not competing estimates of the same property.	Report noise source and intervention point before ranking methods.
"Defense X lowers attack success."	Separates prevention, demotion, detection, traceback, and recovery.	A defense can improve traceback without lowering user-facing attack success, or certify bounded corruption without handling prompt-bearing documents.	Security papers should report outcome type, not only attack success rate.
"Calibration improves reliability."	Separates confidence in answer correctness from evidence sufficiency and action utility.	A calibrated answerer can still answer from insufficient evidence; an abstaining system can lower risk while reducing coverage.	Calibration metrics should be paired with evidence and action metrics.
"Benchmark Y is harder."	Compares evidence state, required action, cost budget, and metric target.	A benchmark can be hard because retrieval fails, because evidence conflicts, or because the correct action is refusal; these failures should not be pooled.	Leaderboards should expose diagnostic sub-scores instead of one robust-RAG score.

Table 18: Distilled classes of non-trivial findings produced by the comparability framework. Main Table 14 gives the concrete paper-pair cases; this appendix table groups them into reusable interpretation rules.

Main claim	Cited evidence basis	What the evidence supports	Boundary
Irrelevant or noisy context can harm generation.	Irrelevant-context robustness, RAAT, RGB, and counterfactual-noise studies.	Noise affects both retrieval quality and generator context use.	Does not imply one robustness score covers all noise types.
Knowledge conflict differs from adversarial corruption.	Conflict and poisoning studies with different evidence origins and evaluation targets.	Natural disagreement and attacker-created contradiction require different assumptions.	Conflict-resolution results should not be read as poisoning-defense results.
RAG attacks require capability normalisation.	Poisoning, backdoor, prompt-injection, traceback, and certification studies.	Attack/defense results require corpus access, query control, retriever visibility, and budget.	The paper does not rank attacks by real-world prevalence.
Calibration, abstention, and retrieval control are distinct.	Sufficient-context, uncertainty, adaptive-retrieval, and calibration studies.	Confidence estimation, evidence sufficiency, retrieval triggering, and abstention optimise different quantities.	Does not validate a single universal control policy.
Action-trace evaluation is a benchmark agenda.	Benchmarks exposing subsets of answer, evidence, action, cost, and threat fields.	Existing benchmarks cover pieces of the proposed schema.	No new leaderboard or empirical system ranking is claimed.

Table 19: Narrative claim-to-source ledger. Expanded and strict citation audits were model-assisted and deterministic; this table records the intended evidence boundary rather than certifying every citation.

Case	Papers / settings compared	Why naive comparison fails	Synthesis from the framework
Noise / distraction robustness	IrrelCtx, RAAT, RGB, Chain-of-Note, Sufficient Context	Some vary retriever noise, some change generator reading behavior, and some decide answerability; robust accuracy, ability scores, and abstention metrics answer different questions.	Treat retrieval imperfection, harmful context influence, and insufficient evidence as separate states; compare methods only after fixing the intervention point and required action.
Faithfulness, conflict, and attribution	RAGTruth, FaithfulRAG, AstuteRAG, Tug-of-War, Chain-of-Note	Span-level support, fact-level conflict, document-level reading, and final answer correctness are different target objects.	Citation faithfulness is not the same as conflict robustness; valid comparisons must state support unit, source reliability assumption, and action policy.
Adversarial corruption and RAG security	PoisonedRAG, BadRAG, certified defenses, traceback methods, prompt-injection defenses	One result may assume corpus write access, another bounded corruptions, another instruction-bearing documents, and another post-hoc observability; attack success rates transfer poorly across assumptions.	Report corpus write access, insertion budget, retriever visibility, generator access, stealth, and whether malicious evidence was retrieved before comparing attacks or defenses.
Calibration, abstention, and adaptive control	Self-RAG, CRAG, CalibRAG, uncertainty RAG, Sufficient Context	ECE, selective risk, retrieval-trigger quality, and abstention precision measure different objects; better confidence does not necessarily repair evidence or choose the right user action.	Separate belief estimation from action selection. A complete benchmark should score confidence, evidence sufficiency, final action, and cost.
Evaluation and benchmark comparability	RGB, RAGTruth, Sufficient Context, query-robustness benchmarks, RAG security arenas	A benchmark can be hard because retrieval fails, evidence conflicts, attribution is unsupported, the item is unanswerable, or an attacker is stronger.	Evaluation resources should expose diagnostic sub-scores and threat/action labels rather than one robust-RAG scalar.

Table 20: Critical case studies showing how the comparability framework changes interpretation of representative robust-RAG papers.

H.1 Evidence-Map Cross-Tabulations

These cross-tabulations diagnose where the released map is dense or sparse. They are preprint-heavy and are not used anywhere as field-wide prevalence estimates.

Dimension	Distribution in the 342-paper adjudicated Map B
Publication year	2025: 170; 2024: 92; 2026: 65; 2023: 12; 2021: 2; 2022: 1
Source / venue type	arXiv/preprint: 207; venue/proceedings-linked: 126; workshop/challenge: 5; other/unclear: 4
Legacy literature-family tag	adversarial corruption: 120; knowledge conflict: 57; calibration/control: 55; evaluation/benchmark: 43; noisy retrieval: 34; distraction: 18; extension layer: 7; survey/background: 6; OOD/uncertainty: 2
Method family	defense/mitigation: 119; attack: 75; benchmark/evaluation: 67; control/calibration: 27; method/evaluation: 21; analysis: 21; survey background: 6; conflict modeling: 4; method: 2
Intervention point	generator: 81; corpus/KB: 67; evaluation: 51; context assembly: 48; mixed: 43; system control: 36; retriever/context: 15; not applicable: 1

Table 21: Descriptive corpus statistics for the canonical 342-row adjudicated Map B. The legacy family tag is broader than the framework's failure ontology; these are map statistics, not field-wide prevalence estimates.

I Background and Preliminaries

RAG systems are often described as a simple two-stage pipeline: retrieve documents, then generate an answer. Robustness failures show that this description is too coarse. In practice, a deployed RAG system includes corpus acquisition, chunking, indexing, retrieval, reranking, context assembly, prompt construction, generation, citation, verification, and sometimes tool use or multi-turn interaction. Each stage can introduce reliability errors. A noisy document can enter during acquisition; chunking can split evidence away from its qualifiers; dense retrieval can surface semantically similar but answer-irrelevant passages; reranking can over-prioritise fluent misinformation; prompt construction can place misleading evidence before correct evidence; and generation can over-trust the retrieved context.

Evidence reliability dimensions. We use *evidence reliability* as an umbrella concept with six dimensions. **Relevance** asks whether the evidence is about the query. **Sufficiency** asks whether it contains enough information to answer. **Faithfulness support** asks whether the final claim is entailed by the evidence rather than merely related. **Consistency** asks whether different evidence sources agree. **Provenance and freshness** ask where the evidence came from and whether it is up to date. **Integrity** asks whether the evidence has been manipulated. Existing work usually studies one or two of these dimensions at a time. For example, irrelevant-context robustness focuses on relevance and distraction (Yoran et al., 2024), sufficient-context work focuses on answerability (Joren et al., 2025), conflict work focuses on consistency (Jin et al., 2024; Zhang et al., 2025d), and poisoning work focuses on integrity (Zou et al., 2025; Xue et al., 2024). A central goal of this survey is to place these dimensions in one framework.

Benign unreliability vs. adversarial unreliability. Some evidence failures are benign: a retriever misses a relevant document, a query is ambiguous, or a corpus contains stale information. Others are adversarial: an attacker plants poisoned passages, modifies prompt templates, or injects hidden instructions into retrievable content. The two regimes share mechanisms. A hard negative and a poisoned passage can both be semantically close to the query; a stale document and a malicious document can both conflict with parametric knowledge; an ambiguous query and an adversarial trigger can both cause retrieval instability. However, adversarial settings require stronger assumptions about attacker capability, corpus mutability, and defense bypass. This is why we treat adversarial corruption as a distinct section while still connecting it to noisy retrieval and conflict.

Why robust RAG differs from robust LLMs. Robustness in closed-book LLMs often focuses on input perturbations, calibration, jailbreaks, or distribution shift. RAG adds an external memory whose content changes independently of the model. This creates three asymmetries. First, the corpus can be updated or poisoned without retraining the model. Second, the retriever can fail even when the generator is strong. Third, the generator may treat retrieved text simultaneously as evidence, instruction, and context. Therefore, robust RAG requires reasoning about the interaction between retriever, corpus, prompt, and generator, not only the generator.

Historical foundations and boundary. Many ingredients of robust RAG predate the current RAG literature. Retrieval-augmented pretraining, dense retrieval, and fusion-in-decoder open-domain QA

established the retrieval–reader/generator interface (Guu et al., 2020; Karpukhin et al., 2020; Izacard and Grave, 2021). Selective prediction and neural calibration provide the conceptual basis for coverage–risk and confidence calibration (Geifman and El-Yaniv, 2017; Guo et al., 2017). Long-context work shows that simply placing relevant evidence in a long input does not guarantee use of that evidence (Liu et al., 2024). These works are background foundations rather than members of the venue-augmented robust-RAG evidence map unless they directly study RAG evidence reliability. We cite them to avoid presenting older IR, QA, calibration, and context-utilisation problems as entirely new.

A running example. Consider a hypothetical clinical-document assistant answering whether a drug is safe for a patient group. A benign failure might retrieve an outdated guideline or a passage about a similar but different drug. A distraction failure might retrieve a high-scoring passage that mentions the drug and patient group but discusses a different contraindication. A conflict failure might retrieve two guidelines with different dates. An adversarial failure might retrieve a poisoned webpage recommending unsafe use. An uncertainty failure might occur when no retrieved passage covers the subgroup. This example is illustrative rather than evidence that the surveyed benchmarks validate clinical deployment. It shows why robust RAG is a control problem grounded in evidence reliability: the system may need to re-rank, check dates, compare sources, abstain, or ask for more information rather than simply answer.

J Practical Guidelines

The literature suggests different design priorities for different risk regimes, but the evidence base is strongest for text QA and document-grounded RAG. This section therefore gives design checklists and reporting guidance rather than validated deployment prescriptions for enterprise, medical, legal, or agentic systems.

For Benchmark Builders Benchmark builders should avoid treating robustness as a single score. A useful benchmark should specify: (1) the corpus construction process; (2) the retriever and generator assumptions; (3) the failure type being tested; (4) whether failures are benign or adversarial; (5) the required system action; and (6) the cost budget. For noisy retrieval, benchmarks should vary the type and placement of distractors, not only the number of negative passages. For conflict, benchmarks should distinguish parametric–retrieved conflict from retrieved–retrieved conflict. For adversarial evaluation, benchmarks should report attacker knowledge, insertion budget, trigger type, and whether malicious evidence is actually retrieved. For uncertainty, benchmarks should include answerable, unanswerable, ambiguous, and underspecified questions.

Benchmarks should also provide evidence-level annotations. Without knowing which passage supports which claim, it is difficult to evaluate faithfulness or citation. Without knowing which passage is distracting, it is difficult to measure negative utility. Without knowing which document is poisoned, it is difficult to evaluate defense recall and false positives. The most useful future benchmarks will resemble diagnostic test suites rather than single leaderboard datasets.

For Method Designers Method designers should state the intervention point. A method that filters retrieved passages should not be compared directly to a method that fine-tunes the generator unless the comparison controls for retriever, generator, corpus, and cost. A method that improves calibration should specify whether it calibrates answer correctness, evidence sufficiency, retrieval usefulness, or abstention. A method that handles knowledge conflict should specify whether it assumes retrieved evidence is more trustworthy than parametric memory.

Methods should also report failure cases. A filter may remove useful multi-hop evidence. A conflict resolver may over-trust internal knowledge when the model is outdated. A poisoning defense may fail against paraphrased or multimodal attacks. An abstention method may over-refuse when retrieval returns all-negative contexts. Reporting these failures is not merely a negative result; it clarifies the method's operating region and helps later work combine complementary methods.

For System Builders System builders should start with risk assessment. If the corpus is closed, curated, and static, benign noise and calibration may dominate. If the corpus is open-web or user-editable, adversarial corruption becomes central. If the domain has high failure cost, answerability, provenance, and escalation should be first-class. If the system uses tools or persistent memory, re-

trieved instructions and memory poisoning require special controls. These recommendations are extrapolations from the surveyed text-RAG evidence and should be validated in the target domain.

A practical robust RAG stack should include at least four components. First, corpus governance: source tracking, ingestion filters, and update logs. Second, retrieval diagnostics: not just top-k scores, but diversity, provenance, and conflict signals. Third, generation safeguards: evidence separation from instructions, claim-level support checks, and conflict-aware prompting or reasoning. Fourth, control policies: calibrated abstention, clarification, re-retrieval, and escalation. The exact implementation can be lightweight or heavy depending on risk, but the architecture should preserve these functions.

K Open Challenges and Future Directions

Unified evidence reliability estimation. Current systems often estimate relevance, faithfulness, safety, and confidence with separate modules. Robust RAG needs a unified evidence reliability model that accounts for relevance, sufficiency, consistency, provenance, freshness, and adversarial risk.

Causal understanding of context influence. We need better tools for measuring how each passage changes the final answer. Passage attribution, counterfactual context removal, hidden-state probing, and answer-flip analysis can help distinguish useful evidence from harmful distraction.

Conflict-aware generation. Fact-level conflict resolution remains immature. Future systems should explicitly represent claims, sources, timestamps, and confidence, and generate answers that expose unresolved conflict rather than hiding it.

Adaptive control under cost constraints. Retrieval, reranking, verification, and human review all have costs. Robust RAG should optimise risk–cost tradeoffs: when is extra retrieval worth it, when is abstention better, and when should a system ask the user for clarification?

Adversarially realistic benchmarks. Many attacks assume simplified access patterns, while many defenses assume narrow threat models. Benchmarks should specify attacker capabilities, corpus mutability, retriever visibility, access control, and deployment constraints.

Multimodal, agentic, and long-context RAG. Robustness becomes harder when evidence includes images, tables, code, web pages, memory, tools, or long-context documents. Agentic RAG introduces persistent memory and tool calls, expanding both the control problem and the attack surface.

Versioned evidence maps. The literature is changing quickly, especially around RAG attacks and defenses. The practical implication is modest but important: survey artifacts should be versioned and machine-readable. Our supplementary CSV files provide a first snapshot of paper categories, source strata, threat-model fields, benchmark dimensions, assumptions, and known limitations; maintaining a living survey would require public versioning, update cadence, and re-coding procedures beyond the scope of this submission.

L Taxonomy Assignment Rules and Operational Pipeline

Part II: Testing and Applying the Taxonomy

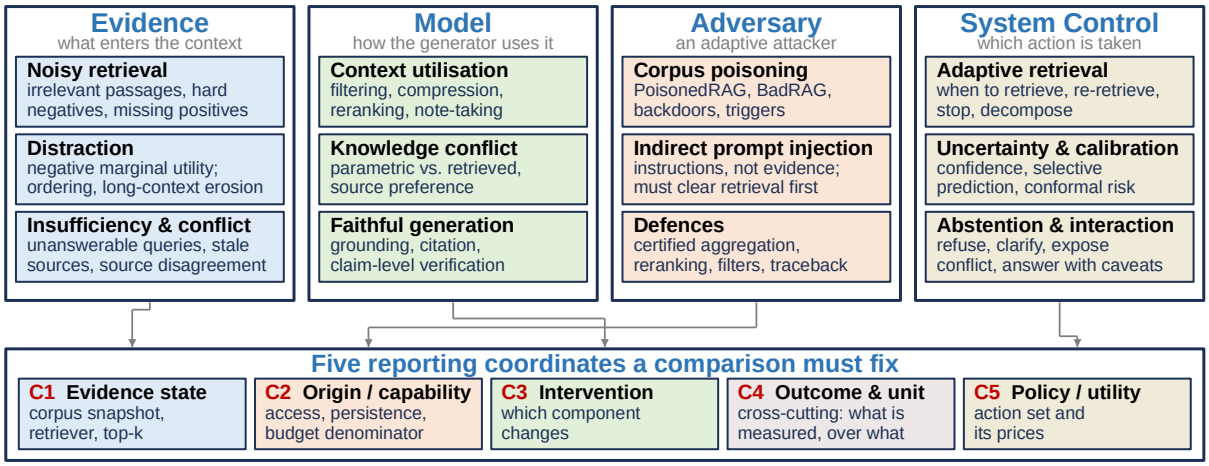


Figure 5: The four failure surfaces and the reduction they support. Each column is one surface of the survey (§4–§7); the band below is the five reporting coordinates of §8, and the connectors show which surface each coordinate collapses. C4 is cross-cutting and has no single parent surface. The organisation is a partition of the *comparison* rather than of the literature: the columns tell a reader where to look for related work, and the band tells a reader what must be stated on both sides before two numbers can be placed together.

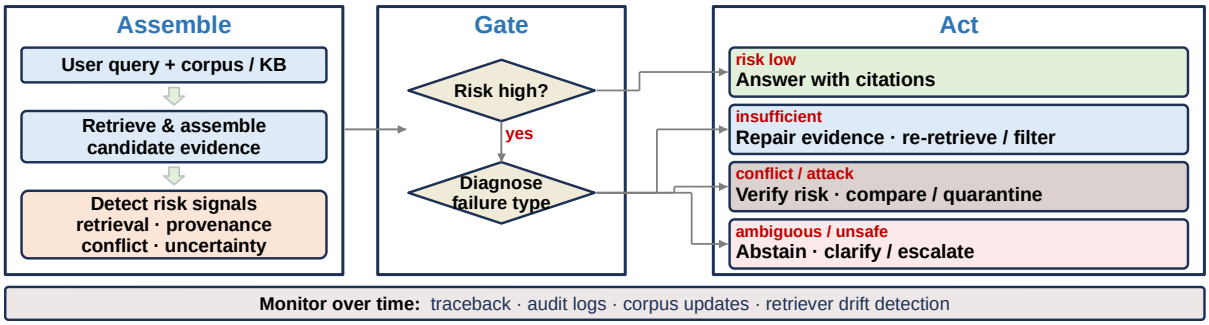


Figure 6: Operational decision flow for robust RAG. Unlike the map in Figure 5, this state-machine view describes the branching actions a deployed system can take when unreliable evidence is encountered: answer, repair evidence, verify risk, abstain, clarify, escalate, or monitor. The branch conditions are the evidence states of §4, and the action set is coordinate C5.

Boundary case	Rule used in this paper	Verdict consequence	Example implication
Noisy retrieval vs. distraction	Retrieval imperfection alone is coded as relevance/noise; harmful influence on generation, confidence, or reasoning is coded as distraction.	Same broad topic but different measured effect is partial.	Random negatives and hard distractors should not be pooled into one score.
Query/OOD vs. evidence state	Query ambiguity, distribution shift, and model uncertainty describe the need or belief state. Code them under origin unless the study labels the resulting retrieved evidence.	Shared uncertainty language does not imply a common evidence state or policy.	An OOD detector is not automatically an insufficient-context detector.
Natural conflict vs. adversarial corruption	Code by the source of contradiction, not by surface form. Naturally occurring or diagnostic contradiction is conflict; attacker-created contradiction is integrity/adversarial corruption.	Natural-conflict and poisoning results are invalid as a direct ranking unless the threat model is redefined.	Faithfulness or source-preference methods are not poisoning defenses by default.
Intervention mismatch	Code the component changed, then inspect the evaluation harness. Different interventions can be compared as alternatives if the remaining pipeline and target are controlled.	Shared harness and end-to-end target can support valid comparison; unmatched pipelines support only a qualified comparison.	A filter and a note-taking prompt can be compared after fixing retriever, generator, evidence states, budget, and action utility.
Metric overlap	Metric targets must match the object being measured. Accuracy, citation support, attack success, calibration, and selective risk are separate families.	Shared dataset but different metric family is partial; mutually incompatible targets are invalid for one-scalar ranking.	A correct answer with unsupported citation should not pass a faithfulness comparison.
Observable but unnamed action	Infer behavior only when action is decision-relevant and the protocol is unambiguous; otherwise use *not applicable*.	Merely omitting the phrase "action policy" does not force under-specification.	An answer-only benchmark supplies *answer* for a policy claim, but a fixed latency comparison needs no action label.
Security capability missing	Mark only decision-critical missing access, budget, objective, or target as unknown. A shared harness may intentionally vary capability.	Unknown capability makes a cross-game rank under-specified; controlled capability variation can support a valid descriptive contrast.	Black- and white-box variants of one attack can be compared inside the same harness without implying cross-paper transfer.
Benchmark vs. method	Code benchmark if the reusable artifact is the main contribution; otherwise code the intervention.	Benchmark-method comparison is partial unless the method is evaluated on the benchmark under aligned assumptions.	RGB is benchmark-first; RAAT is training-first.
Tuple match but control mismatch	After semantic coding, inspect corpus, data, retriever, generator, top-k, budget, baselines, decoding, and uncertainty.	A tuple match is insufficient for direct ranking when these factors confound the reported score.	Two answer-only noise methods on different datasets and generators require a shared rerun or a qualified claim.

Table 22: Boundary rules for applying the revised diagnostic schema. Verdicts are claim-specific and require both semantic and experimental-control checks.

Stage	Question	Typical signals	Representative actions
Detect	Is the current evidence unreliable?	Low retrieval confidence, high contradiction, low sufficient-context score, suspicious document patterns	Flag query/evidence, route to stronger checks, log risk state
Diagnose	What kind of unreliability is present?	Irrelevance, distraction, conflict, poisoning, ambiguity, unanswerability, stale evidence	Assign failure type; separate benign noise from adversarial risk
Mitigate	Can the evidence be repaired before answering?	Alternative retrieval results, filter scores, reranker utility, provenance, source agreement	Re-rank, filter, compress, re-query, retrieve from trusted sources, remove malicious passages
Control	What should the system do now?	Calibrated confidence, selective risk, cost budget, user risk tolerance	Answer, abstain, ask clarification, expose conflict, escalate to human review
Monitor	Did the system fail or get attacked?	User feedback, traceback signals, poisoned-document clusters, drift over time	Update index, quarantine sources, revise prompts, retrain filters, update audit logs

Table 23: An operational pipeline for robust RAG. The pipeline turns the paper's taxonomy into actionable system design steps.

M Comparability Case Ledger

ID	Src	Compared results	Quoted gap	Mismatch	Verdict	Consequence of matching the coordinate
SEC-01	rerun	PoisonedRAG vs. CorpusPoisoning	1% vs 84.2% for the same attack	C1, C2, C4	undefined	C4 differs: 0.01 scores answer manipulation, 84.2 scores top-20 retrieval, so no gap is defined.
SEC-02	invited	PoisonedRAG vs. Phantom	+3.7% ASR in favor of PoisonedRAG	C1, C2, C4	undefined	No matched operating point exists for C4 (targeted answer vs refusal), so no gap is defined at any budget.
SEC-03	no num.	RobustRAG vs. PoisonedRAG	Defense credited with reducing poisoning ASR to <=15%	C1, C2, C4	undefined	No number is transported; RobustRAG re-implements the attack, and its own poisoning ASRs are 37-86.
SEC-04	quoted	BadRAG vs. PoisonedRAG	+1.2% in favor of BadRAG (98.2% vs 97%)	C2, C4, C5	undefined	Objectives differ (denial of service vs targeted answer), so the 1.2-point gap does not rank the two attacks.
SEC-05	rerun	AIP vs. PoisonedRAG	+66.66% ASR in favor of AIP	C1, C2	narrows	At five injected documents on MedSquad the 66.66-point gap stands; it does not transfer to NQ.
NOISE-01	rerun	CRAG (Self-CRAG) vs. Self-RAG 7B	+6.9 accuracy in favor of Self-CRAG	C3	survives	All five coordinates match, so the 6.9-point gap stands as reported.
NOISE-02	rerun	CtrlA 7B vs. Self-RAG 7B	+6.9 accuracy in favor of CtrlA	C1, C3	narrows	Backbone and retriever differ, so the 6.9-point gap has no determined sign.
NOISE-03	invited	RGB (ChatGPT) vs. RAAT (ChatGPT3.5)	+4.53 accuracy/EM in favor of RGB's ChatGPT	C1, C4	undefined	Substring accuracy is weaker than exact match, so the 4.53-point gap has no determined sign.
NOISE-04	invited	Chain-of-Note vs. RAAT (ChatGPT3.5)	+7.55 vs +1.68 EM gain	C1, C3, C4	undefined	The two baselines differ in whether gold context is present, so the two gains are not commensurable.
NOISE-05	rerun	AstuteRAG vs. RobustRAG	+4.4 accuracy in favor of ASTUTE RAG	C1, C5	survives	Model, dataset, retrieval and metric match, so the 4.4-point accuracy gap stands.
FAITH-02	quoted	FaithfulRAG vs. FaithEval (GPT-4o CF)	81.7% vs 47.5% (+34.2)	C1, C4	undefined	Aggregate and single-task scores under different conflict types and models cannot be differenced.
CALIB-03	invited	ADAPT-LLM vs. Self-RAG 7B	+16.75 accuracy in favor of Self-RAG	C1, C4	narrows	A harder test subset and a stricter metric are confounded with the method, so 16.75 is not a method effect.
Within-harness controls (one paper, one harness)						
FAITH-01	invited	RAGTruth (resp-level) vs. RAGTruth (span-level)	78.7% F1 vs span-level	C4	narrows	Response-level and span-level scores are different units, so the contrast holds only within one unit.
CALIB-01	stated	Self-RAG 7B vs. RAG-baseline	+23.9 accuracy in favor of Self-RAG	C3	survives	Within one paper and one harness the 23.9-point gap stands as reported.
CALIB-02	stated	CtrlA 7B vs. FLARE (reimpl.)	+4.0 accuracy in favor of CtrlA	C3	survives	Within one paper and one harness the 4.0-point gap stands as reported.

Table 24: Worked cross-paper comparisons. Of 12 comparisons published papers state (Src=stated) or invite by grouping (Src=invited), 7 have no common scope, 3 survive only on a strict subset and 2 survive as stated; 3 within-harness contrasts are negative controls. *Src* also records how the other side's number reached the citing paper: *quoted* transports it, *rerun* re-implements the other system in this harness, *no num.* asserts a relation without giving the other side's number, and *invited* is a juxtaposition no author stated. Every number is copied verbatim from a locally extracted full text and re-checked at its line by `verify_cross_paper_cases.py`; all 102 provenance, numeral and verbatim-quotation checks pass. C1 evidence state, C2 origin/capability, C3 intervention, C4 outcome, C5 policy/utility; the rule of §8 is applied exactly, so only C1, C2, C4 and C5 threaten a comparison and a C3 mismatch never does. An independent 6-judge panel reproduces these verdicts on 100% of the cross-paper rows for the binary distinction the paper uses and 58% for the three-way label (Table 25), which is why no rate from this table is quoted as a result. The per-case source line ranges, the full side-A and side-B reported values and the coordinate-matched readings are released as `supplement/cross_paper_comparability_cases.csv`.

Case	Kind	Ours	Independent panel	Agree
CALIB-03	cross	narrows	unde 6	0/6
FAITH-02	cross	undefined	surv 3, unde 3	3/6
NOISE-01	cross	survives	surv 6	6/6
NOISE-02	cross	narrows	unde 6	0/6
NOISE-03	cross	undefined	unde 6	6/6
NOISE-04	cross	undefined	unde 6	6/6
NOISE-05	cross	survives	surv 4, unde 2	4/6
SEC-01	cross	undefined	unde 5, surv 1	5/6
SEC-02	cross	undefined	unde 5, narr 1	5/6
SEC-03	cross	undefined	narr 5, unde 1	1/6
SEC-04	cross	undefined	unde 6	6/6
SEC-05	cross	narrows	unde 3, surv 3	0/6
CALIB-01	ctrl	survives	surv 3, unde 3	3/6
CALIB-02	ctrl	survives	surv 6	6/6
FAITH-01	ctrl	narrows	unde 6	0/6

Table 25: The ledger holds up on the distinction it is used for and not on the three-way label. Six judges from three developers re-code every case from the neutral statement of the juxtaposition and the verbatim source excerpts alone, never seeing our coordinate labels, our reading, our matched number or our verdict. On the 12 cross-paper cases the panel majority reproduces our three-way verdict on 58% of cases ($\kappa = 0.32$), and agreement among the six judges themselves is $\kappa = 0.52$. On the binary distinction the paper's conclusion actually uses—does the quoted gap stand as reported—the majority reproduces us on 100% and the panel is unanimous on 67% of cases. This is why we report the ledger's negative statements and not a rate.

N Screening Frame, Census and Codebook Ablation

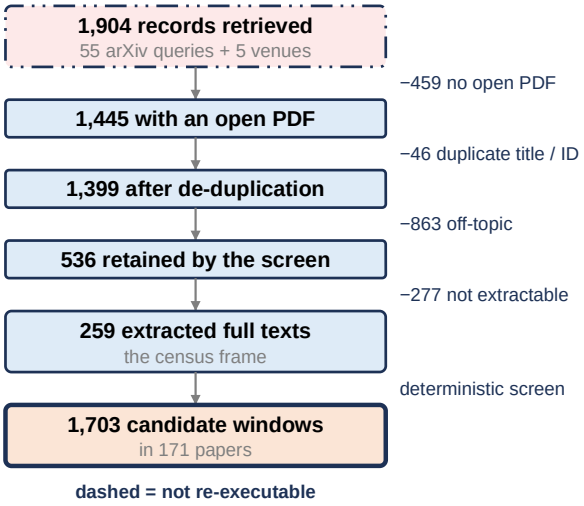


Figure 7: Sampling frame for the census. Solid boxes are stages a third party can re-execute from the release; the dashed box is documented but not re-executable, because per-query hit counts were not logged at the snapshot date and live indices drift. The frame is therefore reconstructible from the 259 released texts downward, and the population those texts were drawn from is not. Every composition figure in this paper is stated for the frame or for the re-weighted window universe, never for the literature.

Prior survey	Refs	In-sc.	Ded.	Scr.	Text	Map
RAG survey (Gao) (Gao et al., 2024)	227	8	0.50	0.38	0.25	0.62
RAG evaluation (Yu) (Yu et al., 2024)	74	2	0.00	0.00	0.00	0.50
Trustworthy RAG (Zhou) (Zhou et al., 2024)	192	28	0.64	0.50	0.46	0.50
Securing RAG (Xu) (Xu et al., 2026)	121	69	0.75	0.64	0.55	0.58
Knowledge conflicts (Xu) (Xu et al., 2024a)	200	3	0.33	0.33	0.33	0.33
Abstention (Wen) (Wen et al., 2025a)	179	1	0.00	0.00	0.00	0.00
Pooled	878	95	0.67	0.56	0.47	0.54
Pooled in-scope recall, Wilson 95% CI						
Deduped (1,399)			0.67 [0.57, 0.76] (64/95)			
Full texts (259)			0.47 [0.38, 0.57] (45/95)			

Table 26: Coverage of the six prior surveys' reference lists against our frame, the reviewer's requested recall-against-a-known-list check. *Refs* is the bibliography Semantic Scholar resolves for each survey; *In-sc.* keeps titles naming both a retrieval and a robustness concept (keyword lists released in `frame_coverage_audit.json`). The stage columns give the share of in-scope references present in the deduplicated release (1,399), the relevance-screened set (536), the extracted full texts (259) and the adjudicated Map B (342); matching is by arXiv id, DOI, ACL id, then normalised title. The 1,904 retrieved and 1,445 open-PDF stages are count-only in the release, so the deduplicated column lower-bounds retrieval recall. Per-survey CIs and unfiltered recall are in the released JSON.

Judgment	Ref. est.	Scored	Acc.	95% CI	Base	Acc. (LOO)
Provenance (transported / re-implemented / unclear)	171	170	0.91	[0.86, 0.96]	0.79	0.89
Verdict (survives / narrows / undefined)	129	114	0.82	[0.74, 0.89]	0.77	0.81
Per-class recall of the deployed panel, provenance						
transported		35	0.80	[0.64, 0.90]		
re-implemented		135	0.93	[0.88, 0.96]		
unclear		0	–	–		

Table 27: The deployed panel scored on *real* frame windows against a privileged-information reference, whose adjudicators read both papers' full texts rather than a 45-line window and whose every counted vote carries a quote located mechanically in the document that adjudicator was shown. "Ref. est." counts windows with a quote-verified majority; those without one are reported, not dropped. "Base" is the majority-class rate. The last column removes the one adjudicator sharing a checkpoint with the deployed panel. Measured sensitivity 1.00 and specificity 0.99 for the transported class make the composition correction non-trivial, unlike the identity map the submitted version used: it moves the re-implemented share to 0.881. This is not human gold.

Worst-/best-case accuracy folding in excluded windows			
Setting	N	Worst	Best
Verdict: scored (93/114)	114	0.82	0.82
over full sample	180	0.52	0.88
over deployed-committed	159	0.58	0.87
reviewer $114{+}51$	165	0.56	0.87
Provenance: scored (154/170)	170	0.91	0.91
over full sample	180	0.86	0.91
over deployed-committed	179	0.86	0.91

No-majority (51) vs. established (129) verdict windows			
Feature	Estab.	No-maj.	Test
Quote-verified ref. votes (med.)	3	1	$p{<}0.001$, $\delta{=}+0.79$
Ref. vote-split entropy (mean)	0.53	0.88	$p{<}0.001$, $\delta{=}-0.38$
Survey-table stratum (share)	0.28	0.51	$p{=}0.007$, $V{=}0.23$
Deployed panel commits (rate)	0.88	0.88	$p{=}1.00$, $V{\approx}0$
Window length (med. words)	85	70	$p{=}0.32$, $\delta{=}-0.09$
Coordinate terms (median)	2	3	$p{=}0.13$, $\delta{=}-0.14$

Table 28: Reviewer Q6: bounds and characterisation for the excluded privileged-reference windows. *Top:* the deployed panel is scored only where the reference reached a quote-verified majority (verdict 93/114= 0.82, Wilson [0.73, 0.88]; provenance 154/170= 0.91, [0.85, 0.94]). Worst case counts the panel wrong on every folded-in window, best case right. The reviewer's $114{+}51{=}165$ denominator overstates the panel's committed set: only 45 of the 51 no-reference-majority windows carry a deployed verdict, so the deployed-committed total is 159, not 165. *Bottom:* the no-majority windows differ from the rest mainly in having fewer quote-verified privileged votes and more split ones, and in stratum mix; they are not longer, denser in coordinate terms, or harder for the deployed panel, which commits on them at the same rate. Continuous features: Mann–Whitney p and Cliff's δ; categorical: χ^2 p and Cramér's V (deployed-commits uses Fisher).

Did the panel execute its own rule?			
	n	κ	95% CI
Verdict as *stated* by the judges	704	0.06	[-0.01, 0.11]
Verdict *composed* from their own coordinates	704	0.09	[0.01, 0.15]
Δ (composed $-$ stated)		+0.02	[0.00, 0.05]

*Rule fidelity, judge by judge: share of codings where stated = composed, and count of **survives** verdicts the judge's own coordinates contradict*		
coderA_gpt5mini	1.000	0
coderB_claudeopus46	0.770	162
coderC_gemini31pro	0.925	53

What the evidence quotes are: 1,879 non-empty quotes classified		
verbatim in the passage shown	1,639	0.872
verbatim after Unicode/de-hyphenation	80	0.043
abridged with an ellipsis, fragments verbatim	76	0.040
table re-serialised from the passage	59	0.031
elsewhere in the same paper	–	0.000
re-serialised from elsewhere	4	0.002
near paraphrase of the passage	–	0.000
absent from the source document	21	0.011

Table 29: Two diagnostics the submitted version did not run, both over data it had already released. *Top:* the decision rule of §8 composed mechanically from each judge's own coordinate labels, beside the verdict that judge stated. Composing the rule does not make the panel reproducible, so the census failure is not the panel failing to execute its instrument. *Bottom:* every evidence quote classified against the passage its author was shown. The submitted version reported a single located rate, which pools an extraction artefact with a fabrication; separating them shows 0.987 of non-empty quotes are supported by the passage shown and 0.011 are absent from the source document altogether.

Field	Maj. share	Obs.	κ	$\kappa_{\max}$	$\kappa/\kappa_{\max}$
C1 evidence state	0.630	0.60	+0.19	0.72	0.26
C2 origin / capability	0.589	0.54	+0.11	0.61	0.18
C3 intervention	0.871	0.75	-0.11	0.42	-0.26
C4 outcome / unit	0.814	0.71	+0.08	0.53	0.16
C5 policy / utility	0.526	0.54	+0.10	0.51	0.19
Verdict	0.507	0.45	+0.06	0.46	0.13
The headline contrast, on matched label spaces					
Census, three classes		0.45	0.06	0.46	0.13
Census, binary		0.52	0.05	0.37	0.12
Constructed (stated), three classes		0.83	0.74	0.81	0.92
Constructed (stated), binary		0.95	0.86	0.88	0.97

Table 30: How much room κ had. $\kappa_{\max}$ is Cohen's maximum attainable κ, averaged over judge pairs: the largest value these judges could have reached without changing how often each of them uses each label (Umesh et al., 1989; Sim and Wright, 2005). A near-constant field caps κ regardless of behaviour, which is the argument §8 makes for C3; the last column applies it to every field instead of one, so the argument is symmetric. It also repairs the headline contrast: matching the label space does not close the gap between the constructed and the published setting, and the right reading of the census verdict is that its judges reach 0.13 of what their own marginals allow where the constructed judges reach 0.92.

Passage discloses ...	Windows	Papers	Obs.	κ	95% CI
0 of four disclosed	44	9	0.66	0.10	[-0.14, 0.33]
1 of four disclosed	23	9	0.42	-0.11	[-0.32, 0.09]
2 of four disclosed	157	25	0.44	0.04	[-0.13, 0.09]
3 of four disclosed	253	60	0.42	-0.02	[-0.13, 0.05]
4 of four disclosed	227	57	0.44	0.08	[-0.07, 0.18]
High ($\geq$ 3)	480			0.03	
Low ($\leq$ 2)	224			0.11	
Difference (high $-$ low)				-0.08	[-0.27, 0.05]

Table 31: The disclosure counterfactual on published text. Disclosure is scored deterministically from a released lexicon over the passage the judges were shown, so the exposure is a document feature and not a model output: regressing agreement on it is not regressing the panel's labels on themselves. Windows whose passages mention more coordinate-relevant terms do *not* yield more reproducible verdicts. This is the observational test of the checklist recommendation, and it does not support it; §9 shows what does, and what does not follow from it.

Panel	J	Common items		All	Own
		AC_1	κ	κ	κ
Submitted panel (1 mini, 2 front.)	3	0.29	0.14	0.33	0.07
Small tier only, 3 developers	3	0.16	0.00	0.29	0.00
Frontier only, 3 developers	3	0.54	0.31	0.51	0.25
Frontier only, 3 dev. $\times$ 2 checkpoints	6	0.47	0.22	0.38	0.23
same dev. (Anthropic)	2	0.56	0.35	0.51	0.49
same dev. (Google)	2	0.54	0.29	0.38	0.24
same dev. (OpenAI)	2	0.53	0.20	0.50	0.21
across dev. (pair A)	2	0.56	0.32	0.53	0.37
across dev. (pair B)	2	0.37	0.07	0.15	0.24

Table 32: Upgrading the judges helps and does not rescue the verdict, on a population that is identical for every row. The same 1,703-window frame is coded by a fully crossed pool of nine judges—a small and two frontier checkpoints from each of three developers—from which these panels are drawn. Because κ is prevalence sensitive, comparing panels each scored on its own majority-eligible subset would confound capability with item composition, so *Common items* fixes one population of 843 windows that a majority of the whole nine-judge pool called eligible, and *All* drops nothing at all: it scores every one of the 1,680 commonly coded windows, carrying ineligibility as its own category. *Own* is each panel's κ recomputed on the majority-eligible subset of *that panel*, which is the population a reader who ran only that panel would see; it is reported because it is what a smaller pool can obtain, and it is the column that must *not* be compared across rows. Conclusions are unchanged across the three: an all-frontier panel of three developers multiplies verdict κ several-fold over the submitted panel and still reaches only *fair*. The three *same dev.* rows are the reference level, two frontier checkpoints from one developer where nothing differs but the checkpoint; they bound what *correlated* judges reach, not what the task affords, and Table 33 shows that one cross-developer pair beats two of them.

Frontier pair	Dev.	κ	AC_1	Elig. κ
opus-4.6 vs opus-4.8	same	0.35	0.56	0.56
opus-4.6 vs gem-3.1p	cross	0.32	0.56	0.61
gem-3.1p vs gpt-5.4	cross	0.32	0.53	0.67
gem-3.1p vs gem-2.5p	same	0.29	0.54	0.50
opus-4.6 vs gpt-5.4	cross	0.29	0.52	0.61
opus-4.8 vs gpt-5.4	cross	0.25	0.50	0.43
gem-2.5p vs gpt-5.4	cross	0.22	0.47	0.52
opus-4.6 vs gem-2.5p	cross	0.20	0.47	0.33
gpt-5.4 vs gpt-5.5	same	0.20	0.53	0.71
gem-3.1p vs opus-4.8	cross	0.19	0.44	0.43
opus-4.8 vs gpt-5.5	cross	0.11	0.50	0.26
opus-4.8 vs gem-2.5p	cross	0.07	0.37	0.16
gem-2.5p vs gpt-5.5	cross	0.07	0.42	0.64
gem-3.1p vs gpt-5.5	cross	0.03	0.34	0.63
opus-4.6 vs gpt-5.5	cross	0.00	0.35	0.49

Table 33: Every frontier-frontier judge pair, on the common item population, ordered by verdict κ. The averaged same-developer advantage is +0.11; its exact randomisation over the 15 ways of matching six checkpoints into three disjoint pairs gives $p = 0.07$, against a smallest attainable value of 0.07. The contrast is also not uniform: 6 cross-developer pairs agree better than the weakest same-developer pair, so sharing a developer is reported as residual structure and not as an effect.

Instrument	AC_1	κ	Unan.	*Survives* share	$\Delta\kappa$
Five coordinates (released)	0.54	0.31	0.48	0.40–0.63	—
C1 and C4 only	0.56	0.34	0.51	0.55–0.62	+0.03 [-0.05, +0.12]
No coordinates, asked directly	0.58	0.31	0.53	0.62–0.73	-0.00 [-0.10, +0.11]

Table 34: The verdict does not become reliable when the instrument is simplified. The same three frontier judges, one per developer, code the same 1,703-window frame three times, changing only the codebook: the released five coordinates, the two coordinates that are not degenerate across systems, and no coordinates at all. All three arms are scored on the same 832 windows. $\Delta\kappa$ is against the five-coordinate arm with a 95% interval resampling source papers. *Survives* share is the range across the three judges of how often each calls a comparison sound; the spread, not the noise, is what fails to port.

Subset	N	Accuracy	95% CI
Panel majority, all items	238	0.945	[0.92, 0.97]
passage declares provenance	79	1.000	[1.00, 1.00]
provenance only from structure	80	1.000	[1.00, 1.00]
recall, reimplemented	80	1.000	[1.00, 1.00]
recall, transported	79	1.000	[1.00, 1.00]
recall, unclear	79	0.835	[0.75, 0.91]
Constant baseline	238	0.336	—

Table 35: The provenance label scored against a reference, not against agreement. Items are constructed excerpts whose provenance is fixed when the item is written; the prompt, the three judges and the majority rule are byte-identical to the ones that produced the census labels. The three reference classes are equinumerous, so the constant baseline is $1/3$. The *implicit* rows carry no declaration sentence at all, so a classifier that keyed on a re-implementation or attribution phrase would score at baseline there. The stratum that asserts a relation while supplying no number for the other side is not scorable under a three-option prompt and is reported separately: the deployed prompt sends 100% of those judgments to *unclear*, while the same prompt with a fourth option recovers them at 100%, so the category is codable and the deployed instrument simply cannot express it.

Eligibility gate	Class.	Re-impl.	Transp.	Unclear	Share	95% CI
Unanimous $(3/3)$	381	328	53	26	0.861	[0.82, 0.89]
Majority $(\geq 2/3)$	610	533	77	44	0.874	[0.85, 0.90]
Any judge $(\geq 1/3)^{\dagger}$	610	533	77	44	0.874	[0.85, 0.90]
No gate†	639	549	90	44	0.859	[0.83, 0.88]
Majority gate, leave one coder out						
drop coder A (gpt5mini)	403	350	53	26	0.868	[0.83, 0.90]
drop coder B (opus46)	532	464	68	33	0.872	[0.84, 0.90]
drop coder C (gemini31pro)	437	375	62	37	0.858	[0.82, 0.89]

Table 36: Reviewer Q7: the re-implementation share is stable under the eligibility gate. *Class.* is the number of windows the provenance panel assigned to re-implemented or transported (its denominator); *Share* is re-implemented over that count with a Wilson interval. The majority row reproduces the published 533/610 $= 0.874$ exactly. Tightening the gate to unanimous or dropping any one coder moves the share by at most 0.02 (leave-one-out range 0.858–0.872). †The provenance pass was elicited only on the majority-eligible ledger, so no window that fails the majority gate carries a provenance label: the any-judge row necessarily equals the majority row over coded windows (374 of the 1032 any-judge-eligible windows have no label and are not invented), and the no-gate row adds only the 29 coded windows now outside the frame.

Instrument (non-LLM unless noted)	Acc.	macro-F1	ΔAcc. [95% CI]	Δmacro-F1 [95% CI]
Verdict (survives / undefined), $N = 114$ windows, 48 papers				
Constant majority (baseline)	0.77	0.44	—	—
TF-IDF + logistic reg.	0.77	0.44	+0.00 [0.00, 0.00]	+0.00 [0.00, 0.00]
Structural feats. + logistic reg.	0.83	0.75	+0.06 [-0.05, 0.17]	+0.32 [0.05, 0.39]
LLM panel majority	0.82	0.78	+0.04 [-0.13, 0.20]	+0.35 [0.13, 0.44]
Provenance (re-impl. / transported), $N = 170$ windows, 61 papers				
Constant majority (baseline)	0.79	0.44	—	—
TF-IDF + logistic reg.	0.79	0.44	+0.00 [0.00, 0.00]	+0.00 [0.00, 0.00]
Structural feats. + logistic reg.	0.78	0.51	-0.01 [-0.06, 0.02]	+0.06 [-0.01, 0.20]
LLM panel majority	0.91	0.92	+0.11 [-0.09, 0.33]	+0.48 [-0.01, 0.57]

Table 37: A non-LLM instrument on the same task. Each instrument is trained on the evidence-privileged reference labels and evaluated with grouped cross-validation that holds out whole citing papers; the LLM panel row is the deployed three-judge majority scored on the same windows and reproduces the published accuracies (0.82 and 0.91). Intervals resample citing papers. On the disputed verdict no instrument – LLM or not – beats the constant baseline on accuracy (every ΔAcc. CI includes 0), and the non-LLM structural model matches the LLM panel; the accuracy ceiling is a property of the text, not of the LLM instrument class. Provenance is the contrast where the LLM panel pulls ahead of the cheap instrument.

Deployed judge (by developer)	Agr. same	Agr. diff	Δ	95% CI
Verdict				
OpenAI	0.33	0.31	+0.02	[-0.05, 0.10]
Anthropic	0.66	0.65	+0.02	[-0.05, 0.08]
Google (shared ckpt)	0.72	0.70	+0.01	[-0.06, 0.09]
Provenance				
OpenAI	0.82	0.80	+0.02	[-0.02, 0.06]
Anthropic	0.98	0.97	+0.02	[-0.00, 0.04]
Google (shared ckpt)	0.87	0.86	+0.01	[-0.01, 0.03]

Table 39: Per-deployed-judge self-preference. For each deployed judge, agreement with same-developer reference adjudicators versus different-developer adjudicators, on quote-verified adjudicator labels; the Google judge shares its exact checkpoint with refC. Every same−diff difference is small and positive with a 95% CI (window bootstrap) that includes 0, for both the verdict and provenance passes: no judge is scored materially more kindly by its own developer.

Reference set	n	Acc.	95% CI	ΔAcc.	flip / lost
Verdict accuracy (published 0.82, N=114)					
All four adjudicators	114	0.82	[0.74, 0.89]	—	—
−refA (Anthropic)	71	0.80	[0.70, 0.90]	-0.01	0 / 47
−refC (Google, shared ckpt)	91	0.81	[0.73, 0.90]	-0.00	0 / 30
−refD (Anthropic)	104	0.79	[0.71, 0.88]	-0.03	0 / 12
−refE (OpenAI)	92	0.79	[0.70, 0.88]	-0.02	0 / 29
−Anthropic (refA+refD)	50	0.84	[0.74, 0.94]	+0.02	0 / 67
−Google (refC)	91	0.81	[0.73, 0.90]	-0.00	0 / 30
−OpenAI (refE)	92	0.79	[0.70, 0.88]	-0.02	0 / 29
Provenance accuracy (published 0.91, N=170)					
All four adjudicators	170	0.91	[0.86, 0.96]	—	—
−refA (Anthropic)	162	0.89	[0.84, 0.94]	-0.02	0 / 13
−refC (Google, shared ckpt)	159	0.89	[0.84, 0.95]	-0.01	0 / 15
−refD (Anthropic)	168	0.89	[0.84, 0.94]	-0.02	0 / 7
−refE (OpenAI)	169	0.91	[0.85, 0.96]	-0.00	0 / 5
−Anthropic (refA+refD)	121	0.88	[0.82, 0.94]	-0.03	0 / 52
−Google (refC)	159	0.89	[0.84, 0.95]	-0.01	0 / 15
−OpenAI (refE)	169	0.91	[0.85, 0.96]	-0.00	0 / 5

Table 38: Developer-independence of the evidence-privileged reference. Accuracy of the deployed panel is recomputed against a reference majority formed after dropping each adjudicator (leave-one-out) and each developer's adjudicators (leave-developer-out); "−Google (refC)" drops the checkpoint shared with the deployed Google judge and reproduces the published leave-one-out number. The last column counts reference windows whose established majority *flips to a different label / loses its majority* under the drop: no drop ever flips a label, so removing an adjudicator only shrinks coverage. Intervals are paper-clustered bootstraps.

Quantity	Estimate	95% CI
Windows in the universe	6,710	—
surfaced by the screen	1,703	—
missed, sampled	420	—
Eligible rate, missed windows	0.336	[0.27, 0.41]
Eligible rate, screened windows	0.444	—
Screen recall, transported	0.498	[0.26, 1.00]
Screen recall, re-implemented	0.276	[0.24, 0.33]
Re-implemented share, in frame	0.874	—
Re-implemented share, re-weighted	0.926	[0.86, 0.96]

Table 40: What the screen misses. The universe is every window under the screen's own windowing that names another paper's system and carries at least one result-formatted number; the screen's second-numeral requirement and its three surface strata are dropped, which are exactly the conditions that select for baseline tables. A uniform sample of the missed windows was coded by the same three judges under the same two prompts. Intervals resample source papers.

O Constructed Instrument Benchmark

Window	12 lines of `poisonedRAG2024.txt`: a baseline table putting 0.97 beside 0.01 for the Corpus Poisoning Attack.
Screen	Gazetteer system absent from this paper's own title region; two result-formatted values in-window; *baseline* stratum. Enters the frame. No judgment yet.
Provenance	*re-implemented*: the 0.01 is this paper's measurement of the other system, not a value copied from it.
Coordinates	C1 **match** (same corpus, retriever, questions); C2 **match** (five injected texts on both sides); C3 **mismatch**, as always—*excluded from the rule*; C4 **mismatch** (target questions answered with the attacker's answer, against queries with an adversarial passage in the top-k); C5 **not determinable** (neither side states an action set or a cost).
Rule	C4 unmatched and no operating point at which the two quantities coincide $\Rightarrow$ **undefined**. Replace with the common scope both texts support; do *not* substitute a computed number.
A panel	Nine judges agree on the provenance row (κ 0.55) and not on the rule row (κ 0.06). On constructed prose that *states* C5 they agree at κ 0.74: the disagreement lives in the coordinate that reads *not determinable*.

Figure 8: One juxtaposition traced end to end. The screen is deterministic; provenance is the judgment automation recovers; the rule row is the one it does not. The C5 label is where the failure of §9 lives: faced with a silent passage, judges answer something rather than *not determinable*.

Prose	Passage	N	Base	Coord.	Rule	Verd.	κ
Cano.	States every coordinate	90	0.33	0.97	0.891	0.822	0.73
Cano.	Withholds one, verdict unaffected	87	0.34	0.89	0.889	0.847	0.83
Cano.	Withholds one, verdict forced	36	1.00	0.85	0.403	0.380	0.26
Natu.	States every coordinate	90	0.33	0.93	0.835	0.791	0.75
Natu.	Withholds one, verdict unaffected	90	0.33	0.85	0.832	0.837	0.85
Natu.	Withholds one, verdict forced	35	1.00	0.81	0.252	0.252	0.21

Table 41: Disclosure crossed with prose realism, each cell scored against *its own* constant baseline (*Base*). *Cano.* states one coordinate per sentence in the codebook's own vocabulary; *Natu.* states the identical facts in paper-style prose, reordered and interleaved with sentences that carry no coordinate information. *Coord.* is mean recovery over C1, C2, C4 and C5. In the second row of each block the withheld coordinate does not determine the verdict, so all three classes occur and the baseline is $1/3$; in the third the verdict follows from the withholding alone, so the baseline is 1.00 and no estimator can beat it.

Estimator	Exact	Range	Weakest judge
Five coordinates (rule)	0.861	0.80–0.92	+0.46
Rule with C2/C5 ablated	0.738	0.70–0.78	+0.37
Two coordinates (rule, C1/C4 prompt)	0.832	0.80–0.87	+0.47
Judge verdict, five-coordinate prompt	0.824	0.76–0.89	+0.43
Judge verdict, C1/C4 prompt	0.832	0.80–0.87	+0.46
Judge verdict, no coordinates	0.796	0.74–0.86	+0.40
Majority class (narrows)	0.336	—	—

Table 42: Every estimator arm, on the 357 non-degenerate items, scored against the constant baseline of that population (0.336; the three verdict classes are equinumerous inside every disclosure arm, so this baseline is not an artefact of pooling). *Range* is across judges and *weakest judge* is the smallest judge margin over that baseline. The degenerate withholding arm, whose reference verdict never varies and whose constant baseline is therefore 1.00, is excluded here and reported separately in Table 41.

P Formal Estimand Definitions

Let reviewer r receive prompt $T_r \in \{0, 1\}$, rate item i, and emit label $Y_{ri}(T_r)$. Let Z_i denote a constructed reference and $s(Y, Z)$ a reference-relative score.

Realized-panel contrast.

$$\Delta_{\text{panel}} = \frac{1}{I} \sum_i \big[s(M_{r \in \mathcal{R}_1} Y_{ri}(1), Z_i) - s(M_{r \in \mathcal{R}_0} Y_{ri}(0), Z_i) \big],$$

where M is a majority rule and $\mathcal{R}_t$ is the realized reviewer panel. This quantity is descriptive for those panels and that reference.

Reviewer-population contrast.

$$\Delta_{\text{rater}} = \mathbb{E}_{r,i,Z}[s(Y_{ri}(1), Z_i) - s(Y_{ri}(0), Z_i)] .$$

This target requires treatment assignment and uncertainty at the reviewer level, plus a defined reviewer population and reference distribution.

Consequence contrast.

$$\Delta_{\text{loss}} = \mathbb{E}[L\{a(Y(1)), \theta\} - L\{a(Y(0)), \theta\}] ,$$

where $a(\cdot)$ is the accept, narrow, reject, or defer decision and θ is the deployment state.

Variance decomposition. Let $\mathcal{R}$ be the realized reviewer composition and $\mathcal{Z}$ the realized reference process. Then

$$\text{Var}(\widehat{\Delta}) = \mathbb{E}\Big[\text{Var}(\widehat{\Delta} \mid \mathcal{R}, \mathcal{Z})\Big] + \text{Var}\Big[\mathbb{E}(\widehat{\Delta} \mid \mathcal{R}, \mathcal{Z})\Big] .$$

Source resampling with reviewers and references fixed estimates the first term. Reviewer assignment and reference construction contribute to the second.

Assignment asymptotics for panel majorities. For three conditionally independent votes with correctness probabilities p_1, p_2, p_3, majority correctness has expectation

$$m(p_1, p_2, p_3) = p_1p_2 + p_1p_3 + p_2p_3 - 2p_1p_2p_3.$$

Let Z be a six-versus-six reviewer assignment and $\mu(Z) = \mathbb{E}(\widehat{\Delta} \mid Z)$ the expected difference between the two scheduled panel-majority scores. With I conditionally independent items (or G independent source clusters of bounded size),

$$\text{Var}(\widehat{\Delta}) = O(I^{-1}) + \text{Var}_Z\{\mu(Z)\}.$$

The source-only standard error targets the first term. If reviewer profiles are heterogeneous, the nonlinear function m generally makes $\mu(Z)$ nonconstant, so the second term does not vanish when more items are added to the same twelve assignments. Conditional item precision can therefore converge to zero while uncertainty over reviewer assignment remains. This characterization explains the direction of the simulation; it does not supply a population model for reviewer abilities.

Q Reviewer Heterogeneity and Assignment-Unit Simulation

Setting	Panel/src	Votes/src	Rev. t	Rev. RI
Size: no prompt effect, $\alpha = .05$				
$\sigma_r = 0.00, G = 80$	4.7	5.1	4.7	4.3
$\sigma_r = 0.27, G = 80$ *	24.2	32.9	4.6	4.3
$\sigma_r = 0.78, G = 80$	62.6	70.1	5.1	5.0
$\sigma_r = 0.27, G = 160$	36.7	46.5	4.4	4.3
Power: true prompt effect 0.4 *logits*				
$\sigma_r = 0.27, G = 80$	78.8	89.0	52.8	51.6
$\sigma_r = 0.78, G = 80$	69.6	77.0	11.4	11.4

Table 43: Rejection percentages over 5,000 replicates with twelve reviewers, six per prompt, six claims per source G, and balanced three-of-six panels. Reviewer abilities are redrawn every replicate, so the target is a reviewer-population effect rather than a realized-panel contrast. σ_r is the between-reviewer ability standard deviation on the logit scale; the starred row is the value estimated from the crossed three-coder diagnostic ($\hat{\sigma}_r = 0.27$, Appendix Table 46). *Panel/src* is the status-quo paired source t test on panel-majority correctness; *Votes/src* keeps the source-level standard error but uses individual votes; *Rev.* t is a two-sample t test on the twelve individual reviewer accuracies; *Rev. RI* refers the same contrast to all 924 arm assignments. The last two are executable retrospectively from released pre-aggregation votes with reviewer identifiers, and retain power. The complete grid over both reviewer-ability profiles (fixed quantiles and resampled), all heterogeneities and all source counts is released as `supplement/retrospective_remedy_simulation.json`.

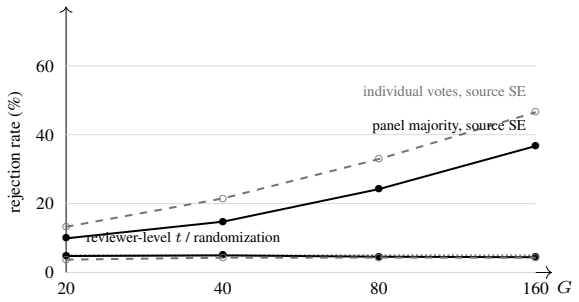


Figure 9: Null rejection rate against the number of sources G at the estimated reviewer heterogeneity $\sigma_r = 0.27$, with reviewer abilities redrawn each replicate. Adding sources makes the two source-level analyses steadily worse, and using individual votes instead of majority labels does not help: the defect is the unit of uncertainty, not the aggregation. The two reviewer-level analyses stay at the nominal level and are executable from released pre-aggregation votes. Table 43 gives the values and the corresponding power.

Pool	Kind	$\hat{\sigma}_r$	95% CI	κ	$\kappa_{\max}$
API-served only	API	0.37		0.05	0.42
Open-weight only	open	0.51		0.09	0.46
Pooled	both	0.50	[0.33, 0.62]	0.10	0.48
Per-judge accuracy on the constructed benchmark, which is what σ_r is computed from					
`insJ_gpt55`	API	0.889			
`apk_gem35flash`	API	0.874			
`insC_gem31pro`	API	0.858			
`apk_opus47`	API	0.838			
`insH_gem25pro`	API	0.828			
`insI_gpt54`	API	0.828			
`apk_sonnet46`	API	0.813			
`apk_gpt5mini`	API	0.801			
`insB_opus46`	API	0.786			
`insG_opus48`	API	0.764			
`apk_haiku45`	API	0.749			
`apk_gpt4o`	API	0.722			
`apk_gpt41`	API	0.709			
`qwen25-72b`	open	0.814			
`qwen3-32b`	open	0.800			
`qwen3-14b`	open	0.794			
`phi4`	open	0.786			
`qwen3-8b`	open	0.728			
`mistral-small-24b`	open	0.719			
`glm4-9b`	open	0.603			
`olmo2-13b`	open	0.559			
`internlm3-8b`	open	0.554			

Table 44: Widening the reviewer pool. The submitted version estimated σ_r from six API-served judges, giving an interval spanning a factor of four. Adding 9 open-weight checkpoints, run under greedy decoding in one harness on byte-identical prompts, brings the pool to 22 and the interval to a factor of 1.9. It also shows the narrow pool was biased downward rather than merely imprecise: the API judges occupy a narrow accuracy band, and checkpoints reaching lower accuracies raise the estimate. The last two columns show the census result is not a property of one model class: the open pool fails to agree on the verdict as thoroughly as the API pool, and not because its labels are too uniform for κ to find signal.

Crossed design	Rev.	R	Items	$\hat{\sigma}_r$	Boot.
Ours: census judges (this task)	model	9	840	0.38	[0.18, 0.51]
Ours: constructed benchmark	model	6	432	0.38	[0.12, 0.48]
Ours: model panel	model	28	600	0.54	[0.31, 0.75]
DynaSent (sentiment)	human	674	122,117	0.78	[0.74, 0.81]
GoEmotions (neutral)	human	76	56,958	0.46	[0.37, 0.54]
LeWiDi (offensiveness)	human	116	8,377	0.45	[0.37, 0.53]
MultiPref (quality)	human	160	6,502	0.31	[0.24, 0.37]
Wiki Detox (toxicity)	human	3,944	159,686	0.55	[0.53, 0.56]

Table 45: Between-reviewer heterogeneity estimated with one moment estimator on every crossed reviewer-by-item design we could obtain. R is the number of reviewers retained (≥ 30 judgments for the public corpora). The first two rows are the estimate this paper needs and the literature does not supply: between-reviewer heterogeneity on cross-paper comparability adjudication itself, once on the real corpus against a leave-one-out majority of the other judges and once on the constructed benchmark against a reference fixed before any reviewer saw the item. Public human corpora are scored against a leave-one-out majority of the other annotators; the model panel is scored against the deterministic minimum of a displayed loss rule, which no reviewer supplies. The last column is the 2.5–97.5 percentile interval of a reviewer bootstrap (resampling reviewers with replacement, $B \geq 2{,}000$ draws), which targets the correct unit of replication for a between-reviewer variance parameter. An item bootstrap and a parametric bootstrap are reported in the supplement JSON for comparison.

Quantity	Value
Crossed design	3 coders $\times$ 80 claims
Coder accuracy vs. adjudicated	0.850, 0.775, 0.725
$\hat{\sigma}_r$ (moment estimate)	0.271
Reviewer bootstrap 95%	[0.000, 0.347]
Item bootstrap 95%	[0.000, 0.522]
Parametric bootstrap 95%	[0.000, 0.732]
Parametric bias of $\hat{\sigma}_r$	-0.044
Identification from published reporting	
Widest admissible σ_r set	100% of the grid
Point identified by reporting?	no

Table 46: Reviewer heterogeneity is estimable only from a crossed reviewer-by-item design, and three coders do not estimate it. All three resampling schemes are shown because they disagree about the width and not about the conclusion: every interval contains zero, so this design alone cannot distinguish a badly miscalibrated panel from a homogeneous one. The reviewer bootstrap is the scheme reported for all seven designs in Table 45, because the reviewer is the unit the parameter is about; the item bootstrap resamples the wrong unit and is shown only for comparison. A parametric bootstrap puts the estimator's bias at -0.044 here, against at most 0.012 in every large design, which is what two degrees of freedom costs. The published agreement rates located in the audited papers place no usable restriction on σ_r: at the label prevalence those papers imply, every value on a 0–1.2 grid reproduces the reported agreement for some item dispersion. Separating the two components requires the item-by-annotator assignment, which none of the nine audited papers reports.

Setting	Source Panel RI	Rater RI	SD/SE
$\sigma = 0.20$, $G = 20$	7.6 (0.4)	4.9 (0.3) 3.7 (0.3)	1.14
Homog., $G = 80$	4.9 (0.3)	4.5 (0.3) 4.5 (0.3)	0.99
$\sigma = 0.20$, $G = 80$	17.0 (0.5)	5.3 (0.3) 4.6 (0.3)	1.44
$\sigma = 0.20$, $G = 160$	25.9 (0.6)	4.9 (0.3) 5.2 (0.3)	1.77

Table 47: Oracle design benchmark, retained for reference. Entries are rejection percentages with Monte Carlo standard errors in parentheses, and SD/SE is the empirical standard deviation of the D$-$C contrast divided by the mean source-only standard error, so a value above one marks an optimistic standard error. Each row uses 5,000 replicates, 12 auditors (six per prompt), six claims per source, and balanced three-of-six panels; G is the number of sources and rater ability follows 12 fixed normal quantiles on the logit scale around 75% mean agreement. Source t and Panel RI use the identical studentized panel-majority statistic; Panel RI refers it to all 924 six-versus-six assignments under a full-vote oracle with an arm-independent schedule, and Rater RI uses individual accuracy. Panel RI is not implementable retrospectively, which is why Table 43 replaces it with two analyses that are, and evaluates them at the estimated σ_r rather than at an assumed one. The complete null-simulation sweep, including the $\sigma_r = 0.10$, $\sigma_r = 0.30$ and $G = 40$ profiles and the raw-contrast randomisation sensitivity, is released as `supplement/assignment_mismatch_simulation.json`.

R Crossed Deployment-Panel Details

Panel	Field	N	κ	$\kappa_{\max}$	AC_1
Three human coders	verdict	80	0.35	0.49	0.57
	C1	80	0.15	0.44	0.34
	C4	80	0.36	0.78	0.68
Deployed model panel (3)	verdict	79	0.21	0.71	0.23
	C1	79	0.08	0.41	0.28
	C4	79	0.31	0.54	0.55
Second model panel (3)	verdict	80	-0.11	0.12	0.02
	C1	80	-0.01	0.15	0.23
	C4	80	-0.00	0.18	0.19
All model judges	verdict	79	0.11	0.42	0.14
	C1	79	0.12	0.36	0.31
	C4	79	0.17	0.38	0.36

Table 48: Reliability on the same 80 published comparative claims, under the same frozen codebook. The human panel is three independently recruited coders (210 person-hours) who were additionally instructed to open each source; the model panels read the quotation and its context. The deployed panel is the census panel of §10 (gpt-5-mini, claude-opus-4.6, gemini-3.1-pro-preview); the second panel is a disjoint checkpoint from each of the same three developers, so the comparison does not rest on one triple. κ is Fleiss' κ over the panel, $\kappa_{\max}$ the Umesh ceiling those raters' own marginals permit, AC_1 Gwet's prevalence-robust coefficient. The second panel's near-zero κ should be read against its ceiling of 0.12, not as noise: two of its checkpoints sit at opposite ends of the threshold range (0%–90% of claims called sound), so their marginals barely overlap and almost no κ is attainable. That is the threshold heterogeneity of §11 in its sharpest form. Intervals in `human_anchor_results.json` resample sources.

Population	N	census-eligible	obs. agr.
Published claims	80	5–8%	0.98
Screen windows	704	100% (by selection)	0.45

Table 49: Why the human anchor is not transferred to the census. The same three deployed judges apply the same census codebook to the 80 published claims the human panel coded and to the 704 eligible screen windows of §10. Only 5–8% of the claims are eligible under the census rule, which demands a number credited to a different paper, so the two populations barely intersect and the high agreement in the first row is judges defaulting to one label on ineligible items, not an easier task. We therefore report the human result as evidence about adjudicating published claims and compare no κ across the two rows.

Developer (models)	Probe	C2 Δ	C5 Δ
01.AI (1)	+12.3	-5.2	+2.5
AI2 (2)	+2.3	-4.3	-1.6
Alibaba (5)	-0.2	+10.4	+2.1
Anthropic (2)	+0.4	-2.3	-7.0
Google (2)	+0.9	+6.6	-3.2
HuggingFace (2)	-0.1	+2.7	-3.7
IBM (2)	-3.9	-1.3	+0.1
Meta (1)	-4.8	+5.6	-3.5
Microsoft (2)	-0.4	+10.2	+3.1
Mistral (2)	-7.0	-0.6	+0.0
NVIDIA (1)	+1.0	+4.7	-2.0
TII (2)	-1.8	+1.7	+5.0
Zhipu (1)	-0.8	-4.7	+5.4
Pooled (random effects)	-0.4	+3.1	+0.1
95% CI	[-1.2, +0.4]	[+0.7, +5.5]	[-1.3, +1.5]
95% pred.	[-4.4, +3.6]	[-9.8, +16.0]	[-7.2, +7.3]
I^2	90%	98%	97%
dev. level	+0.1	+1.7	+0.4

Table 50: Controlled consequence test, *full* minus *placebo* in accuracy points, aggregated by developer. Checkpoints are nested in developers, so the developer-level row averages within developer before pooling. The prediction interval, not the confidence interval, is the quantity relevant to a claim about a model population; it contains zero on both substantive strata. The competence probe is the negative control. Per-model contrasts, Holm-adjusted values, paraphrase spread and the decision-normalisation report are in Appendix Table 51.

Model	Stratum	Abl.	Plac.	Full	F−P	F−A	Para. SD	McNemar p	Holm p
Qwen2.5-14B	all	60.9	59.0	61.6	+2.6	+0.7	3.3	0.008	–
Qwen2.5-14B	probe competence	94.8	91.9	92.3	+0.4	-2.5	5.5	0.031	–
Qwen2.5-14B	c2 capability	59.9	56.6	61.4	+4.8	+1.5	5.8	0.031	0.000
Qwen2.5-14B	c5 utility	44.9	44.9	46.5	+1.6	+1.6	2.4	0.001	0.068
Qwen2.5-7B	all	60.0	56.0	59.8	+3.7	-0.2	2.4	0.003	–
Qwen2.5-7B	probe competence	89.2	85.4	82.1	-3.3	-7.1	8.8	1.000	–
Qwen2.5-7B	c2 capability	66.1	59.0	70.1	+11.1	+4.0	7.0	0.000	0.000
Qwen2.5-7B	c5 utility	39.3	38.3	38.2	-0.1	-1.0	2.1	0.383	1.000
Qwen3-14B	all	54.1	54.8	61.3	+6.5	+7.2	4.3	0.000	–
Qwen3-14B	probe competence	94.4	96.5	97.3	+0.8	+2.9	2.8	1.000	–
Qwen3-14B	c2 capability	53.5	51.7	65.9	+14.3	+12.4	5.1	0.000	0.000
Qwen3-14B	c5 utility	34.5	37.2	38.8	+1.6	+4.3	5.1	0.002	0.253
Qwen3-8B	all	59.7	60.0	65.9	+5.9	+6.2	4.2	0.000	–
Qwen3-8B	probe competence	69.0	69.2	70.0	+0.8	+1.0	4.0	1.000	–
Qwen3-8B	c2 capability	74.0	74.5	80.5	+6.0	+6.6	7.9	0.016	0.000
Qwen3-8B	c5 utility	40.8	40.9	49.2	+8.2	+8.3	8.6	0.000	0.000
OLMo-2-13B	all	47.4	46.0	47.8	+1.8	+0.4	5.8	0.016	–
OLMo-2-13B	probe competence	69.6	65.0	70.6	+5.6	+1.0	10.5	0.016	–
OLMo-2-13B	c2 capability	50.7	48.3	53.1	+4.8	+2.4	3.1	0.000	0.000
OLMo-2-13B	c5 utility	32.9	34.2	31.0	-3.1	-1.9	9.2	0.021	0.000
OLMo-2-7B	all	41.5	40.2	41.7	+1.4	+0.1	7.8	0.500	–
OLMo-2-7B	probe competence	43.1	48.1	44.8	-3.3	+1.7	4.5	0.500	–
OLMo-2-7B	c2 capability	44.8	46.9	44.3	-2.6	-0.5	5.5	1.000	–
OLMo-2-7B	c5 utility	37.5	29.7	37.5	+7.8	+0.0	15.6	1.000	–
Granite-3.1-8B	all	53.8	54.8	53.7	-1.2	-0.1	2.3	0.011	–
Granite-3.1-8B	probe competence	81.9	85.2	78.1	-7.1	-3.7	7.7	0.003	–
Granite-3.1-8B	c2 capability	55.9	56.4	57.6	+1.2	+1.7	3.5	1.000	0.200
Granite-3.1-8B	c5 utility	37.5	38.1	37.5	-0.6	+0.0	0.7	1.000	0.213
Phi-4-mini	all	61.5	60.8	62.0	+1.2	+0.5	0.7	0.000	–
Phi-4-mini	probe competence	67.3	69.4	69.8	+0.4	+2.5	7.5	1.000	–
Phi-4-mini	c2 capability	75.8	73.2	74.0	+0.7	-1.9	4.7	0.024	1.000
Phi-4-mini	c5 utility	44.2	44.0	46.0	+2.1	+1.9	2.1	0.000	0.213
Falcon3-7B	all	58.4	58.1	59.1	+1.0	+0.7	2.1	0.860	–
Falcon3-7B	probe competence	66.7	66.7	66.7	+0.0	+0.0	0.0	1.000	–
Falcon3-7B	c2 capability	68.8	68.3	69.0	+0.6	+0.2	1.2	1.000	0.901
Falcon3-7B	c5 utility	44.0	43.5	45.5	+2.0	+1.6	4.3	0.678	0.302

Table 51: Per-model, per-stratum results for the version-2 controlled experiment. Accuracies are percentages averaged over four paraphrases; F−P and F−A are the full-minus-placebo and full-minus-ablated contrasts in percentage points; *Para. SD* is the standard deviation of the full-minus-placebo contrast across paraphrases. McNemar tests use per-item paraphrase-majority decisions. Holm adjustment covers gate-passing models on the two substantive strata.

S Framework Details and Positioning

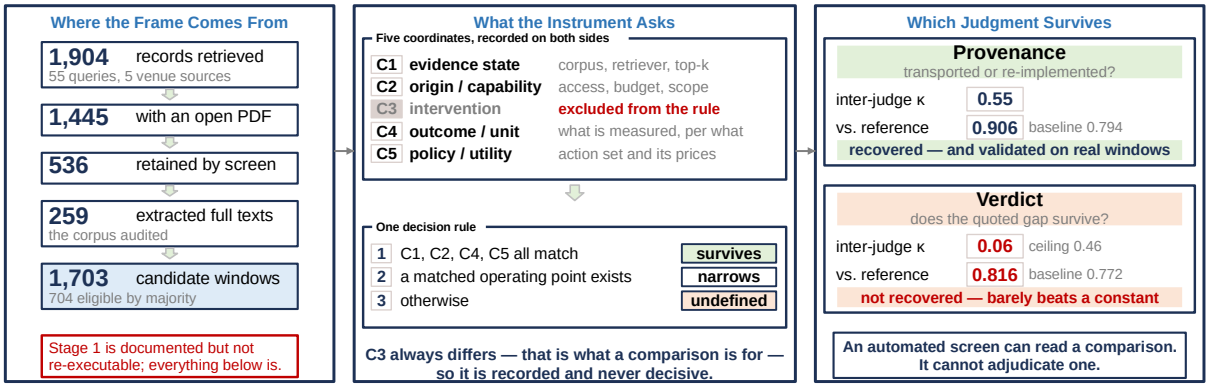


Figure 10: The taxonomy as an instrument. *Left:* the documented screening flow from retrieval to the 259 extracted full texts the survey reads and the audit runs over. *Middle:* the five coordinates and the one decision rule they feed, with the intervention coordinate recorded but excluded. *Right:* which of the instrument's two judgments survives contact with published text.

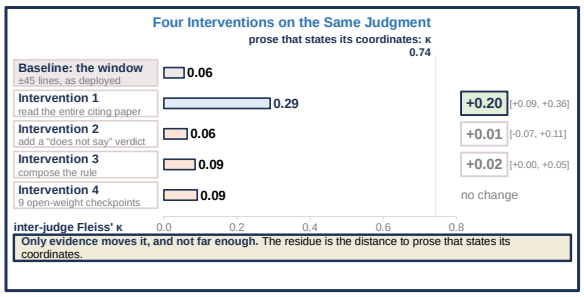


Figure 11: Why adjudication is not recoverable, established by intervention rather than by inference. Judges, codebook and items are held fixed and one thing is varied at a time. Only supplying the papers themselves moves inter-judge agreement, and it stops well short of the level the same instrument reaches on prose written to state its coordinates.

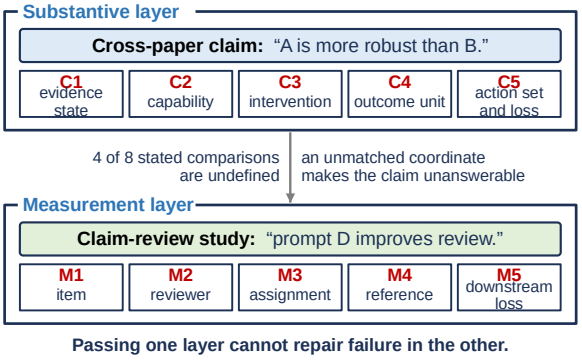


Figure 12: The two layers answer different questions about the same sentence. The substantive layer asks whether two published results concern one decision problem; Table 24 applies it to real comparisons and supplies the four-of-eight count. The measurement layer asks what a study of claim review estimates; Table 43 shows which analyses of such a study are valid. Passing one layer cannot repair failure in the other.

Adjacent line	Primary unit	Unresolved comparison question	This paper's increment
General RAG surveys (Gao et al., 2024; Sharma, 2025)	Architectures, modules, applications	Whether similarly named gains concern the same evidence state, outcome, and experimental game	Adds a claim-level comparison lens and an estimand audit; does not claim a venue-complete census.
Evaluation surveys and unified comparisons (Yu et al., 2024; Wang et al., 2024b; Alushi et al., 2026)	Metrics, benchmarks, and common-harness reruns	Which controls are required by a specific cross-study or mechanistic inference when rerunning is impossible	Separates scientific coordinates from assignment, participant, item, and reference units; the simulation quantifies one resulting failure mode.
Trustworthy and security RAG surveys (Zhou et al., 2024; Ni et al., 2025; Xu et al., 2026)	Trust dimensions, attack surfaces, and defenses	Whether benign conflict and attacker-created corruption share a transferable capability envelope	Makes origin/access/budget explicit and shows why a prompt-overlapping reference cannot establish independent criterion validity.
Conflict and abstention surveys (Xu et al., 2024a; Wen et al., 2025a)	Conflict types and refusal behavior	Whether a confidence or sufficiency signal identifies the action that maximizes utility at a matched operating point	Separates signal, action, and utility and releases a consequence-defined counterfactual target without claiming an evaluated policy.

Table 52: Positioning relative to adjacent surveys and empirical comparison work. Each adjacent line partitions the literature by mechanism, which is what a reader needs to *find* related work. None partitions the *comparison*, which is what a reader needs when two numbers appear together, and none is accompanied by evidence that its categories can be read off a paper. The increment is a reporting-coordinate reduction that is tested (§9) and applied at scale (§10); it is not a venue-complete systematic review, and it does not supersede these surveys for discovery.

Field	Question	Representative values	Why it changes a comparison
Evidence state	What is observable at the decision boundary?	Supported; distracting; insufficient; conflicting; stale/weak provenance; adversarially manipulated; not evidence-conditioned.	Query ambiguity and model uncertainty are not silently mixed with properties of retrieved evidence.
Origin / capability	How did that state arise, and what can an attacker observe or change?	Benign retrieval; constructed stressor; natural drift/conflict; distribution shift; attacker-created evidence with access, budget, and target.	Transfer depends on the data-generating process and the attacker's capability envelope.
Intervention	What part of the system changes?	Corpus/index; retriever; reranker; context assembly; generator; verifier; system control; evaluation artifact.	Components may be alternatives in a shared harness or complementary stages in a pipeline.
Outcome	What object and unit are measured?	Answer correctness; retrieval quality; claim support; attack success; privacy leakage; calibration; selective risk; cost.	Identically named datasets do not make different target quantities interchangeable.
Policy / utility	What response is selected, under which costs and stakes?	Answer; verify; re-retrieve; expose conflict; clarify; quarantine; recover; abstain; coverage–risk utility.	A signal becomes operational only through an action set and a declared utility context.
Control check	Does the experiment support the intended inference?	Dataset and corpus snapshot; retriever; top-k; generator; decoding; budget; baselines; uncertainty.	Tuple compatibility alone cannot remove experimental confounding.

Table 53: The diagnostic schema. The five fields identify whether two results address a compatible question; a claim-specific control check then determines whether their reported numbers support the intended inference. The fields are not assumed orthogonal or sufficient.

Coordinate	Controlled values	Equivalence / boundary rule	Verdict effect
Evidence state	Supported; irrelevant/distracting; insufficient; conflicting; stale/weak provenance; adversarially manipulated; not evidence-conditioned.	Code what is observable at the decision boundary. Query ambiguity, OOD input, confidence, and attacker access are not evidence states by themselves.	A direct performance claim normally requires a common state or an explicit cross-state estimand.
Origin / capability	Benign retrieval; constructed stressor; natural conflict/drift; query or distribution shift; attacker-created evidence; shared clean harness.	Record how the state arose. For security, include corpus write access, query control, retriever visibility, generator access, insertion budget, objective, and target.	A cross-source rank over two different games narrows at best and is otherwise undefined; a shared harness may deliberately vary capability as the controlled factor.
Intervention point	Corpus/index; retriever; reranker; context assembly/compression; generator prompting or training; verifier; system-control policy; evaluation-only artifact.	Code the component changed. Different interventions may be compared as alternatives when a shared end-to-end harness controls the remaining pipeline; otherwise they provide component-specific or compositional evidence.	C3 never enters the decision rule. Comparing two different systems means the intervention differs, so a C3 mismatch is the normal state of a comparison and cannot by itself make one undefined; it is recorded so a reader knows what changed.
Outcome / unit	Answer correctness; retrieval quality; evidence support; attack or defense utility; privacy leakage; calibration; selective risk/coverage; cost/latency; human preference.	Record definition, direction, aggregation, and unit. Relative gains over different baselines are not automatically a shared estimand.	A one-scalar ranking requires a common target. Without one there is no operating point to re-read at, so the gap is undefined rather than narrowed.
Policy / utility	Answer; abstain; re-retrieve; filter; verify; expose conflict; clarify; quarantine; recover; not applicable; unreported.	Use *not applicable* for fixed descriptive comparisons. When policies compete, report action set, stakes, preferences, coverage, and cost.	Different actions can be ranked only under common utility; a signal alone does not establish the preferred response. A stated price at the other side's setting narrows the comparison.
Second-stage controls	Dataset/corpus snapshot; retriever/top-k/chunking; generator/decoding; compute budget; baselines; uncertainty/statistical design.	Select controls according to the stated inference. The tuple cannot remove confounding from unmatched experimental systems.	Tuple compatibility is necessary diagnostic information but not sufficient for a direct empirical ranking.
Final verdict	Survives; narrows; undefined.	Exactly the rule of §8, and the only verdict vocabulary used anywhere in this paper or its artifacts. *Survives*: C1, C2, C4 and C5 all match, so the quoted gap stands. *Narrows*: a threatening coordinate differs but a side reports an operating point at the other's setting, or the comparison holds on a strict subset; report that matched number. *Undefined*: no such operating point exists, or a decision-critical field is unavailable after full-text inspection.	Verdicts attach to comparison claims, not permanently to paper pairs, and a new shared-harness experiment can change the supported verdict. An undefined comparison is never replaced by a substitute number computed across the unmatched coordinate.

Table 54: The decision procedure of §8 in full. The five coordinates diagnose semantic compatibility and claim-specific controls determine whether the claim as written is warranted, but only C1, C2, C4 and C5 enter the verdict: C3 is the thing being compared and is recorded rather than tested. The three verdicts here are the same three used in the body, the census codebook and every released artifact.

Boundary case	Five-coordinate diagnosis	Second-stage control check	Supported conclusion
Different interventions, shared harness	A retriever filter and a generator-side reader address the same evidence state, metric target, and answer policy; intervention differs.	Same corpus, retriever output before intervention, generator, budget, metric, and seeds.	A valid end-to-end comparison is possible. Intervention mismatch alone does not force *partial*.
All five fields match, systems differ	Two answer-only methods target benign retrieval noise and answer correctness at the generator.	Different datasets, retrievers, model scales, top-k, and compute budgets.	The papers are conceptually comparable but their reported numbers are not directly rankable without a shared rerun or adjustment.
Action unnamed but observable	A QA benchmark never uses "action policy" but requires one answer for every item and gives no abstention channel.	Evaluation code and output schema confirm answer-only behavior.	Code *answer* when policy is decision-relevant; use *not applicable* for a fixed descriptive comparison.
Same pair, different claim	An attack artifact and a defense share a threat model but optimise attack success and defended utility.	A common attack–defense harness can evaluate both roles; a paper-only scalar comparison cannot.	Valid as components of one evaluation, partial as substitutable methods, invalid if ranked under incompatible objectives.
Same label, incompatible capability	Two poisoning studies both report attack success, but one assumes bounded corpus insertion and the other controls retrieved instructions at query time.	Capability, budget, retrieval probability, and objective do not align.	A direct ranking is invalid even if intervention, metric name, and answer behavior appear similar.

Table 55: Counterexamples delimiting the schema. Verdicts depend on the stated comparison and experimental controls; the five coordinates are diagnostic fields rather than a necessary-and-sufficient classifier.

T Reporting Checklist

Required field	What to report	If omitted	Example consequence
Evidence-state definition	Supported, irrelevant/distracting, insufficient, conflicting, stale/weak provenance, adversarially manipulated, or not evidence-conditioned.	Papers with the same robustness label may test different evidence properties.	Noise training is ranked against sufficiency prediction.
Origin / capability	Benign retrieval, constructed stressor, natural conflict/drift, distribution shift, or attacker access, budget, objective, and target.	Stressors are not portable.	Natural conflict is compared with poisoning.
Intervention boundary	Corpus/KB, retriever/context, context assembly, generator, system control, or evaluation artifact.	Pipeline modules are treated as substitutable.	A benchmark is ranked against a mitigation method.
Retrieval setting	Corpus snapshot, retriever, top-k, chunking, reranking/compression, and index drift controls.	Results cannot be reproduced or normalised.	A filter looks better because its retriever retrieved easier negatives.
Outcome and unit	Answer correctness, support, attack success, privacy leakage, selective risk, calibration, cost, or action utility, including definition and aggregation.	Different objectives collapse into one score.	Lower attack success hides clean-utility loss.
Policy and utility	Answer, abstain, retrieve again, filter, verify, expose conflict, clarify, quarantine, or recover, plus stakes, preferences, coverage, and cost when actions compete.	Accuracy and confidence are over-interpreted.	A calibrated answerer may still answer from insufficient evidence.
Model and comparison controls	Dataset, generator, decoding, compute budget, baselines, seeds, uncertainty intervals, and significance procedure.	Tuple-aligned papers can still be empirically confounded.	A method appears better because it uses a stronger generator or a larger budget.
Uncertainty / abstention handling	Coverage, selective risk, confidence target, abstention reward, and re-retrieval cost.	Abstention can hide retrieval failure.	Refusal is rewarded when re-retrieval should be the correct action.
Security threat model	Attacker access, query control, retriever visibility, insertion budget, objective, and adaptive-defense assumptions.	Security numbers are not comparable.	Prompt injection defenses are claimed to protect against corpus poisoning.
Provenance of every quoted number	Whether the other side's figure was *transported* from its paper or *re-implemented* in this harness, and at which configuration.	The one property an automated audit recovers reliably (κ 0.55) is lost, and the stratum where incomparability lives cannot be found.	A re-implementation at a strengthened attack budget reads as a defeat of the published attack.
Reviewer/judge panel and unit map	Who or what adjudicated, how many, which items each saw, and how any prompt or instruction was assigned across them.	The unit of assignment is unrecoverable, so uncertainty cannot be computed at the unit that was varied.	A source-level test rejects a true null 38.6% of the time at the heterogeneity measured here.
Pre-aggregation votes	Per-reviewer labels with stable reviewer identifiers, before adjudication or majority voting.	Only reviewer-level analyses hold their nominal size, and they need these votes.	A reported agreement rate cannot be converted into a valid reviewer-level test after the fact.

Table 56: Reporting checklist for robust RAG under unreliable evidence, in the order of §11. The first block is the five coordinates and the controls that accompany them. §9 shows why they head the list: a comparison whose answer turns on a coordinate the paper does not state is the one case in which a panel neither agrees nor is right, and C1 and C4 are the two fields whose absence produces that case most often. The last block is what the audit of the audit requires.

U Scoping-Map Audit Artifacts

Evidence layer	Unit and size	Structured coverage	Supported use and provenance boundary
Scoping corpus	536 broadly included papers	Metadata and broad relevance	Documentary recall pool; single model-assisted broad screen, not duplicated and not prevalence-estimating.
Map B	342 paper rows	Literature family and intervention; free-text metrics; no structured evidence origin or policy	Source-stratified landscape synthesis; adjudicated subsets, but not a five-coordinate map.
Claim inventory	289 claims from purposive fixed sections	Model-coded coordinates, evidence flags, and claim-specific controls	Source-linked hypothesis index; its verdict distribution and internal ablations are not validated results.
Instrument benchmark	360 constructed comparability items	Five coordinate values and a verdict fixed when each item is written; six judges from three developers under three codebooks	The instrument claim of §9. Templated prose, so it bounds what the schema can do on text that states it, not on any particular published passage.
Cross-paper census	1,703 frame windows	Nine judges, fully crossed, one codebook	The prevalence and reliability claims of §10. No verdict-based rate is an estimate.

Table 57: The released evidence layers answer different questions. Larger model-assisted layers do not validate claim-level verdicts; the human layers test an annotation procedure rather than literature prevalence. No layer is treated as an independently validated gold standard.

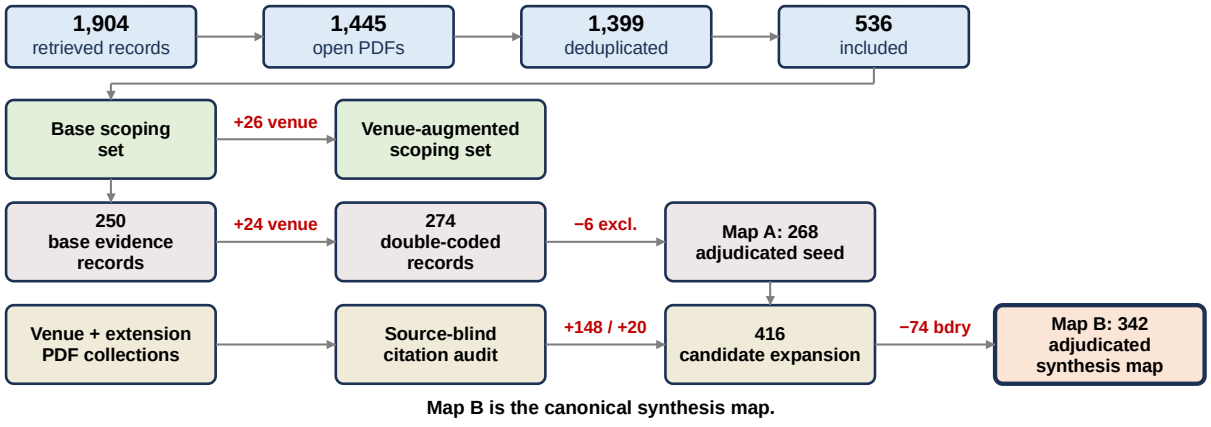


Figure 13: Corpus lineage. Red deltas on the connectors are, in order, $+26$ and $+24$ venue-linked additions, -6 exclusions, $+148$ new and $+20$ enriched rows from the citation audit, and -74 boundary rows removed at adjudication. The 416-row venue-expanded candidate map is second-coded and adjudicated into Map B (342 rows), the canonical synthesis evidence map. Map A (268 rows) is the double-coded adjudicated seed used for IAA and adjudication audit; the base and venue-augmented scoping sets are used only for boundary checks, while a separate 300-paper source-blind citation audit over the venue and extension PDF collections decides which papers enter the main text, case studies, or appendix.

Stage	Count	Rule
Initial retrieval	1,904 records	Keyword search over arXiv, ACL Anthology, OpenReview, USENIX, AAAI/ACM landing pages where open versions were available, plus curated robust/trustworthy RAG paper lists; search cutoff 2026-06-03.
Open full-text corpus	1,445 PDFs	Downloaded only open PDFs; removed failed downloads, inaccessible files, and non-PDF artifacts.
Deduplicated candidates	1,399 papers	Deduplicated by normalised title and arXiv/proceedings identifiers; retained the most complete metadata record when preprint and proceedings versions overlapped.
Relevance screening	536 included papers	Included papers that study RAG robustness, reliability, uncertainty, calibration, faithfulness, attack/defense, or evaluation under unreliable evidence; excluded application-only RAG papers.
Core pool	112 papers	Papers directly shaping the taxonomy, threat/failure model, intervention mechanisms, or evaluation framework.
Core reading set	100 papers	Core papers used as main narrative anchors; the remaining 12 core-pool papers stay in the broader scoping set.
Base scoping set	300 papers	Core papers plus representative supporting work across all taxonomy branches before venue follow-up triage.
Venue-augmented scoping set	326 papers	Base scoping set plus 26 venue-linked additions after follow-up triage.
Candidate evidence records	274 papers	Base evidence map of 250 papers plus 24 venue-linked additions, double-coded and adjudicated.
Map A adjudicated seed	268 papers	Candidate evidence records retained after final adjudication; used for IAA/adjudication provenance.
Map-B candidate expansion	416 papers	Map A plus 148 source-blind human-citation-audit additions and 20 enriched original rows; used for second-coding review.
Map B adjudicated synthesis map	342 papers	Map A plus 74 retained additions after second coding and final adjudication; frozen for current descriptive statistics. Post-freeze discoveries enter a recall log until a new version is coded.

Table 58: Screening-flow summary. Search ends on 2026-06-03; coding and adjudication artifacts freeze on 2026-06-15. The supplement provides query strings, source counts, rules, exclusions, hashes, and a post-freeze recall log.

Artifact / set	Count	Count definition	Use in this paper
Map-A double-coded candidates	274	Candidate records coded by two human sources before final inclusion adjudication.	Used for human-human agreement and adjudication; six records were excluded from Map A.
Map A adjudicated seed	268	Included records after adjudication: 42 venue/proceedings-linked, 222 arXiv-only or emerging preprints, 2 preprints with year tags, 1 review-platform record, and 1 other/uncategorized record.	Used as the IAA/adjudication provenance snapshot, not the current synthesis denominator.
Venue-augmented scoping set	326	Original base scoping set plus 26 triaged venue-linked additions.	Used for broader scoping and boundary checks; not all rows are evidence-map records.
Source-blind citation audit	300	Human citation decisions over venue PDFs and extension PDFs.	Determines which audited papers enter the main text, case studies, appendix, background, or exclusion set; distinct from the base scoping set.
Map-B candidate expansion	416	Map A plus 148 candidate additions supported by the source-blind citation audit; 20 original rows enriched with citation-audit metadata.	Pre-adjudication expansion candidate map used for second-coding review.
Map B adjudicated synthesis map	342	Map A plus 74 retained additions after second coding and final adjudication.	Current evidence map for source-stratified synthesis and Map-B cross-tabs.

Table 59: Reconciliation of reported corpus counts. The 342-row adjudicated Map B is the canonical synthesis map; Map A and the 274-candidate set are retained to report double-coding and adjudication provenance.